

POWER SUPPLY

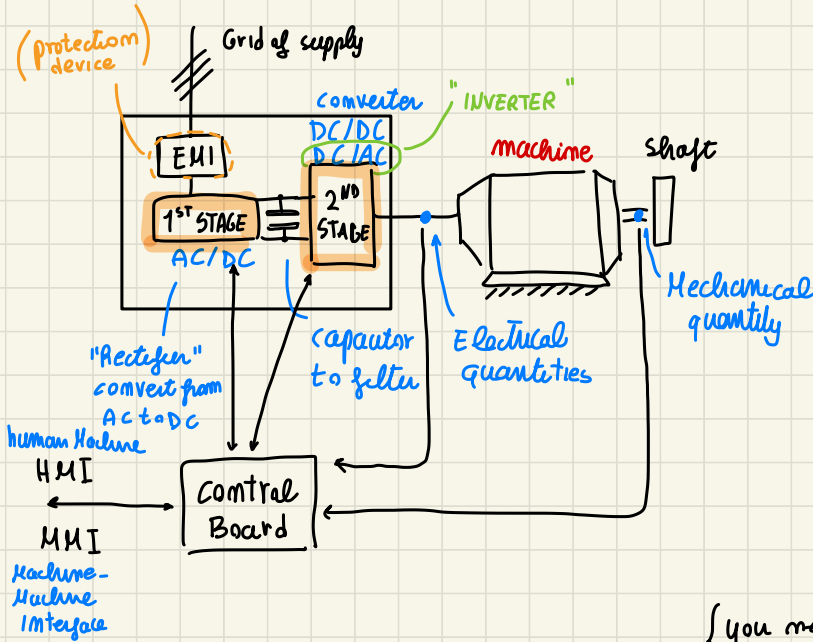
[NOT asked on Exam]

[NOT SUBJECT of the exam] → some problems related to POWER SUPPLY (affecting the operating Regions for example)

POWER ELECTRONICS

↓ problems related → which affect electrical drive operating region

ELECTRICAL DRIVE scheme

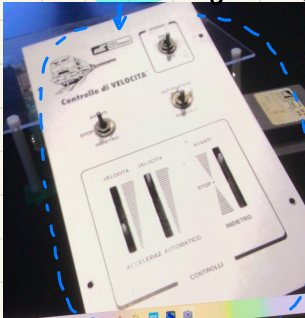


● focus of today! 1st/2nd stage of power supplier

{ you need to realize }
the machine

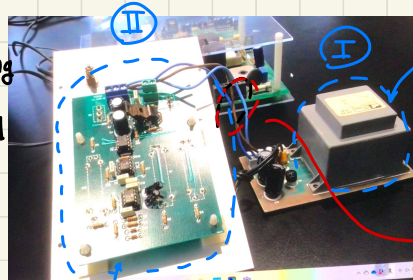
EXAMPLE 1

HMI of a toy panel .. made of 2 stages



human-machine interface
To control a small toy → change V Reference

Looking
→ behind
it..



DC/DC conversion
(II STAGE)

connection to grid with
transformer to reduce Volt.

(I STAGE)

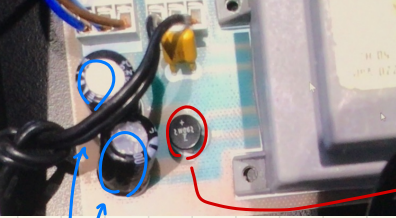
(TRANSFORMER)
produces DC
voltage, AC/DC
rectifier

3 connectors

{ for middle
point capacitor
connection

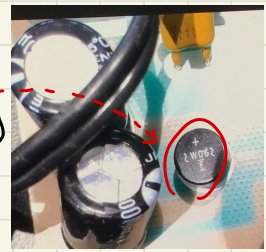
(low cost II STAGE)

I STAGE



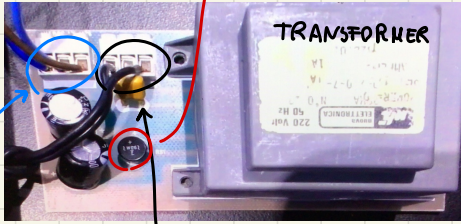
{ 4 PIN device }

DIODE BRIDGE
(2 diode connected)



AC voltage input
and (+)(-) DC output,
made out of 4 diodes.

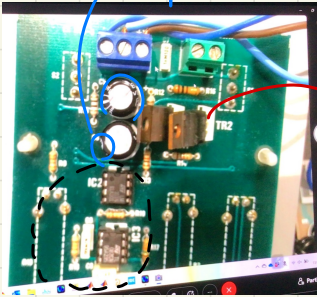
capacitors
in series
properly connect
(OUTPUT)



(INPUT)

II STAGE

Capacitor in series, to supply properly → for FILTERING

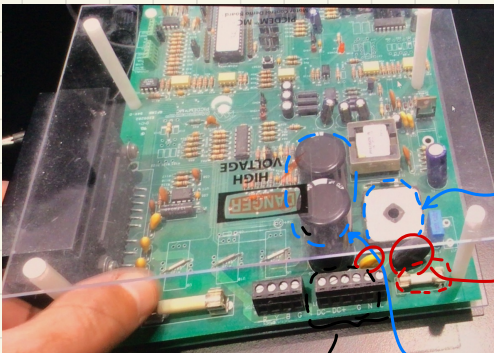


2 MOSFET (switching device)
able to close/open the signal

+ Reference
managing ⇒
device

EXAMPLE 2

other device

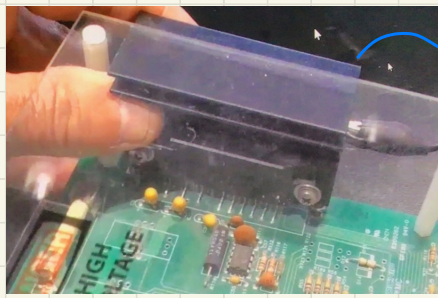


(looking from under the board)
{ 4 terminal diode bridge } → 4 diode bridge

to reduce in RUSH current

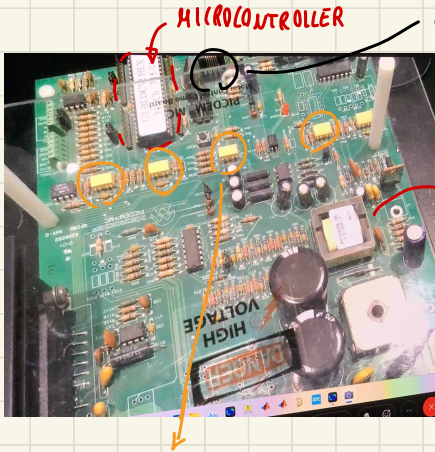
capacitor to limit voltage oscillation

connection
to the grid



INVERTER with lots of PINS
+ cooler device on it
(2 switches in series) 3 LEGS
inverter

+ some fuses
to reduce I_{mRUSH} current



connection
to programm
microcontroller

**INTERNAL
TRANSFORMER**
to realize
all voltage
needed
inside
(Auxiliary voltage)



+ additional
connection and
bottoms to set reference
+ connection for
sensors (encoder/hall effect)

control boards connected to
power electronics by this objects

(translate control board voltage on
different reference)



→ insulation between
control board and power

With different reference ground of power different respect other device GND



transform control voltage to different reference → microcontroller reference GND

→ DIGITAL SIGNALS

with current sensor low cost (i) sensor

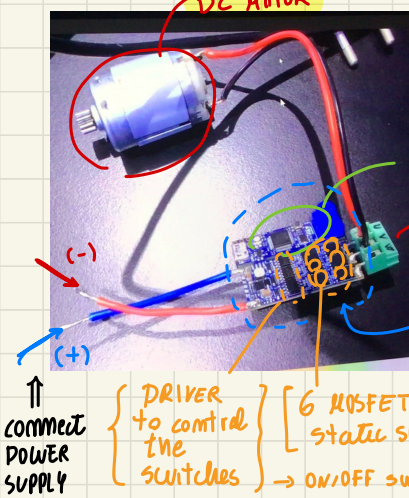
+ current sensors (low cost) used on that board

EXAMPLE 3 control of a DC motor by a drive control board

PROGRAMMER part (change swing microcontroller)

Using 2 of 3 terminals to connect it to DC machine (3 LEGS)

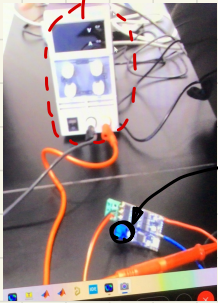
CONTROL BOARD to drive the motor



↑ CONNECT POWER SUPPLY
 { DRIVER to control the switches } [6 MOSFET static SWITCHES]
 → ON/OFF switches

↓ supply that control board

AC/DC done by POWER SUPPLY (1st STAGE)



IF I supply the system I can control the DC motor speed Ω_m
 Using the potentiometer $\Rightarrow$ by means of REFERENCE, With same voltage I'm able to reduce the speed

↓ move as I wish!

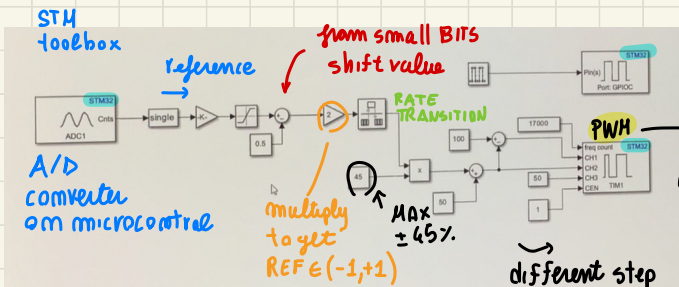
[moving on the desired direction and speed]

Open Loop approach $\uparrow$ by changing the reference: NO CONTROL SCHEME

{ able to drive 15A, }
 $\approx 600W$ SMALL
 $\downarrow$ connecting interface using position sensor INPUT

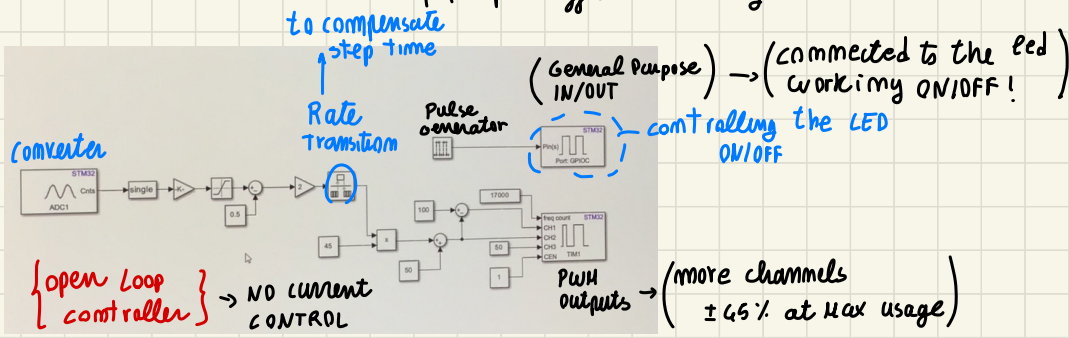
↓
 MATLAB / SIMULINK: able to produce "embedded code" to realize the CONTROL SCHEME (by proper toolbox) (with PI controller)

$\uparrow$ With CAN field bus to interface, and proper INPUT of position sensor



With 3 channels input, connected to switches of driver algorithm working on different sequence together

→ (respect different working time)



⇒ dependy on Ref I'm able to drive the Board With a certain % of Voltage!
input

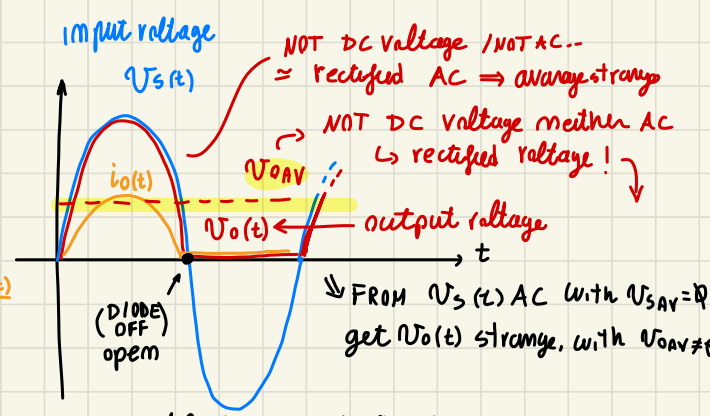
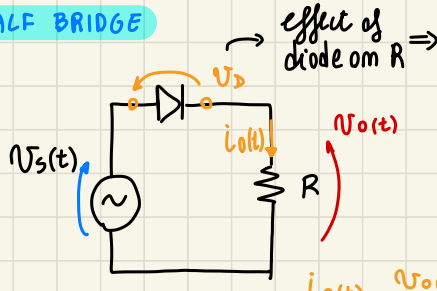
{ in MATLAB you can also work on hardware and } → translate simulink
{ translate code as "C" code from simulink } model into "C" language
easy change on the system

↓
usefull to test algorithm (and see more easy information) → simple algorithm description

↑ { STM32 64xx ecc... Board } compatible with different (good toolbox and Arduino sw to control) circuit board
{ ADC, Encoder manager, Timer, PWM, ... } lot features inside simulink Board Libraries toolbox
↳ (to design my control Algorithm!)

ELECTRONICS AC/DC CONVERTER (I STAGE)

HALF BRIDGE



effect of diode on R ⇒

$V_O(t)$

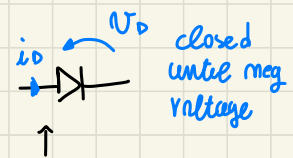
$$i_O(t) = \frac{V_O(t)}{R}$$

NOT DC Voltage / NOT AC...
= Rectified AC ⇒ average strange

NOT DC Voltage neither AC
↳ Rectified Voltage!

output voltage

FROM $V_S(t)$ AC With $V_{S,AV} = 0$
get $V_O(t)$ strange, with $V_{O,AV} \neq 0$

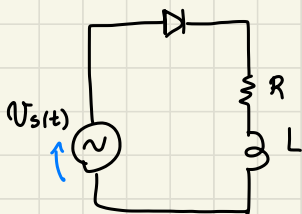


DIODE IS ON until the current goes Neg..
until OPEN, and stay OFF until $V_D > 0$, NO current $V_D = 0$

(you suppose the diode work condition) to solve the behaviour

When diode open $V_S = V_D$ because $V_D = 0$ for $i_D = 0 \leftarrow$ diode as open circuit, no current on R
 KVL $\rightarrow$ all voltage drop on diode (respect previous one)

- With RL CIRCUIT the output voltage is shifted! manage it (supply same voltage) consider a transient of 1 order $\rightarrow$ solution of 1st order transient



homogeneous + particular solution = overall sol.

[related to τ , disappear @ regime]

[solution in steady state when homogeneous, complementary sol $\rightarrow$ 0]

$\leftarrow$ complementary solution of homogeneous eq

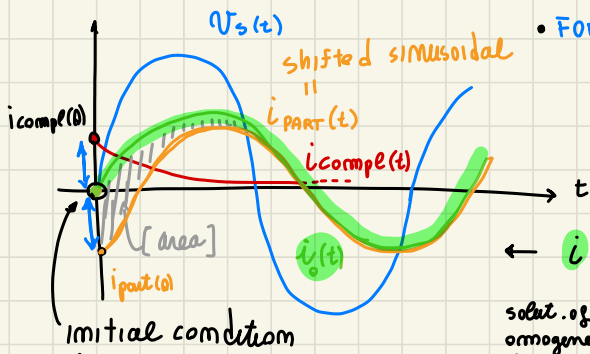
particular solution $\rightarrow$ on STEADY STATE: sinusoidal response (when supply sinusoidal $\rightarrow$ on Regime we have sinusoidal)

- For complementary sol, from dyn equation:

$$V_S = Ri + L \frac{di}{dt} \quad [1^{st} \text{ ORD}]$$

$$R + L\lambda = 0 \Rightarrow \lambda = -\frac{R}{L} = -\frac{1}{\tau}$$

{ inverse of time constant ! }



$$i = i_{\text{comple}} + i_{\text{PART}}$$

sol. of homogeneous eq: exponential

depend on initial condition!

Initial condition

$$i(t=0) = 0 \text{ @ STARTING transient}$$

$$\text{meaning } i_{\text{comple}}(0) = -i_{\text{PART}}(0) \rightarrow i(0) = 0$$

area = difference $i(t) - i_{\text{PART}}(t)$ is complementary sol (goes exponential)

$V_D(t) \neq 0$ until diode ON, remains 0N until $i_D(t) < 0$

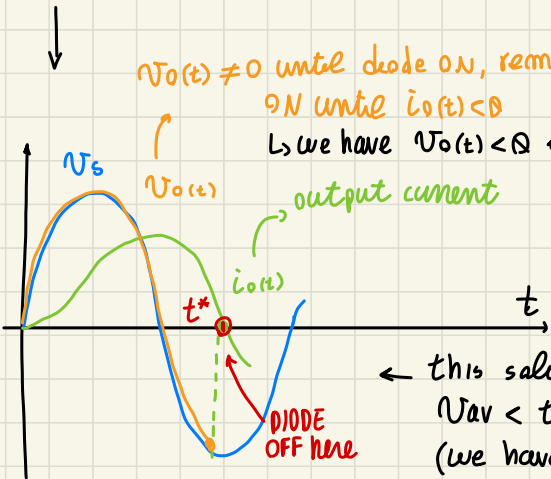
$\rightarrow$ we have $V_D(t) < 0$ on a certain range

NOT a Good Rectifier!!

(size depending on inductance)

V_{AV} depends on the t^* when $i_D(t) < 0$: which depends on shift given by impedance

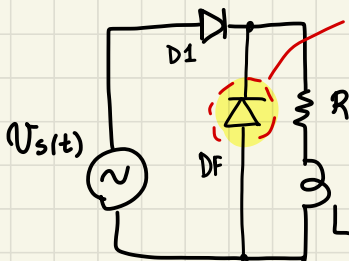
$\leftarrow$ this solution provide $V_{AV} <$ than previous rectifier (dependence on the load)
 (we have some $V_D(t) < 0 \rightarrow$ reducing average)



⇒ to solve this we introduce the free wheeling diode

FREE WHEELING DIODE

NOT both D1, DF closed or we short circuit V_s causing damage! (come on, other off)

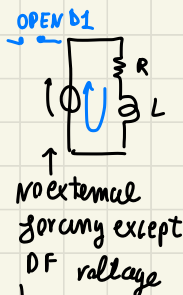
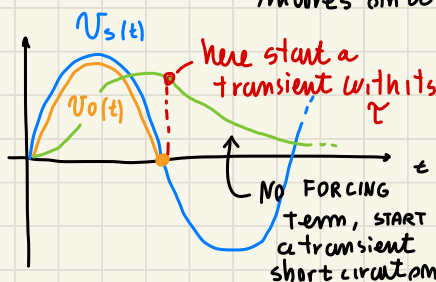
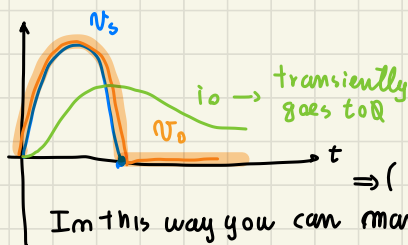


With this additional element (sign change on V_o transient internal without forcing, system works well)

With this diode, when V_o tend to be < 0 , V_{DF} applied to diode DF change sign → DF closed, all current moves on it!

New behaviour! ⇓

V_o in this condition:



⇒ (used for low cost application)

In this way you can manage a good V_{avg} independent from LOAD

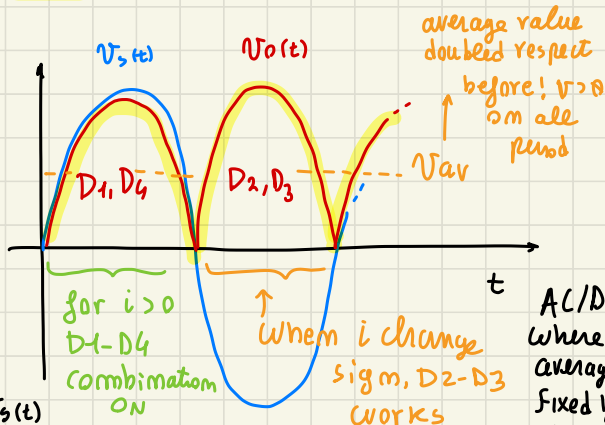
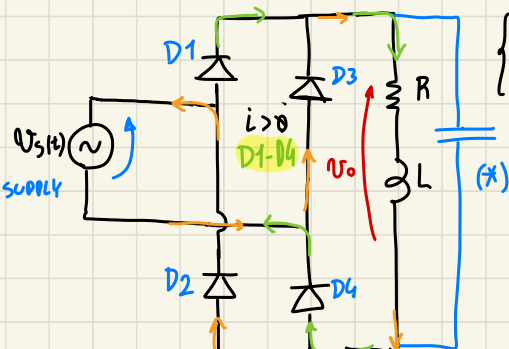
⇒ on our case we have different configuration (more complex solution.)

4 DIODE BRIDGE

Possible combination:

neither both open or just $i = 0$ useless

$\begin{cases} D1-D4 \\ D3-D2 \end{cases} \rightarrow$ [neither D1-D3 closed together neither D2-D4 or short circuit V_s !]



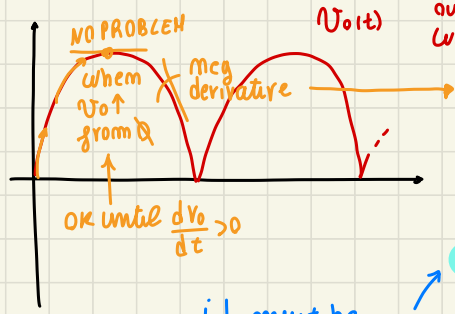
$i < 0$ (opposite respect D1-D4)

AC/DC converter

With output voltage average fixed given $V_s(t)$ [compute as integral]

AC/DC where V_o average fixed! NOT changeable!

⇒ (*) introducing the capacitor → it acts as **FILTER**



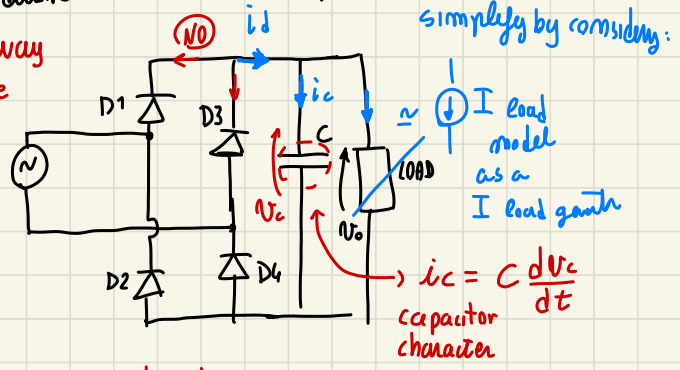
output voltage without capacitor --

What happens when $\frac{dV_o}{dt} < 0$

i_d need to satisfy
 $i_d = i_c + i_{load} \geq 0$
 condition

i_d must be positive! $i_d < 0$ NO! no way to bring current inside bridge! (DIODE D1, D3 NOT take enter current)

CIRCUIT



When $\frac{dV_o}{dt} < 0 \rightarrow \frac{dV_c}{dt} < 0$!
 $i_c < 0$!

BUT you need to guarantee $i_d = i_c + i_{load} \geq 0$
 $\leftarrow \text{When } dV_c/dt < 0$

↳ $|i_c| > |i_{load}|$ When $i_c < 0$ bring $i_d < 0$ ISSUE!!

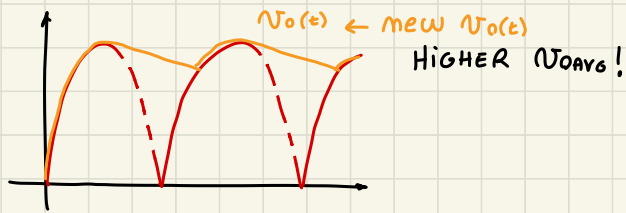
diode bridge open, disconnected from LOAD → D1, D3 open

Diode bridge
 ← get disconnected at this point!

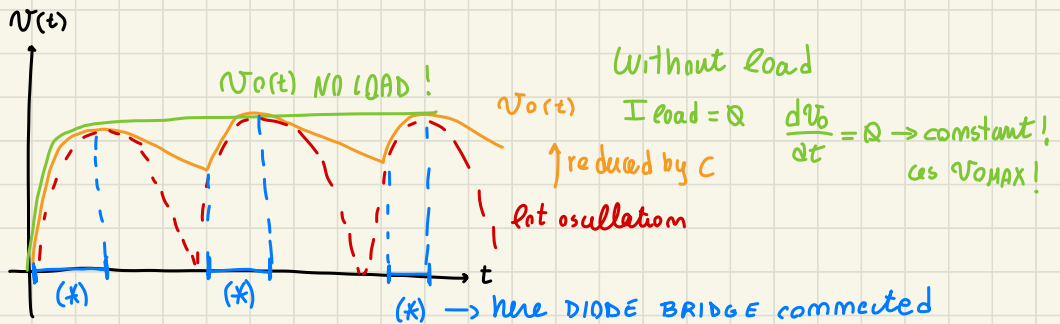
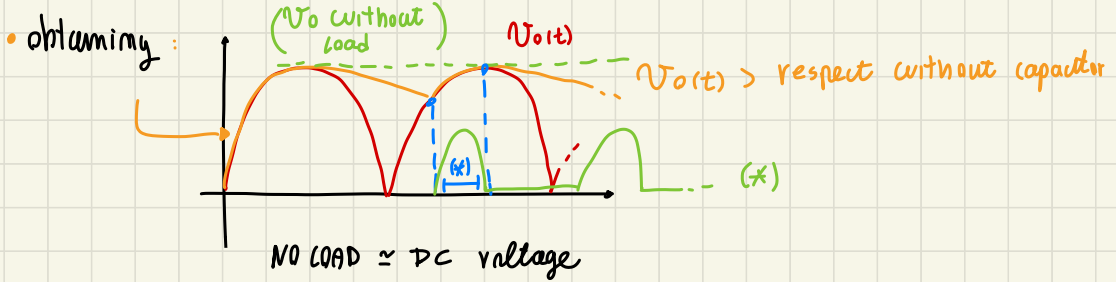
⇓ LOAD supplied only by capacitor!

So after $i_c < 0 \rightarrow$ disconnect → $V_o(t)$ goes as linear decreasing behaviour

$i_c(t) = I_{load}$ constant → $\frac{dV_o(t)}{dt} = \text{const}$
 V_o linear



disconnect load from bridge $\rightarrow$ load supplied by capacitor



$\Rightarrow$ capacitor reduce Ripple! $\Rightarrow$ less oscillation, reduce Ripple by Filtering

[classical AC/DC with capacitor] output voltage depends on the LOAD still..

$\Downarrow$

PROBLEMS on Electrical Drives:

1) DIODE BRIDGE connected to the LOAD only for short time period (x)

meaning GRID connected to LOAD for small $\Delta t \rightarrow$ so the CURRENT absorbed by the GRID appears only during those period, NOT sinusoidal

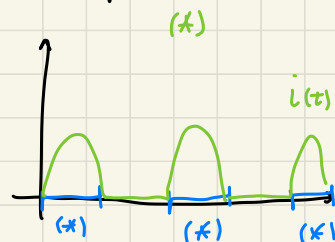
$\downarrow$

pick current!

NOWADAYS we use some

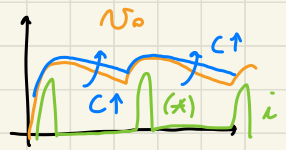
(PF) (Power factor) device, otherwise you disturb the grid by spike request

$\hookrightarrow$ interference on the GRID

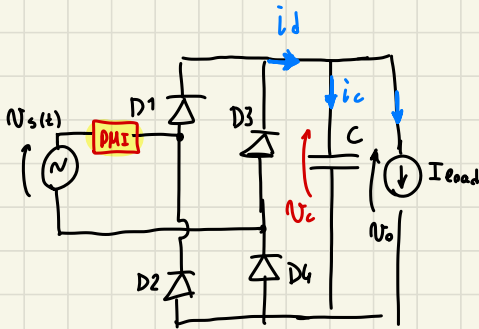


• What IF I want to DECREASE the RIPPLE:

1) I can try to increase capacitant: $C \uparrow$ more capacitant



↓
slope decrease (depends on C) → RIPPLE DECREASES (OK) ⇒ BUT this means you
MOVE energy from GRID
to load on smaller Δt !
short period but same power average in less time, ← so greater current peak (x)
higher peak ⇒ that's why you need to
introduce **PHI FILTER** to limit this effect

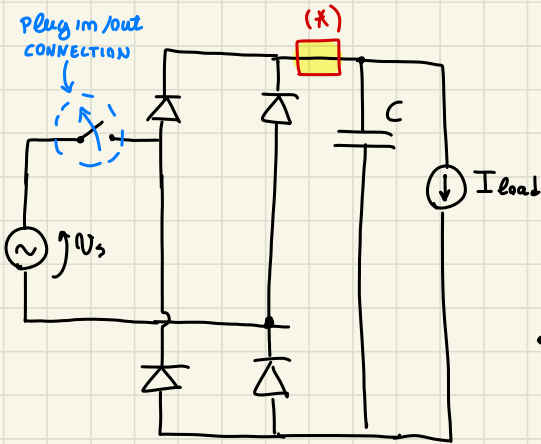


2) another problem is related to the fact that you plug your system to put GRID terminals

↳

I don't know WHEN I plug into the GRID!

Plug in/out CONNECTION



I DON'T KNOW WHEN I PLUG-IN

- IF When closing $V_s = 0$ NO PROBLEM
↓

Im general $V_c = 0$ @ Beginning

- IF closing when $V_s \neq 0$
↓

you fix $V_c = V_s$ to capacitor!

DANGEROUS! ↑ VOLTAGE on C

is the state variable

Destroy capacitor IF try to fix V_c instantly

$$\Leftrightarrow \begin{cases} dV_c \rightarrow \infty \\ i_c \rightarrow \infty \end{cases} \Rightarrow P \rightarrow \infty$$

NO instant change!



IF open when C charged, because $V_c(t)$ is state variable!

⇒ NO instant change on V_c possible, theoretically tends to ∞ power!



attention to starting condition when PLUG IN system



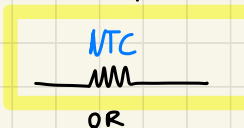
SOFT START device (*) → to LIMIT inrush current (coming from the GRID)
to LIMIT: inrush current < that can destroy circuit)

{ Resistance limiting
the current, in series of C }

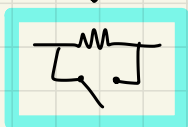


(SOFT START)

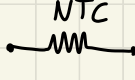
↓ realized as a Resistance in series to capacitor



Negative Temperature Coefficient Resistance,
1Ω IF cold → for $T \uparrow R \downarrow$ to 0.1Ω
(Not used to limit inrush current on device)



← industrial application, RESISTOR LIMITING current with mech switches closed when C reach regime

OR · Neg temp resistive device ... $\rightarrow$  NTC

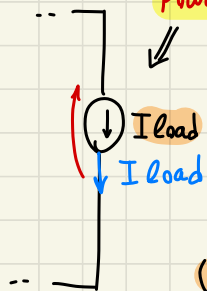
device a bit used to limit inrush current

sometimes (PTC) used, like used on FRIDGE $\rightarrow$ to manage supply machine
INDUCTION MACHINE

$\downarrow$ place this **SOFT-START** on your inverter!

3) problems: related to the FLOW of POWER

POWER FLOW like a motor { GRID $\rightarrow$ LOAD }

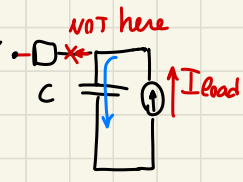


IF you try to break your system $\rightarrow$ in the power flow POV
the voltage doesn't change sign... $\rightarrow$ while
while I_{load} try to move opposite but
circulate only on capacitor (Diode open)

(**Breaking Energy**)

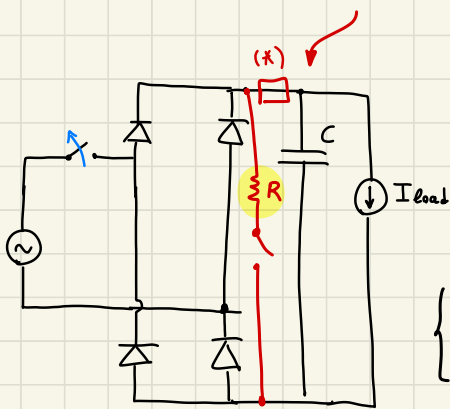
$\downarrow$ all current on,
 $V_c = \int i_c \uparrow$ increasing

$\Leftarrow$ voltage on capacitor

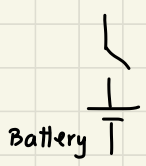


Voltage become very high
(GRID rectifying bridge don't accept
breaking energy $\rightarrow$ all on capacitor!) $\rightarrow$ to avoid this issue **we add RESISTANCE!**

on METRO train, to **reduce energy we dissipate it by Breaking resistance**



OR with a Battery sometimes

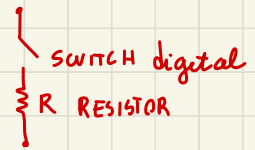


use
breaking
energy to
charge it

move energy
of breaking
to Recharge
the battery

{ Breaking energy dissipated on
Breaking resistance }

on TRAM: lots of RESISTOR used to
introduce this feature not used also in TRAINS



sometimes on power converter you have 2 terminals to connect external Resistance

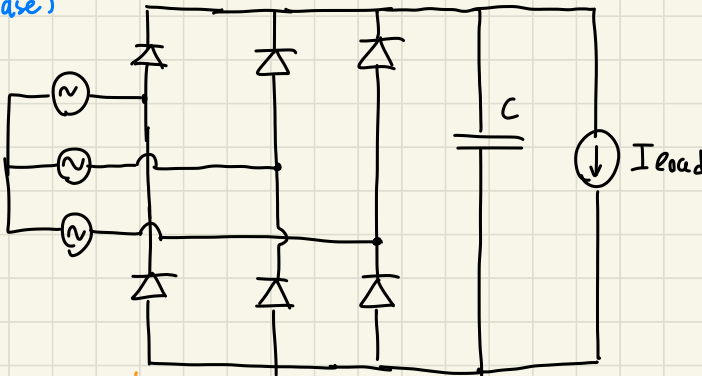
on Breaking Resistance that can be added to avoid breaking energy issue

sometimes limited BATTERY, cannot accept too much breaking energy $\rightarrow$ you dissipate it by resistance

• Another solution (usually INDUSTRIAL usage) $\rightarrow$ for $P > 6 \text{ kW}$ INDUSTRIAL

(3phase)

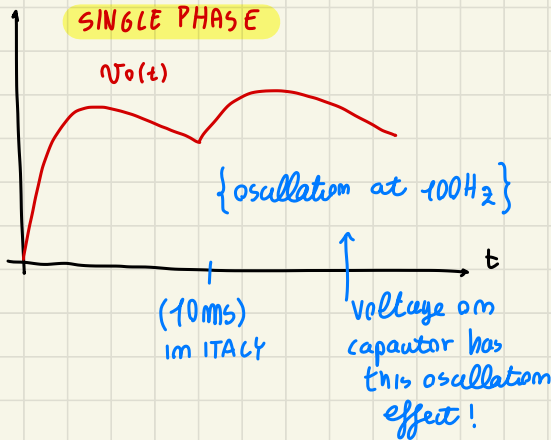
$\rightarrow$ you need to introduce 3 phase connection to the grid



6 DIODES bridge with effect similar to single phase 4 DIODES bridge

$\downarrow$ main difference between 3phase and single phase...

SINGLE PHASE

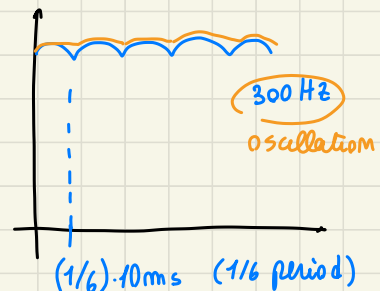


[this one Disturbance !]

$\hookrightarrow$ overall, lower oscillation, and disturbance oscillation @ 300 Hz: easier to FILTER!

(Vs)

THREE PHASE



due to capacitor $\downarrow$ oscillation easier to filter

• IF power supply done by DIODE BRIDGE, it provide oscillation $V(t)$ on my control system

I have a control voltage V_a oscillating

↓
I need to face problem that my input supply is oscillating at 100 or 300 Hz!
(NOT purely DC)

IF AS II STAGE I use DC/DC converter with 100% Duty cycle (permanently closed) on machine I see NOISE oscillation → DISTURBANCE! sometimes you need to properly compensate this disturbance by good control strategy

DC/DC converter (II STAGE)

from 1st stage I get: DC voltage (V_{DC}) as input of II STAGE

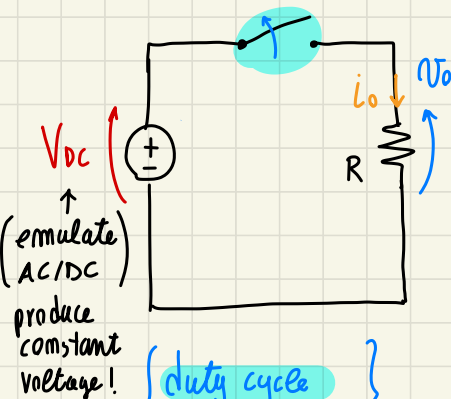
by switch you decide when to apply the voltage to the load!

⊙ AC/DC → V_{DC} with some oscillation that I neglect...

↓ consider I STAGE WORKS well!

- LOW POWER: 200KHz
- HIGH POWER: limit to 1KHz

"silicon" switch at high frequency to OPEN/CLOSE dynamically the system (NOT mechanical → SILICON, work at high freq 5KHz and so...)

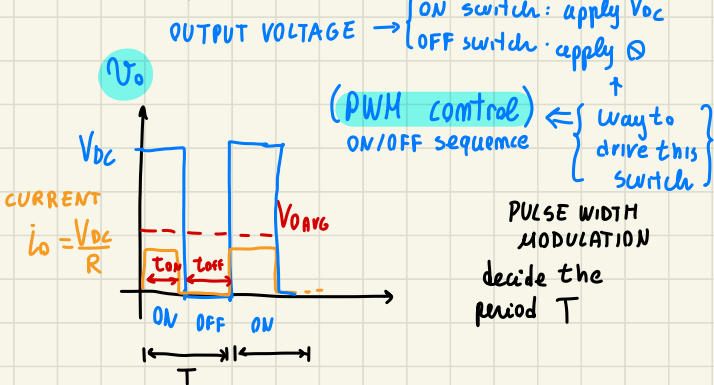


duty cycle
 $\delta = \frac{t_{ON}}{T}$
(percentage of ON respect total period)

NOT continuous voltage supply, But by δ you change the AVERAGE value of V_O

↓
 $V_{OAVG} = \delta \cdot V_{DC}$
Load Voltage

Drive average voltage value by acting on δ ← properly decide δ by PWM technique



(PWM control) ON/OFF sequence
Way to drive this switch

PULSE WIDTH MODULATION

decide the period T

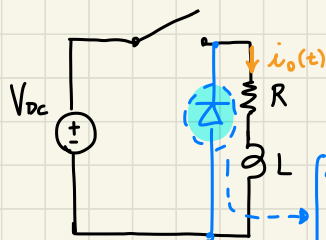
T period divided into two sub period
 $T = t_{ON} + t_{OFF}$

$f_{sw} = 1/T$ switching frequency
← (T divided into ON/OFF)

IF you INTRODUCE an **inductance** (in the LOAD...)

↳ **i_L is state variable** $\Rightarrow$ you cannot change ON/OFF the switch instantaneously!

↓ **PROBLEM on Inductance...**



$$V_L = L \frac{di_L}{dt} \rightarrow +\infty \text{ if } i_L \text{ changes instant}$$

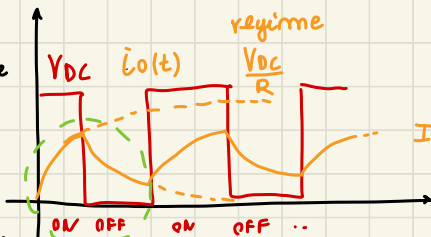
$\Rightarrow$ NO! you cannot open switch instant.
(you see a **SPARK** on circuit)

So we have to introduce a **DIODE** to drive current without damage!

FREE WILLING DIODE

When SWITCH CLOSE, work as usually
When SWITCH OPEN, current goes through DIODE

Some technique of DRIVING Voltage (PWM)
each time ON/OFF switch you go to new regime...
move to steady state

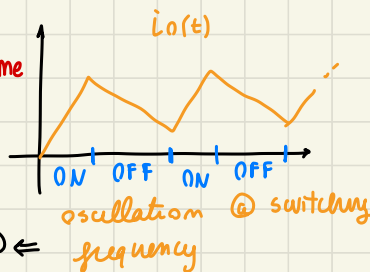


I ORD transient of current

usually $L/R = \tau \gg T$

load time constant $\gg$ switching period

goes as small piece of exp ~ time
 $\lim_{t \rightarrow 0} e^{-\lambda t} \approx \lambda t$ linear part $\approx$ TRIANGULAR signal



(DC machine noise) $\leftarrow$

$T_e(t) \sim i_o(t)$ torque affected by this noise!

one possible

expression of oscillation is

$$\Delta I = \frac{(1-\delta) V_{dc}}{f_{sw} \cdot L}$$

[current oscillation]

inverse proportional to f_{sw} and L !

NON linearity due to duty cycle

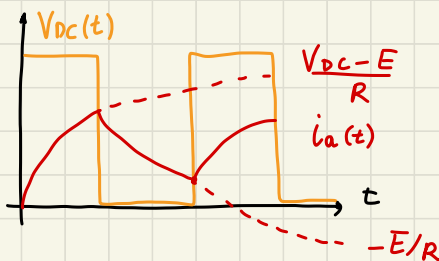
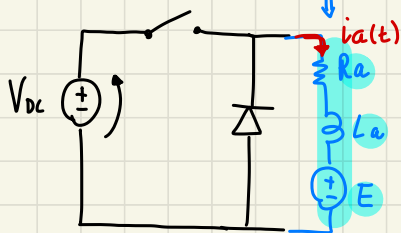
external inductor to reduce ΔI sometimes introduced
($L \uparrow$ to reduce ripple $\Delta I \downarrow$)

on "Railways": We call it **CHOPPER**

$\hookrightarrow$ "step down" / "BUCK" converter

- This WORKS for different load ... → armature coil of DC machine:

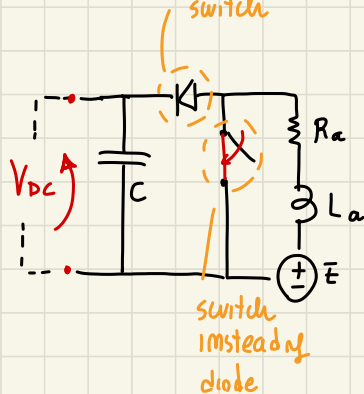
DC machine ARMATURE → (armature winding of separately excited DC machine)



(similar to previous load with different regenerative values)

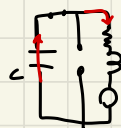
- considering the condition:

diode instead of switch ← exchange SWITCH ↔ DIODE on configuration



closing switch →
 opening switch → i_a move to capacitor
 "BOOST CONVERTER" / "step-up"

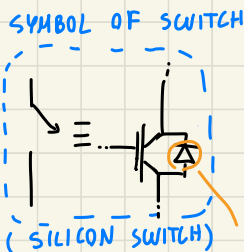
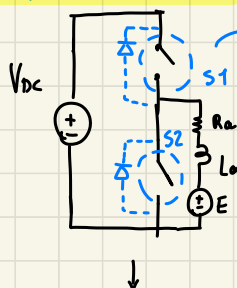
E drives current inside!



$(V_{dc} > E)$ ↓ moves towards different voltage V_{dc} from E
 ↑
 move energy from a voltage lower than output one!

another configuration: CONVERTER

HALF BRIDGE
(DC/DC CONVERTER)



used to move the machine like a motor

anti-parallel DIODE

2 quadrant

S1 close [voltage and current positive, like MOTOR]

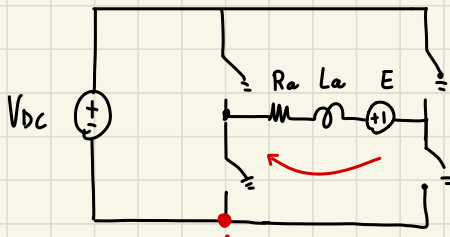
(Used to reverse the current direction)

OR S2 close [current neg: reversed, moved to DC BUS: braking! move and break]

WORKS as BOOST

OR, Best solution

FULL BRIDGE (DC/DC)



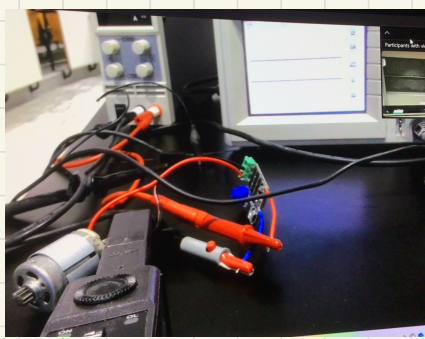
(*) (measure this voltage) positive voltage applied

With proper Switch OPEN/CLOSED you realize desired behaviour
• you can apply $V_{load} \geq 0$ (negative load voltage feasible)

+ other possible configurations
↓
use as boost or properly close to circulate energy inside cell

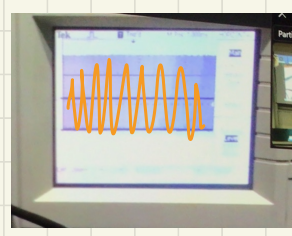
[you are able to work with $\geq$ Torque...] $\hookrightarrow$ 4 quadrant
pos/neg Voltage, also braking $T \geq 0, -\Omega_m \geq 0$

on that case... EXPERIMENT
↓

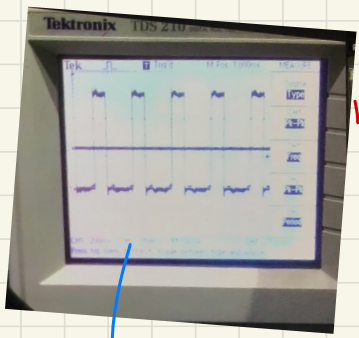


Let's see FULL BRIDGE configuration of $i(t), V(t) \hookrightarrow$ I can provide $V \geq 0$

CONVERTER V_{dc} given by P.S.
using 2 of the 3 legs of the FULL BRIDGE to control the permanent magnet DC machine
(*) voltage measure

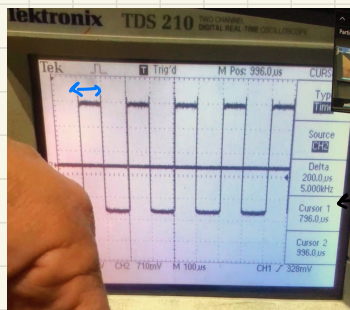


lot of noise... (noise!)



VOLTAGE
with limited speed to limit the noise

(hard measure to) select GND
I need differential probe to measure real V applied



(100 μ s of each segment)

↓
measuring overall time...

I have $T = 200 \mu s \rightarrow 5 kHz = f_{sw}$

← changing Voltage reference

I get $\delta = 50\%$ duty cycle

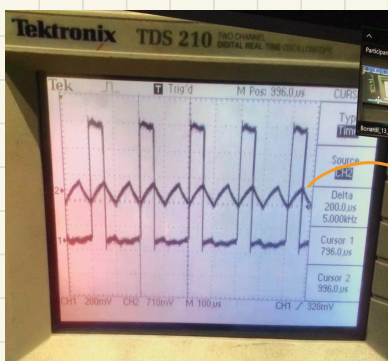
if $\Omega \uparrow$ change of voltage sign! $\Rightarrow$ so speed sign change

with different motion direction $\rightarrow$ Voltage sign change... Give more

USING:

• Unipolar technique

of CONTROL $\rightarrow$ twice current frequency respect voltage



(some oscillation visible...)

controller scheme with unipolar technique

(with proper sequence: during a period I give

$V > 0, V = 0, V > 0 \rightarrow 2$ times:

freq of current = twice voltage freq!

With transient NOT triangular

& low! comparable to T , you see

exponential ripple, if τ HIGH you don't see

$i \approx 0$ (without mechanical load)

current $i \geq 0$ oscillation! $\rightarrow$ disturbance and $T_e \uparrow$

$i(t)$



(disturbance due to OPEN LOOP APPROACH)

operation point

aliasing oscillation

(disturbance at a given freq)

torque increases $\rightarrow$ operation point as matching point

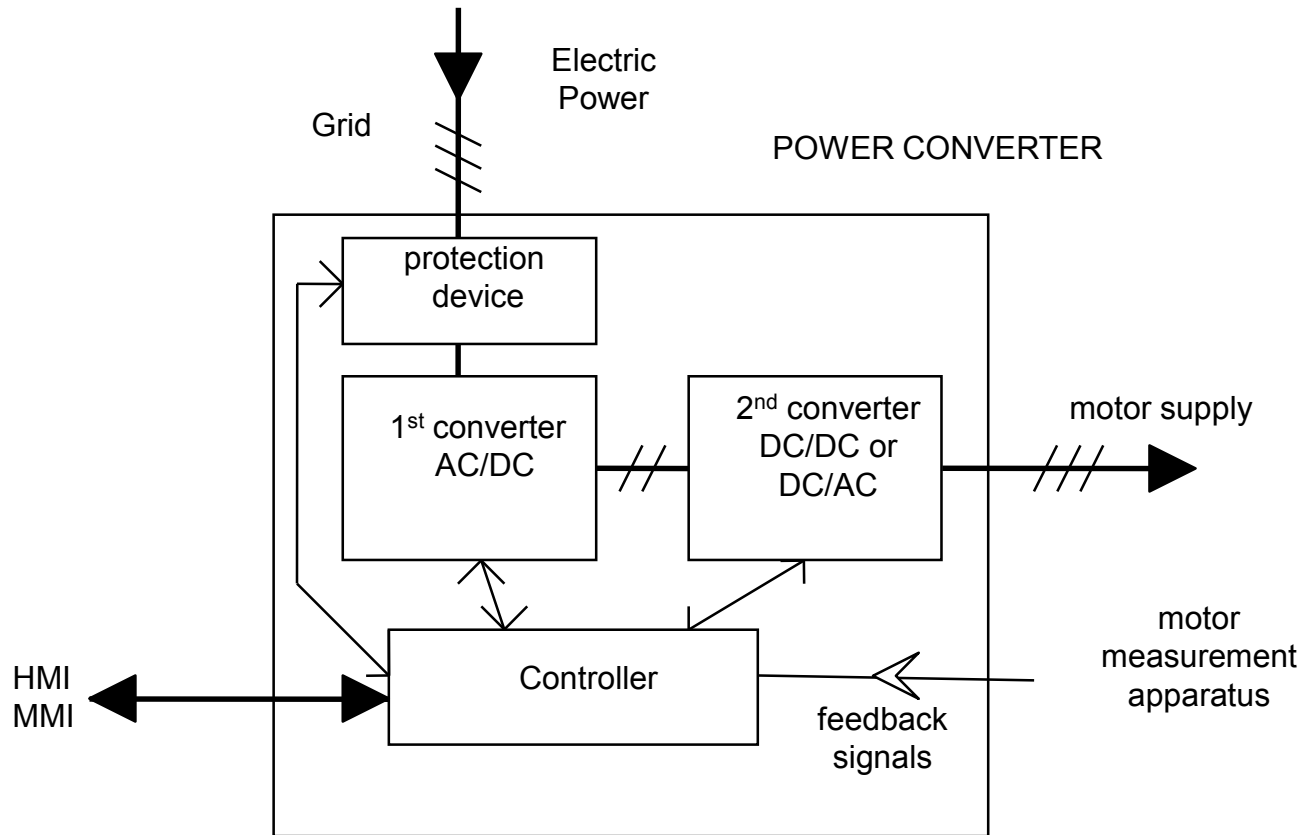
{ from a step you have current transient ! }

oscillation around acceleration torque $i \geq 0$

$i \approx 0$

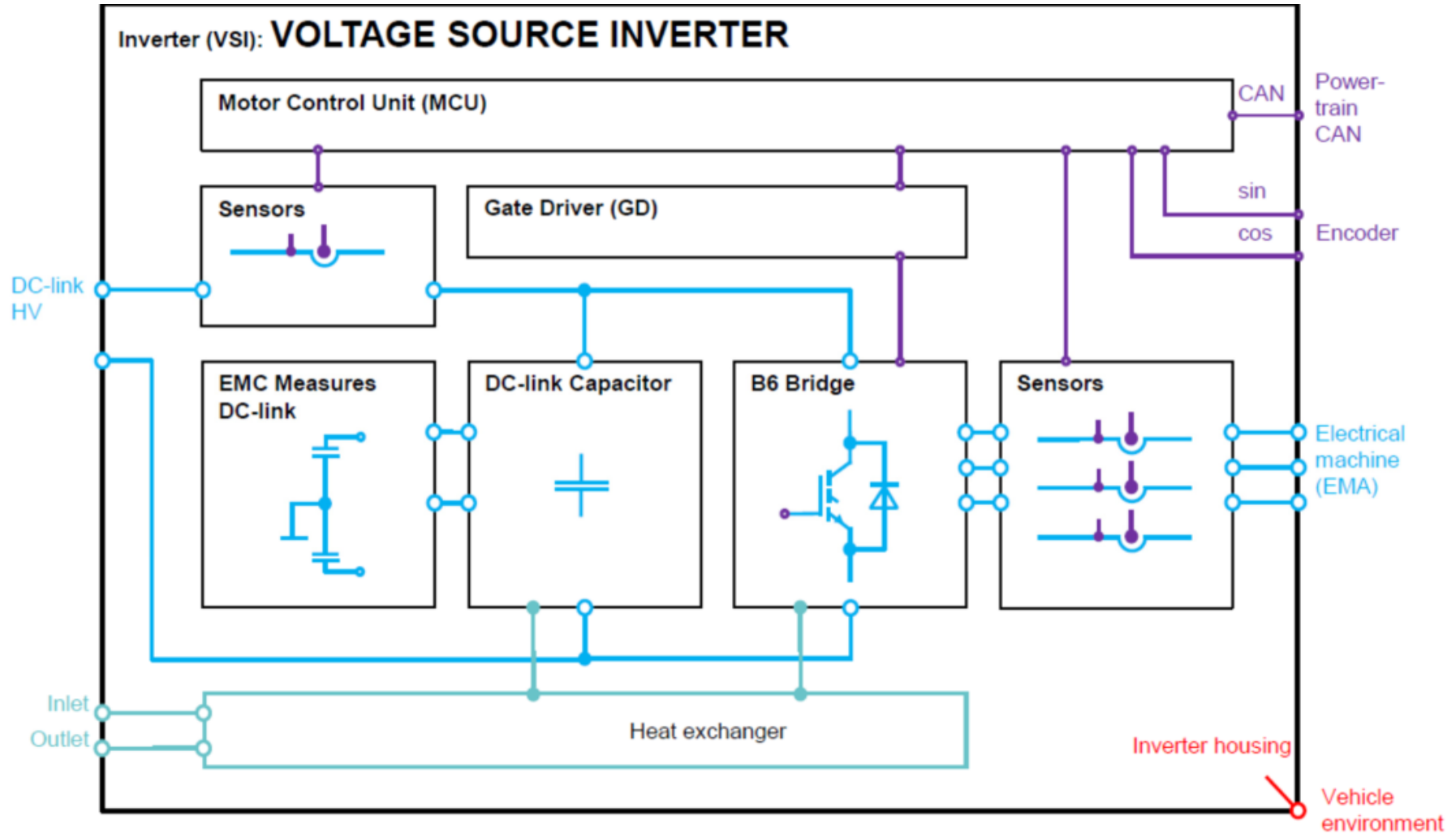


AC/DC conversion system

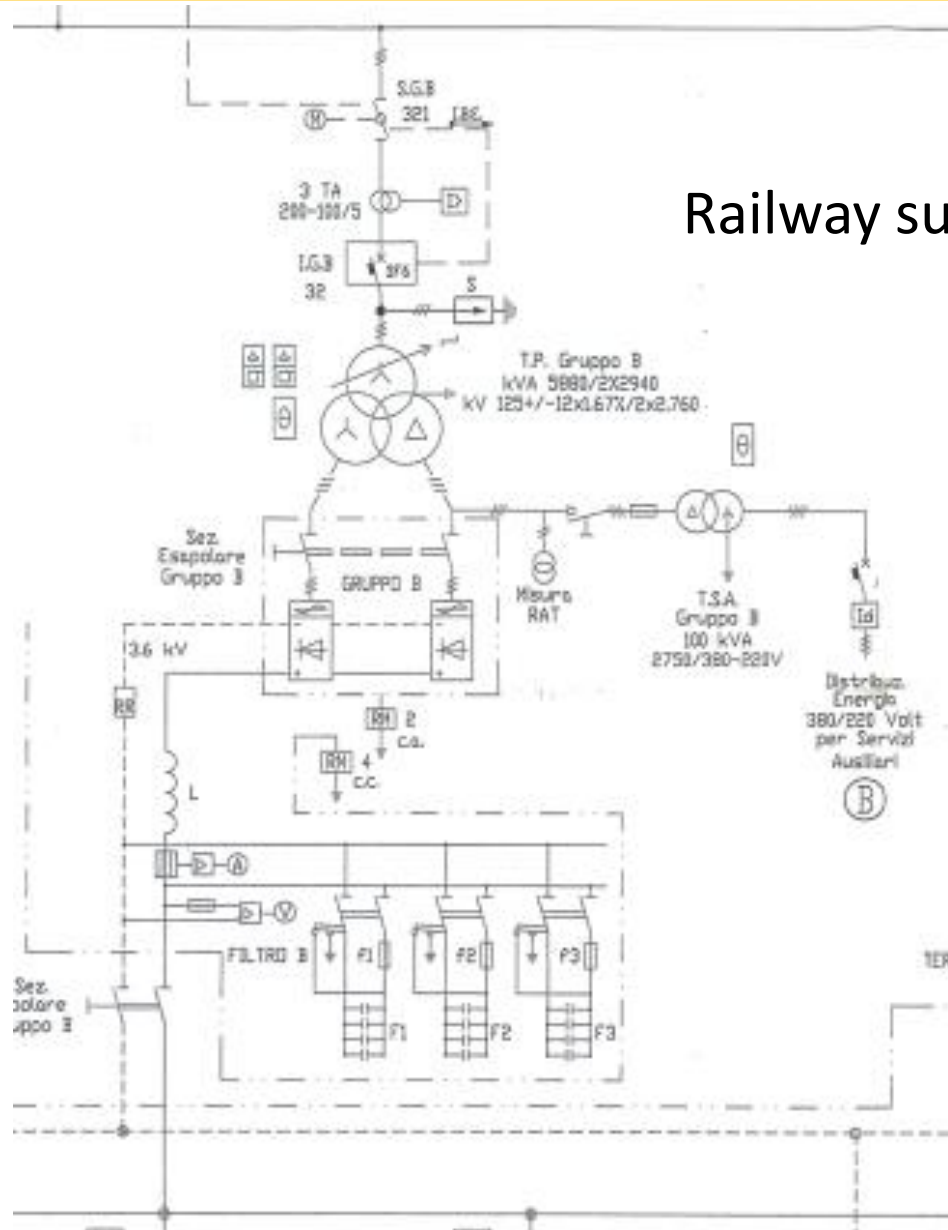


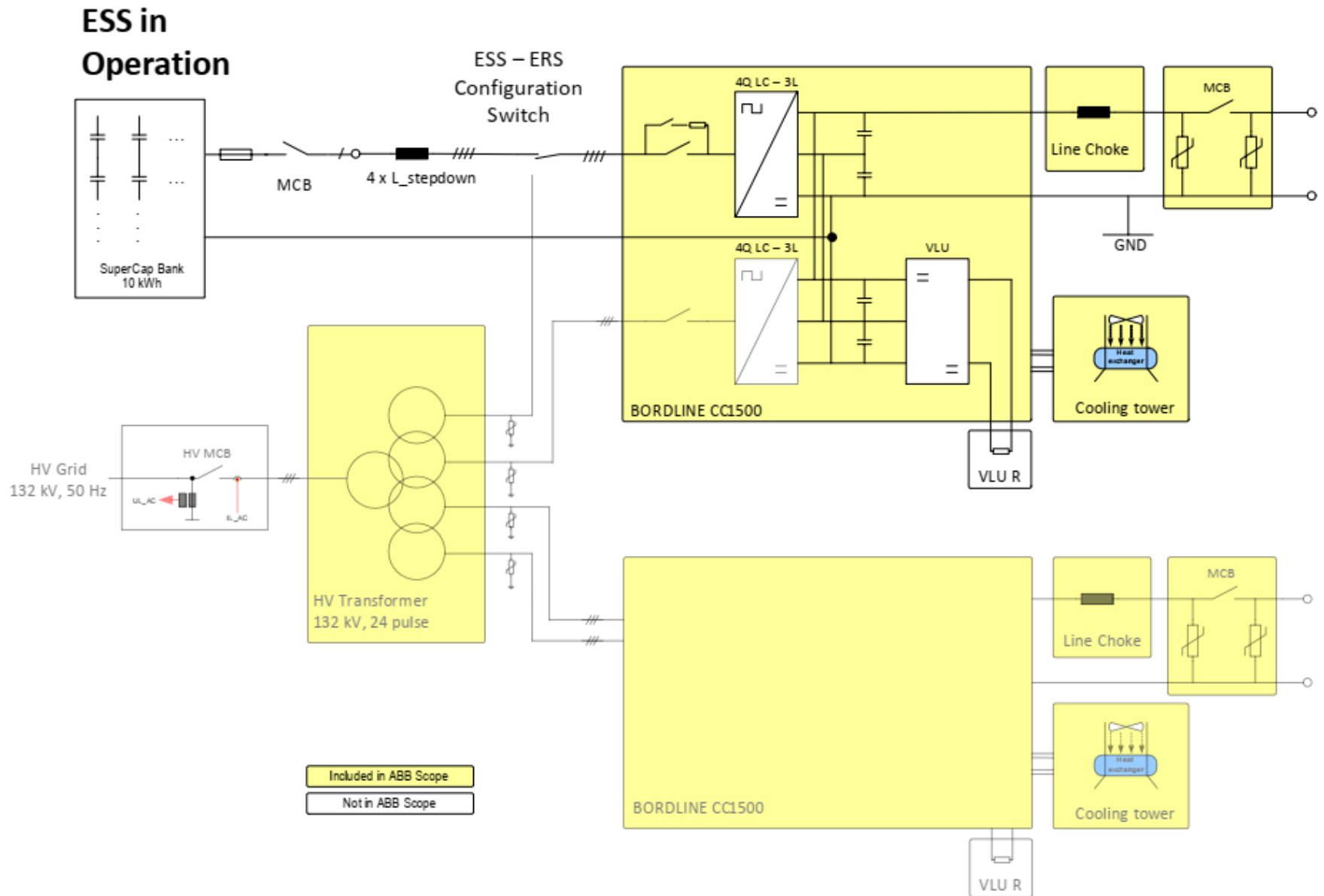
- Charging stations
- AC 25kV Trains
- Railway substations

EV power converter

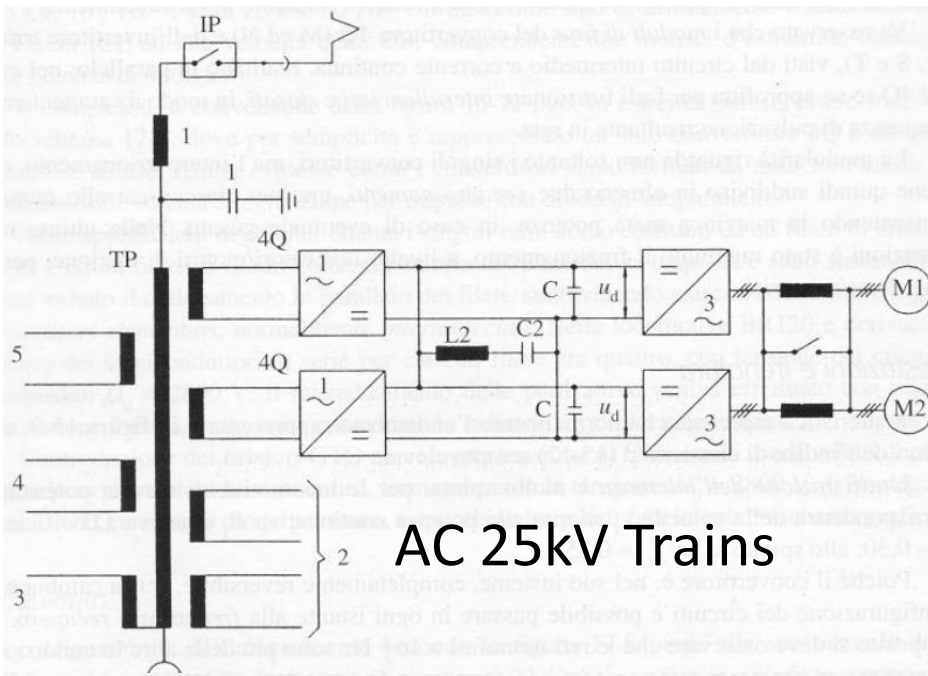


Railway substations (Forlì)

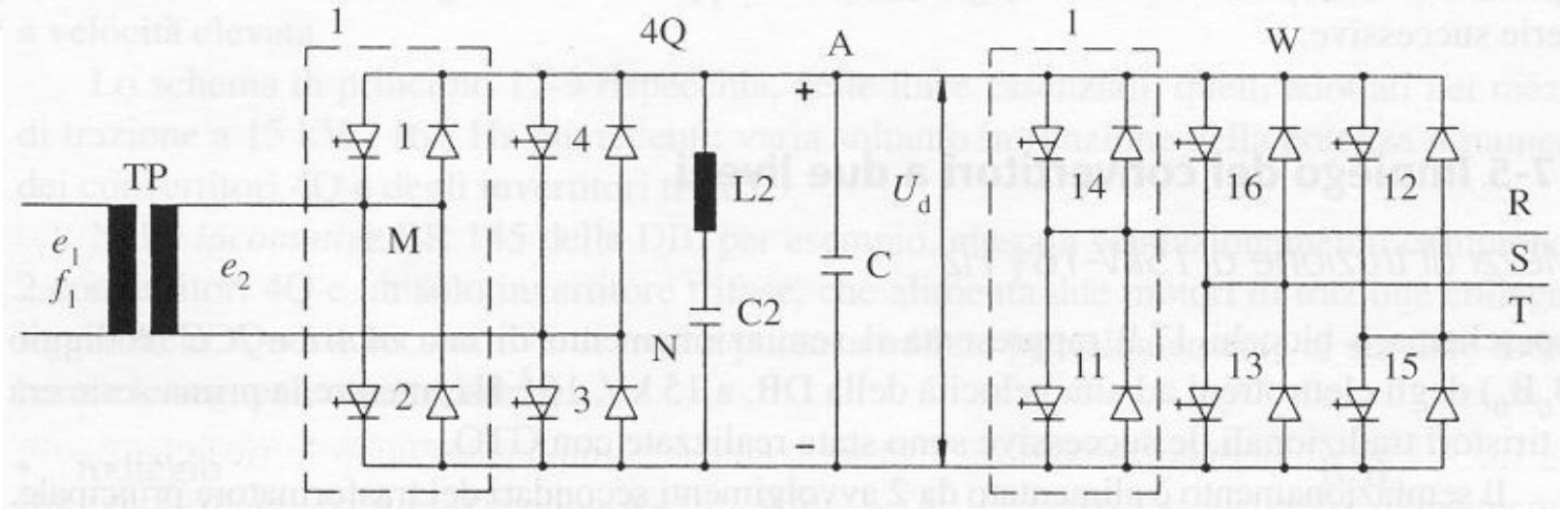
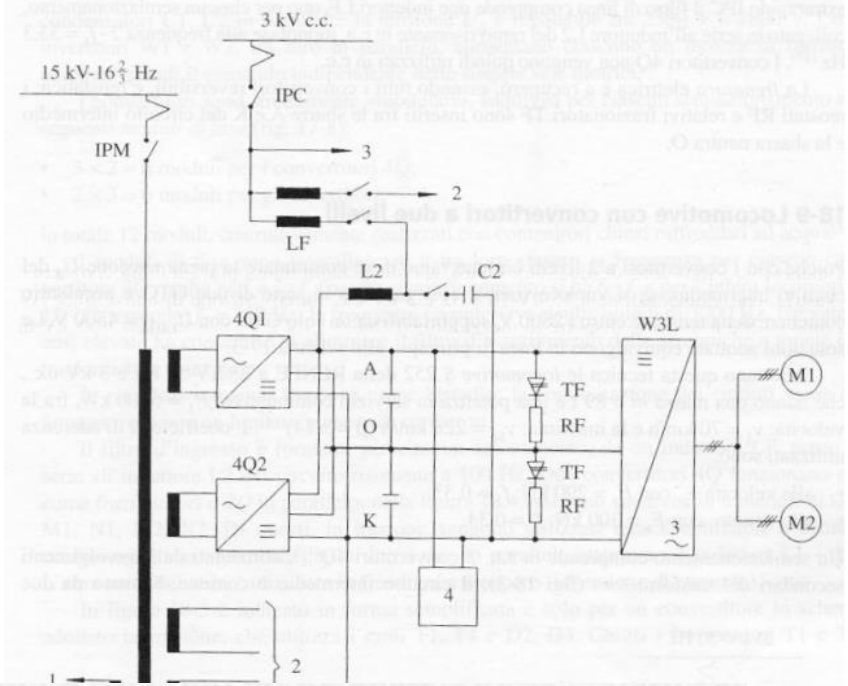




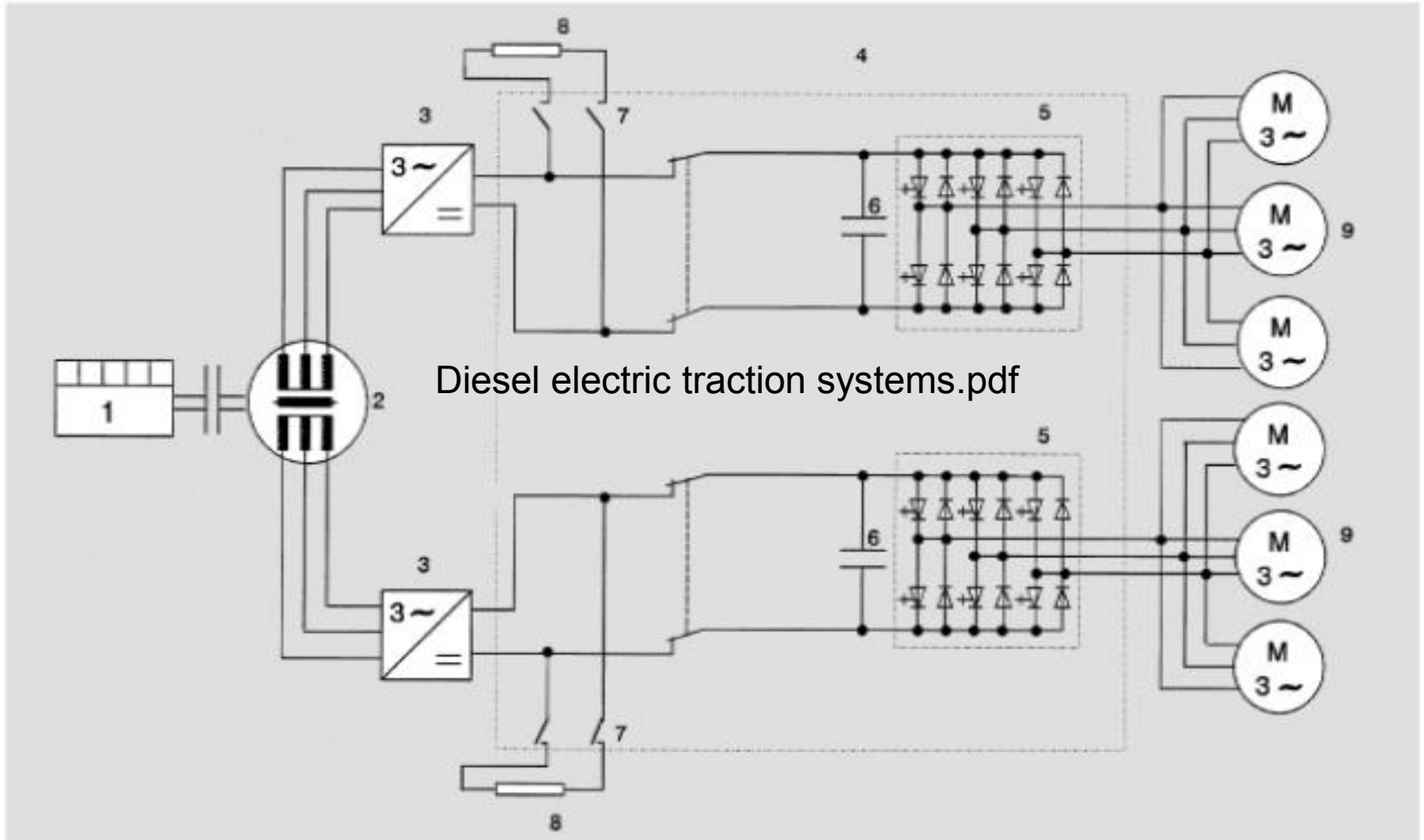
Four-quadrant rectifier



AC 25kV Trains



Diesel-electric traction systems (train, ship)



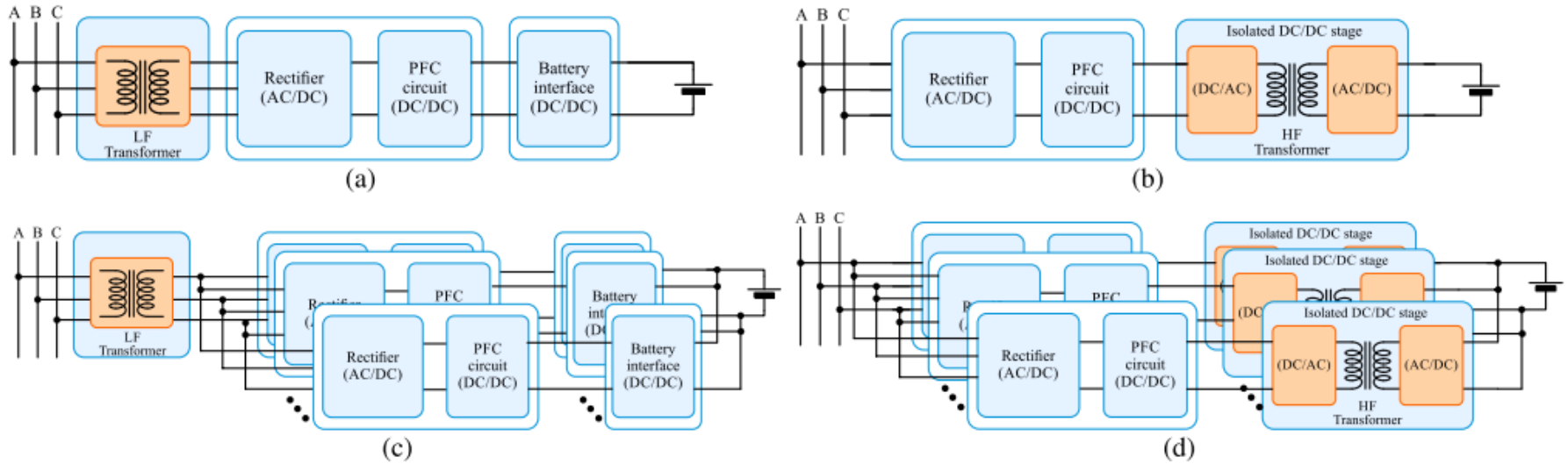


Fig. 1. Simplified block diagram of conventional dc fast charger power conversion systems. (a) Single-module charger with a nonisolated dc/dc converter. (b) Single-module charger with an isolated dc/dc converter. (c) Multiple paralleled modules shown in (a). (d) Multiple paralleled modules shown in (b).

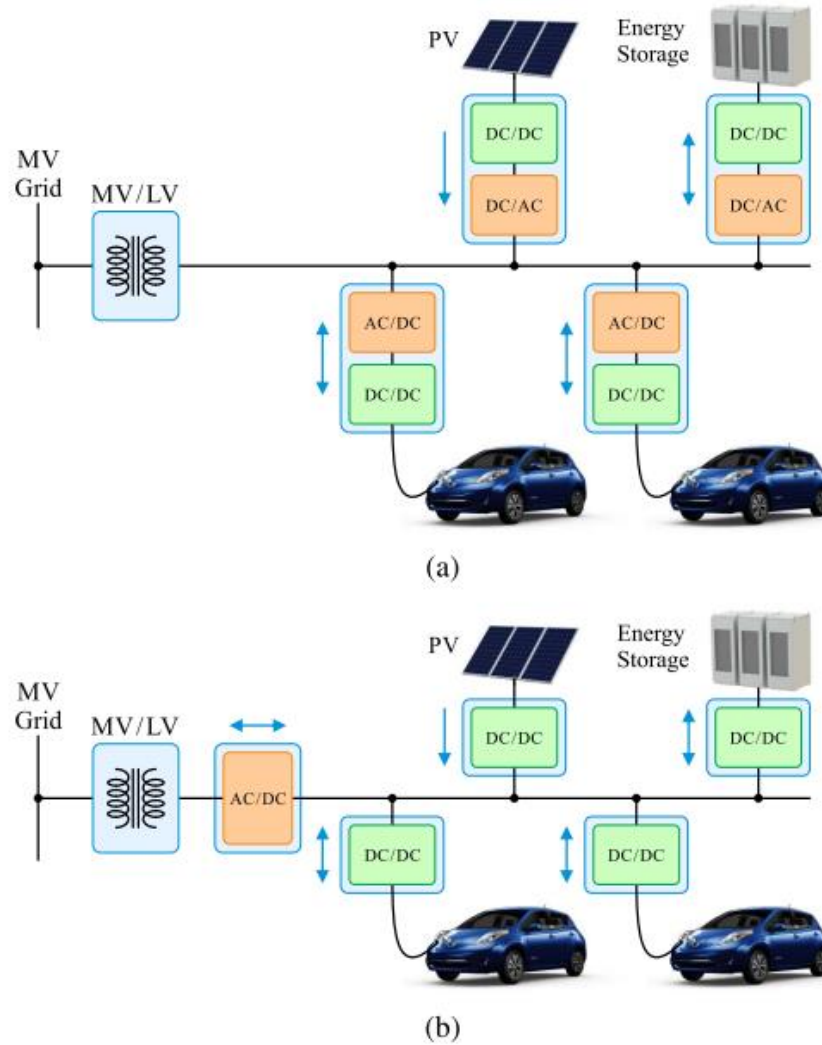
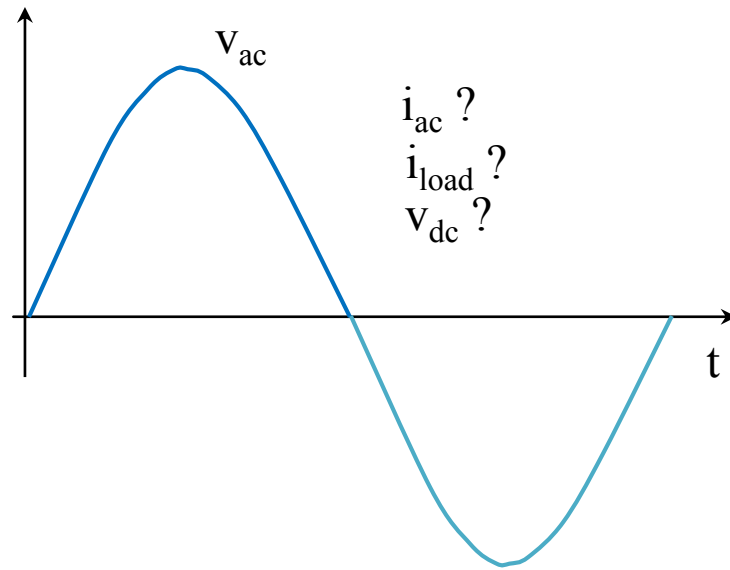
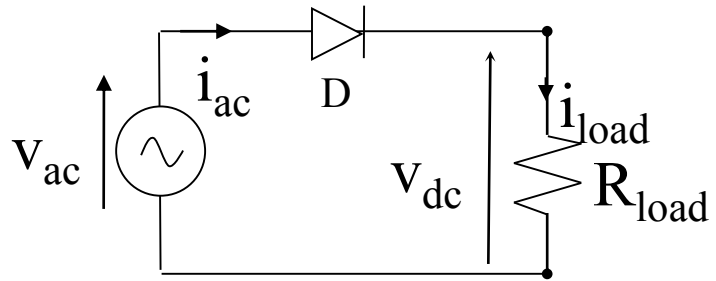
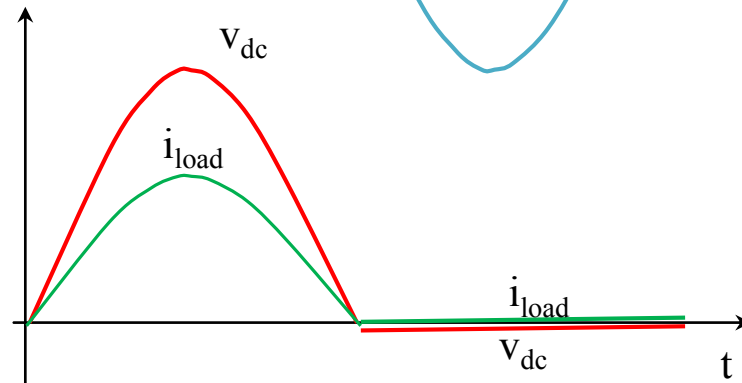
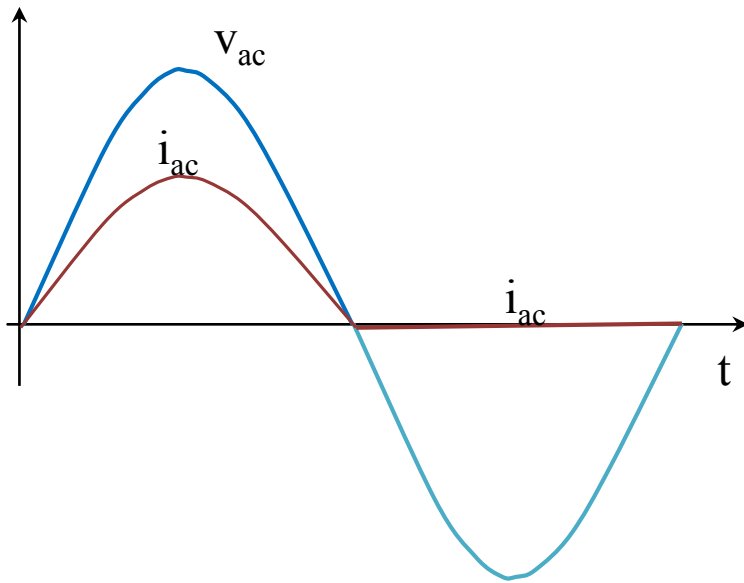
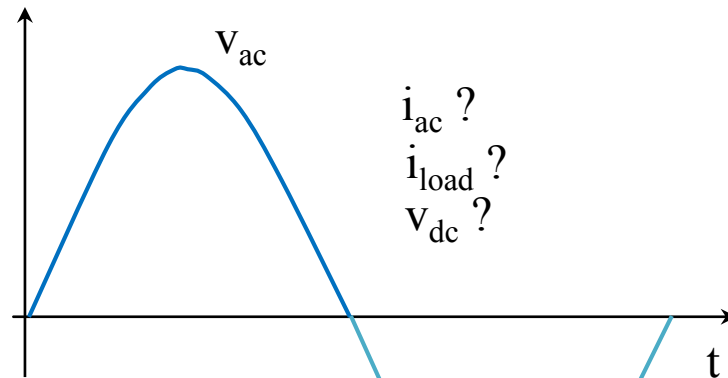
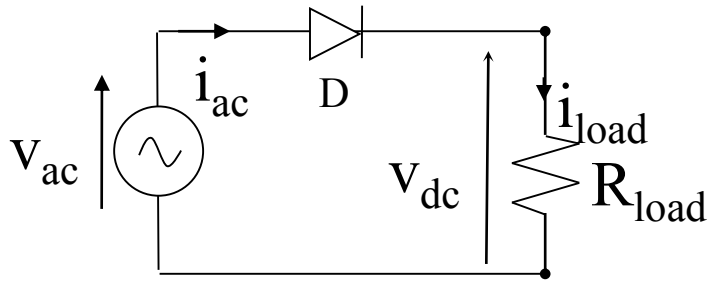


Fig. 3. Configurations for XFC stations. (a) AC-connected system. (b) DC-connected system.

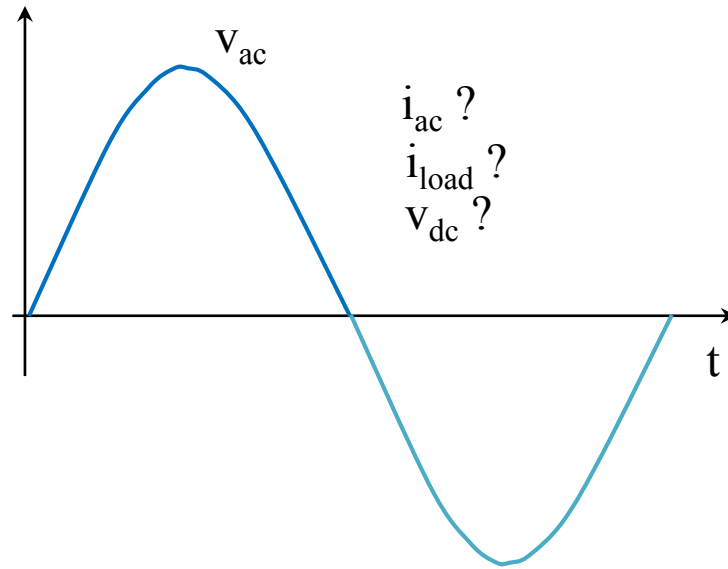
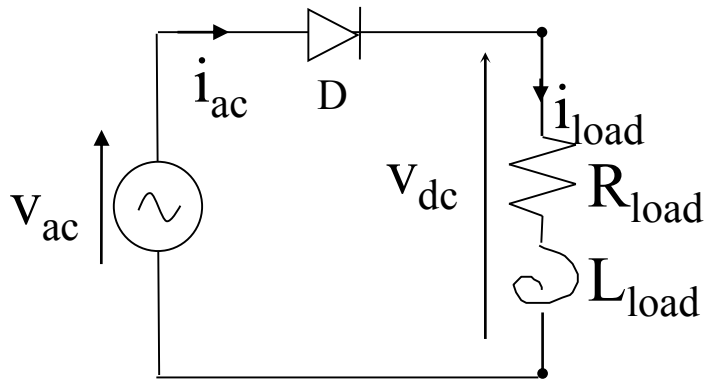
AC/DC: Test



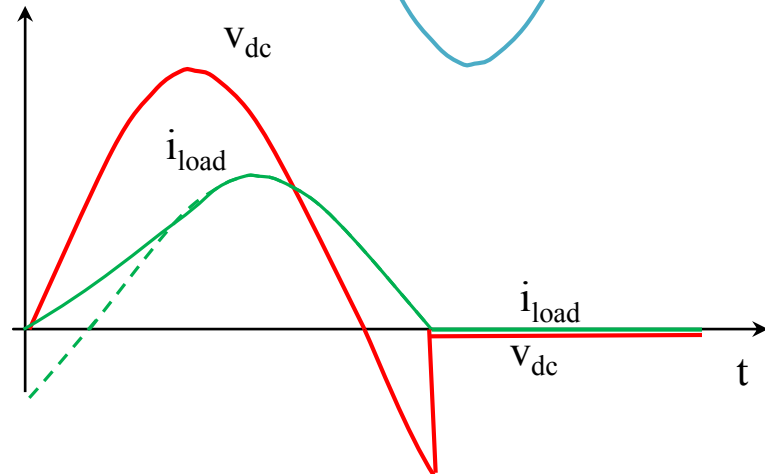
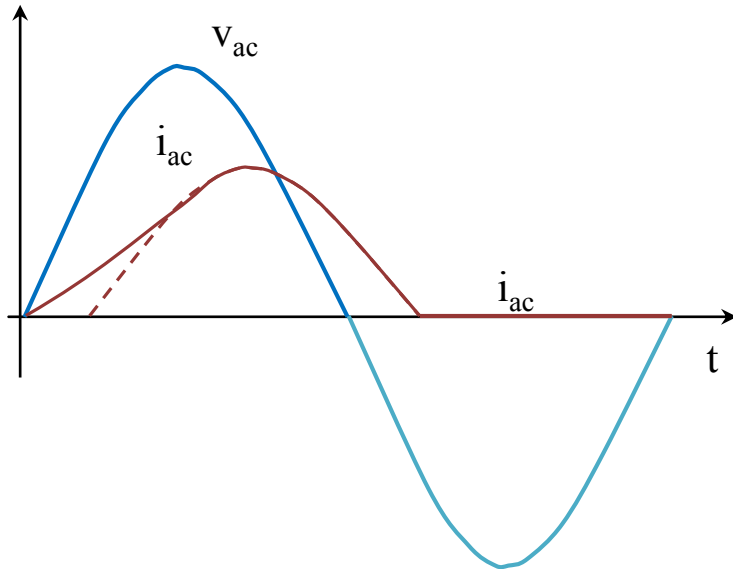
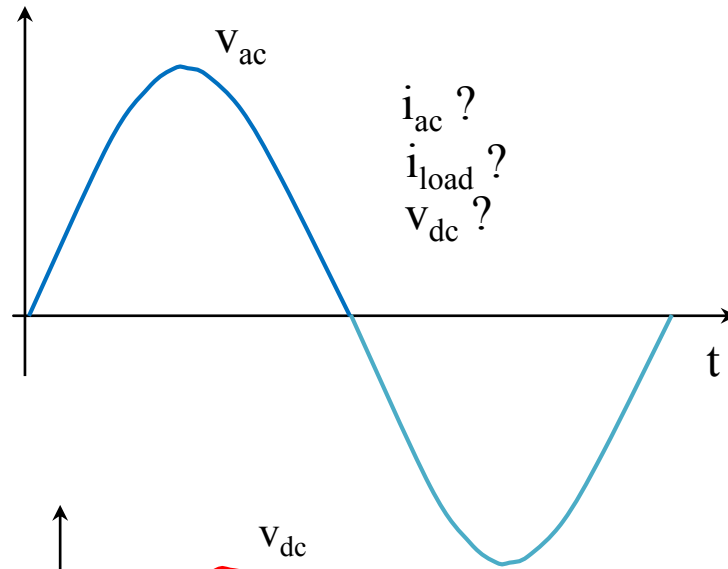
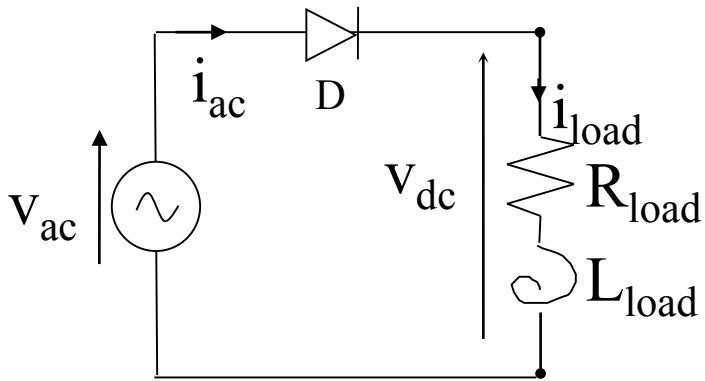
AC/DC: Test



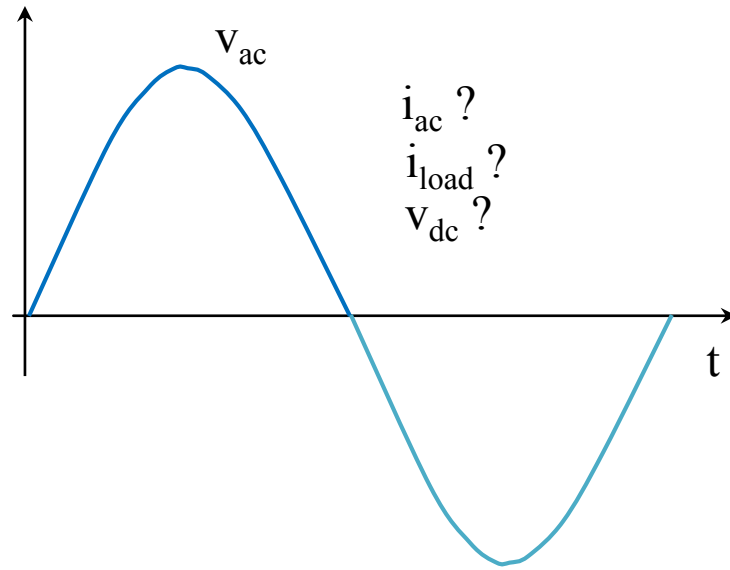
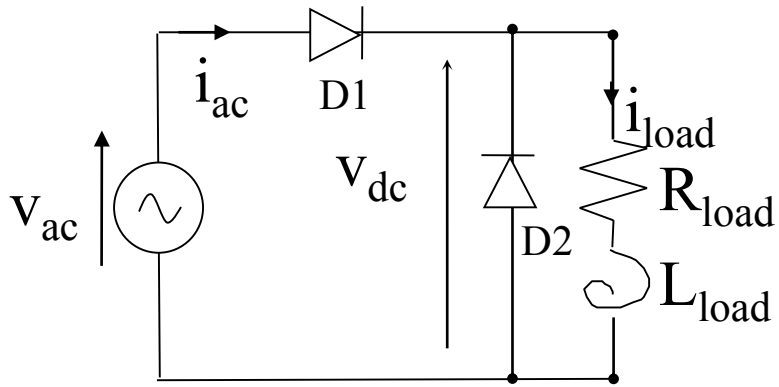
AC/DC: Test



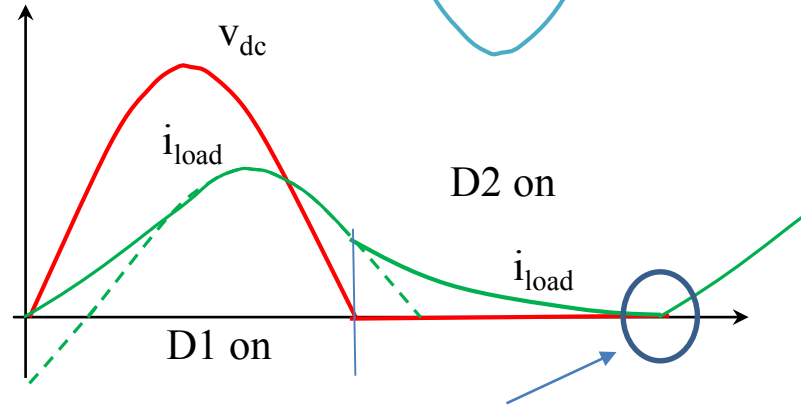
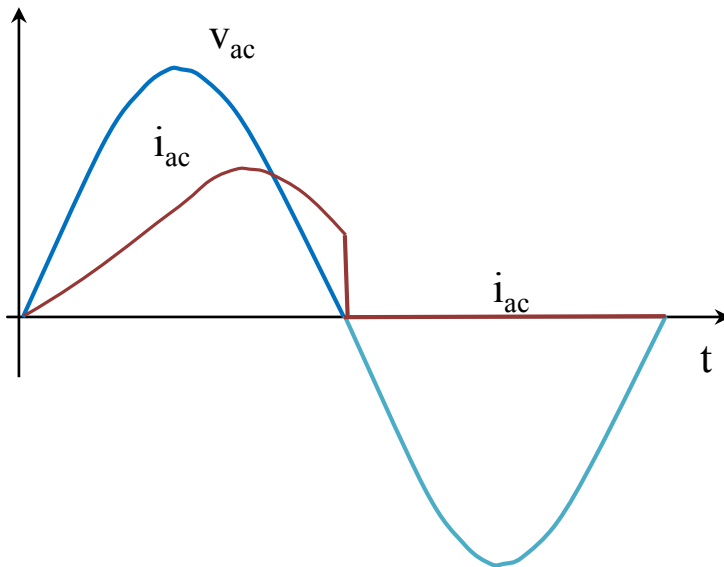
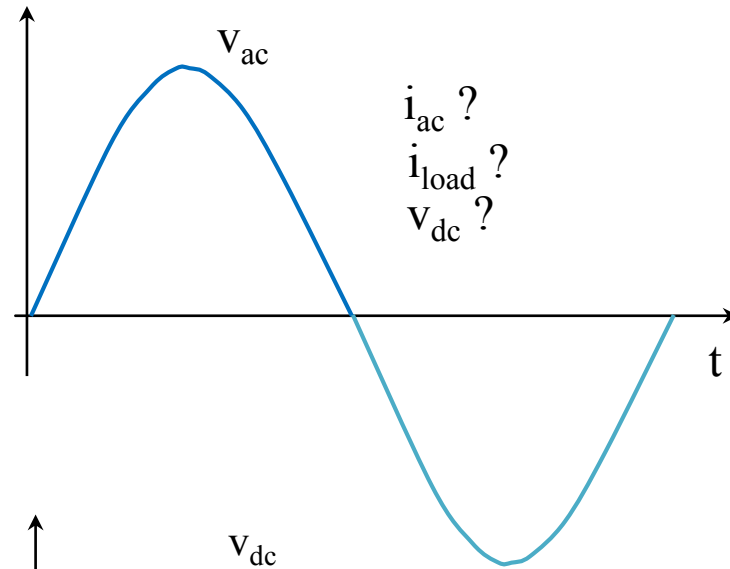
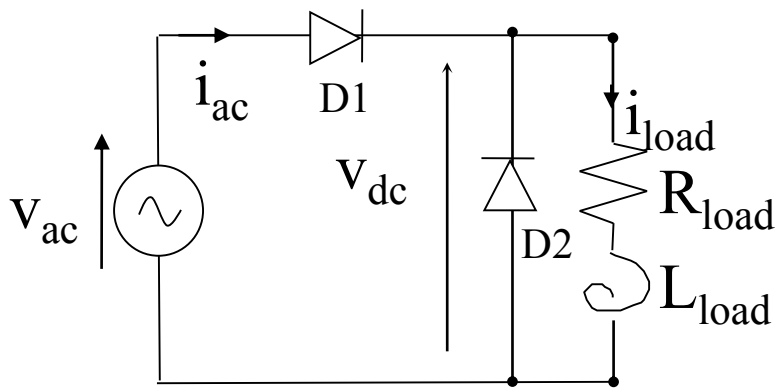
AC/DC: Test



AC/DC: Test

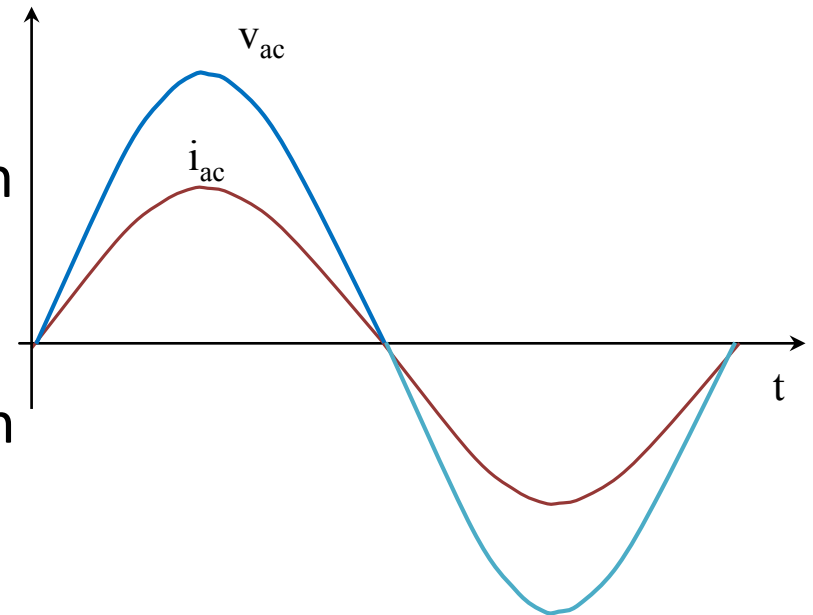
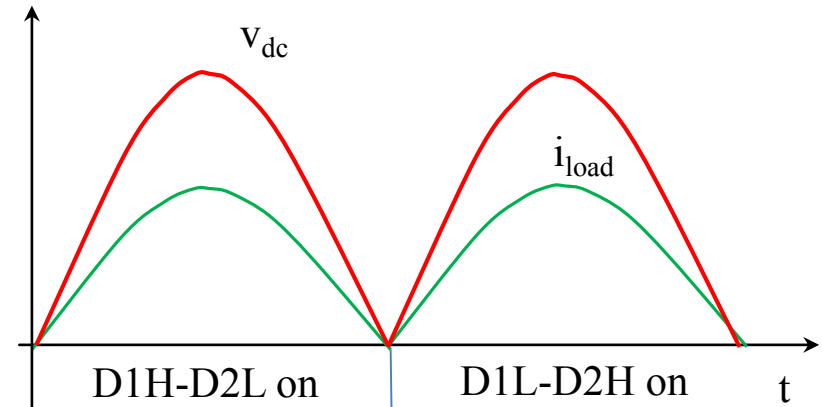
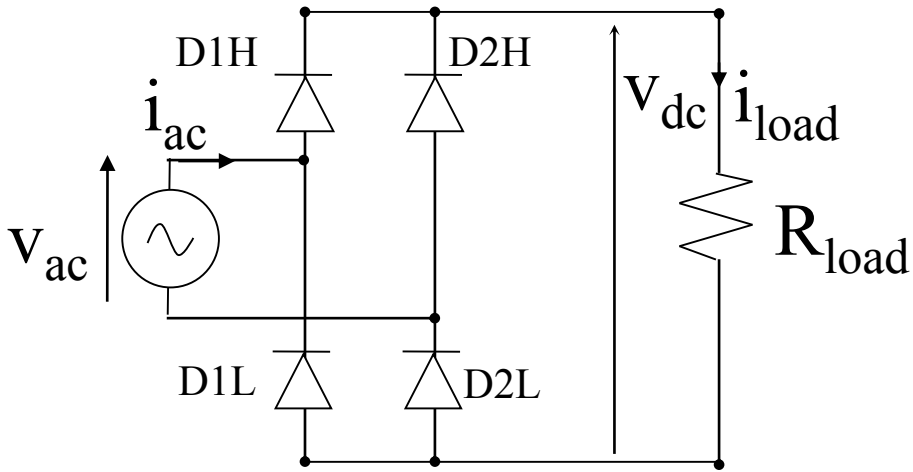


AC/DC: Test



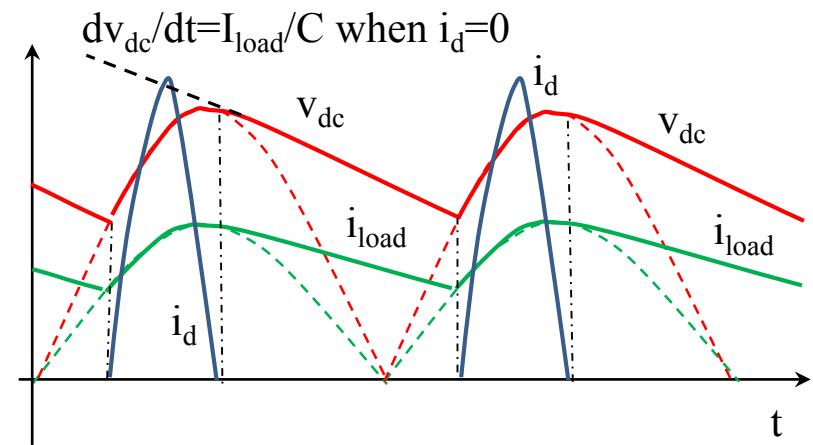
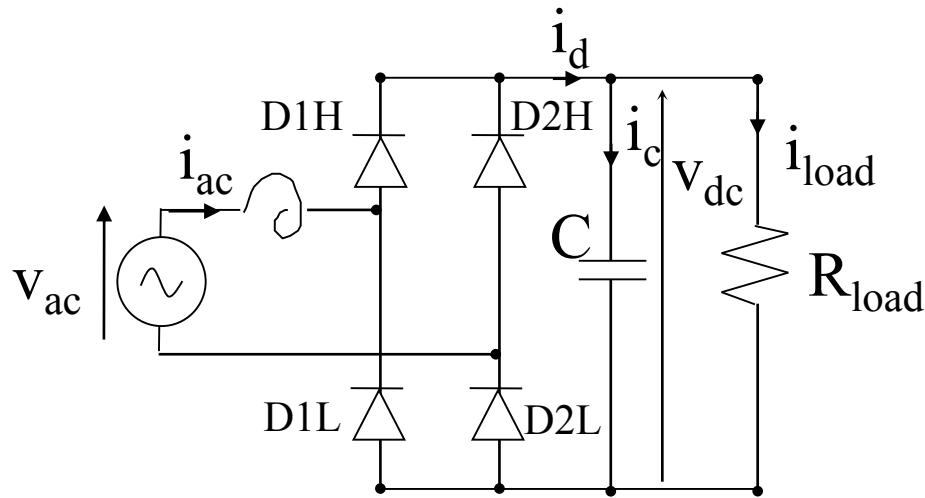
Continuous or Discontinuous mode?

Diode bridge



- Between N diodes with a common cathode, it conducts the one with the highest anode voltage
- Between N diodes with a common anode, it conducts the one with the lowest cathode voltage

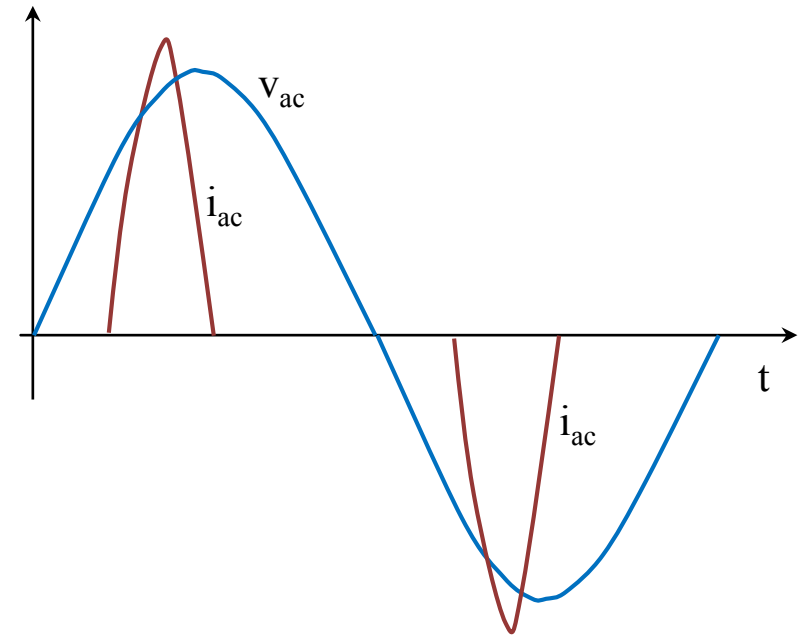
Diode bridge with a dc-bus capacitor



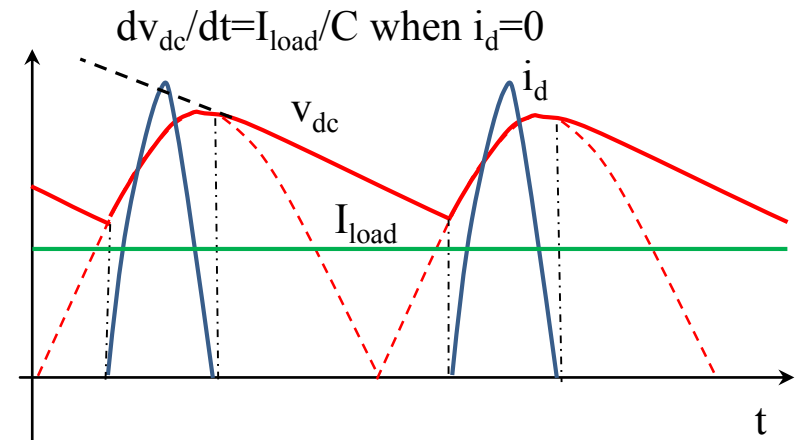
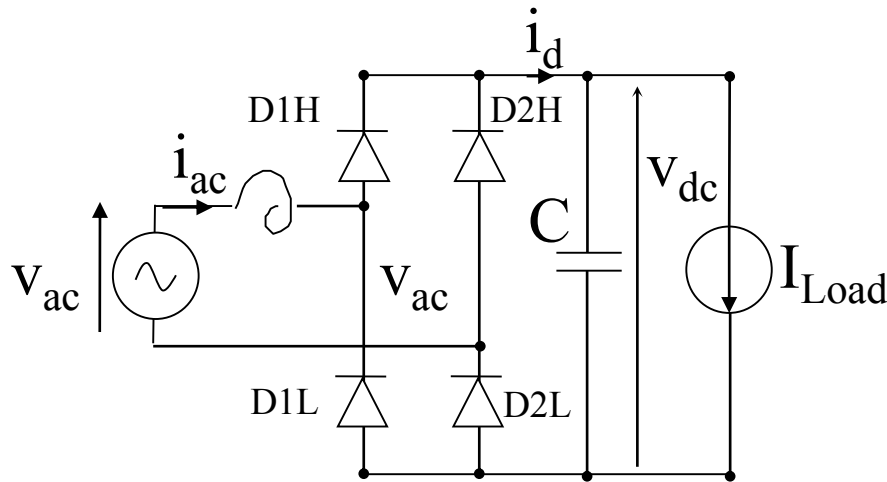
The diode bridge is disconnected from the load when i_d tries to become negative:

$$i_d = i_c + i_{load} = C \frac{dv_{dc}}{dt} + i_{load}$$

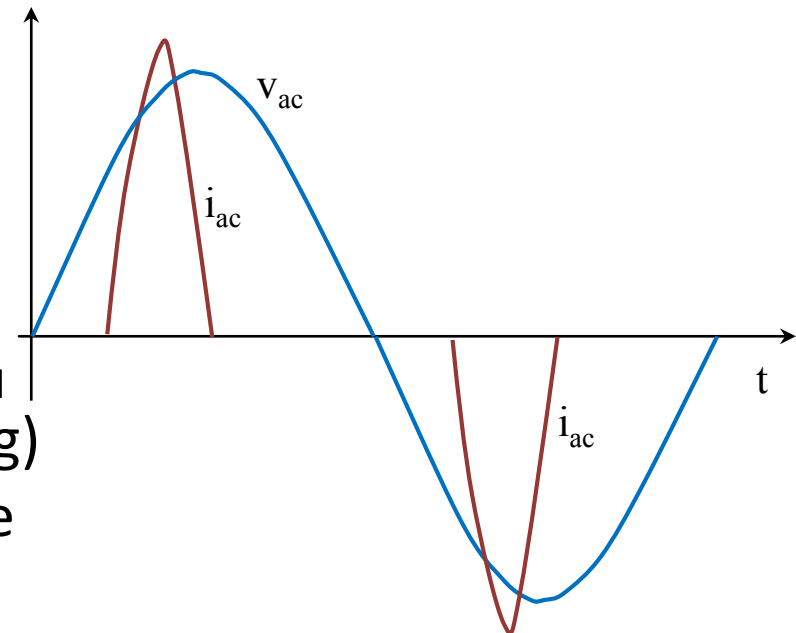
The period during which the grid is connected to the load is less than a semiperiod of the ac voltage



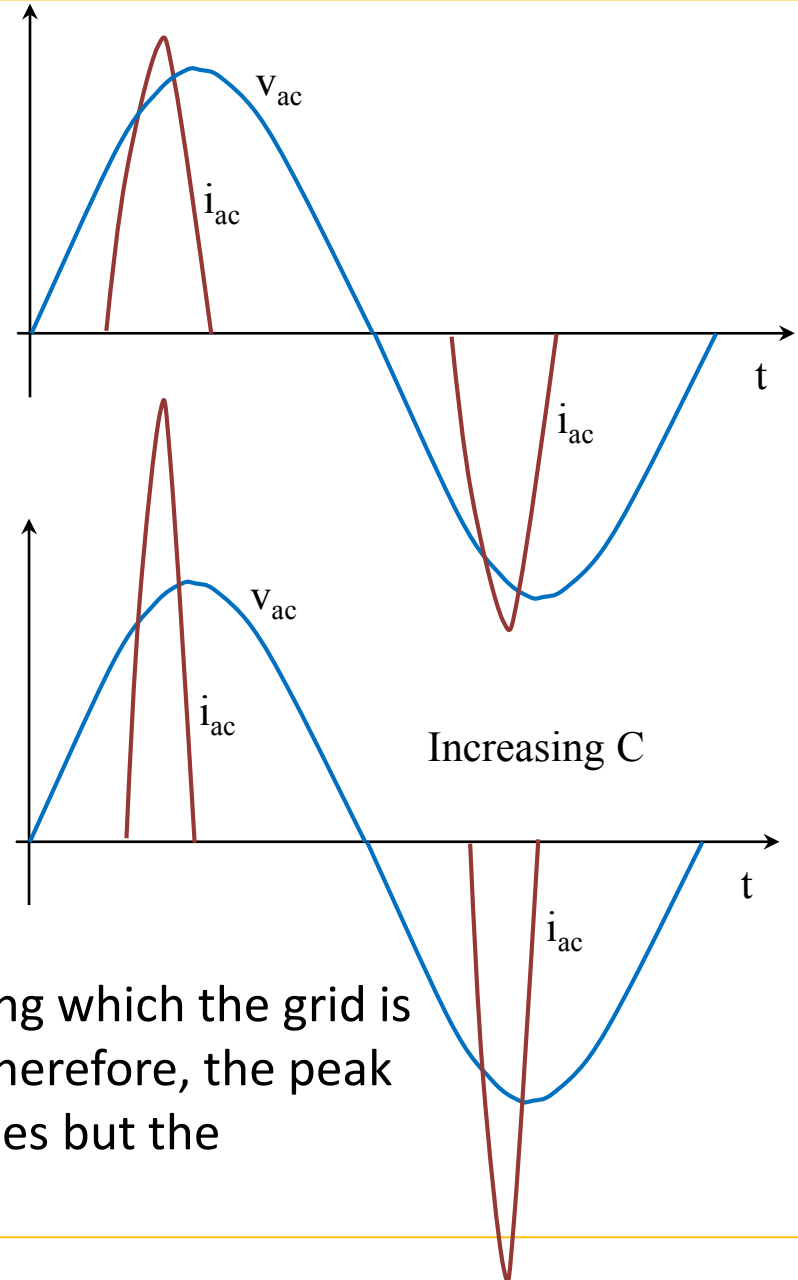
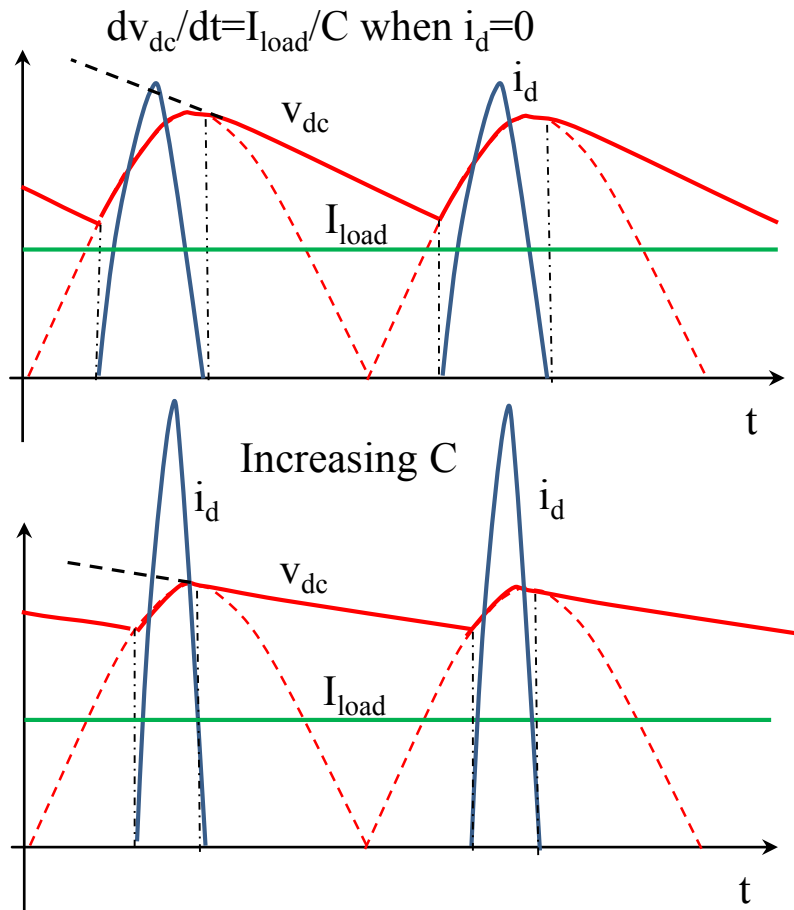
Diode bridge with a dc-bus capacitor



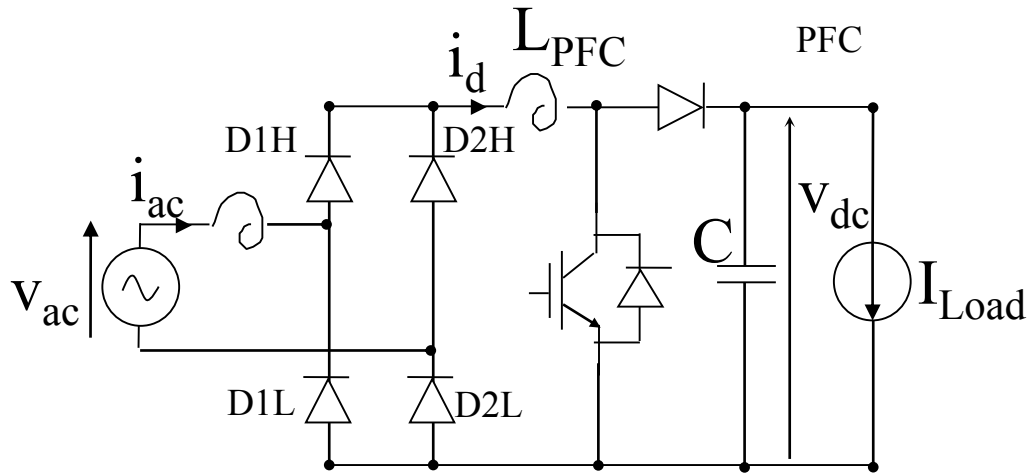
The period during which the grid is connected to the load is less than a semiperiod; therefore, given that the average value of i_d is equal to I_{load} , the peak of i_d is much higher than I_{load} and may produce distortion (notching) in the grid (EMI filter is required). The ac current is leading the voltage.



Increasing the capacitance of the dc-bus capacitor

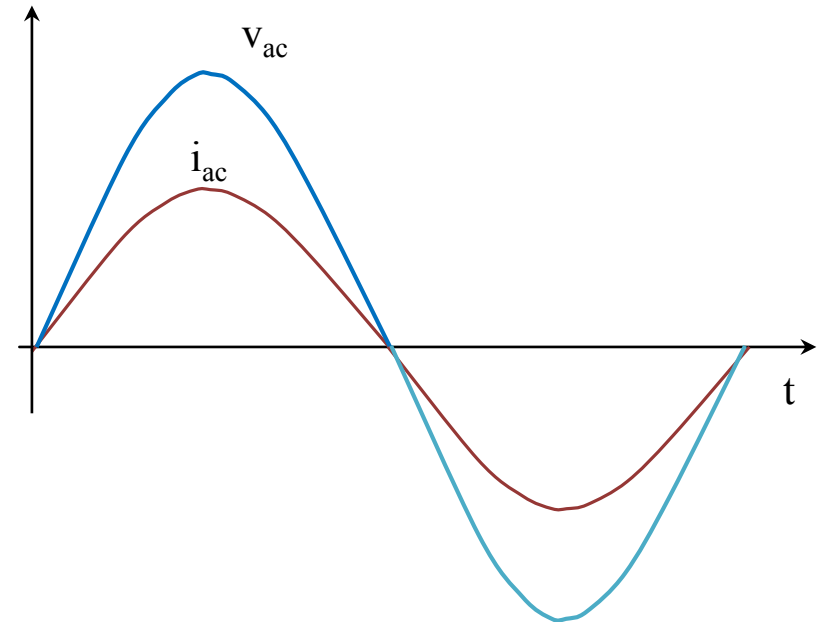
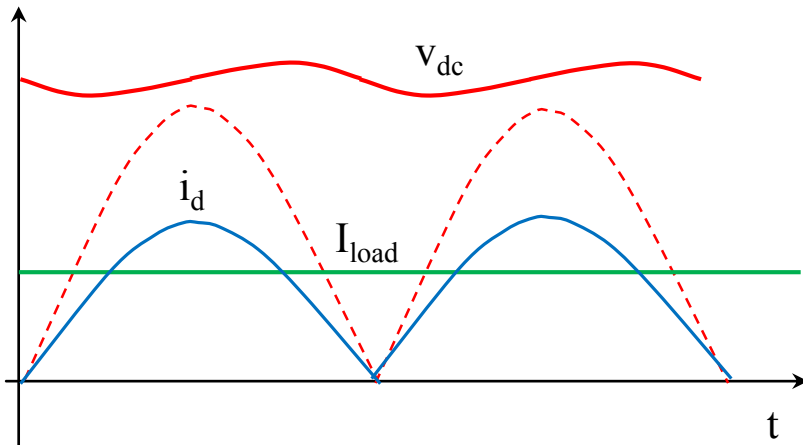


Increasing the capacitance, the period during which the grid is connected to the load is less than before; therefore, the peak of i_d is higher. The dc voltage ripple decreases but the distortion effect on the grid increases.



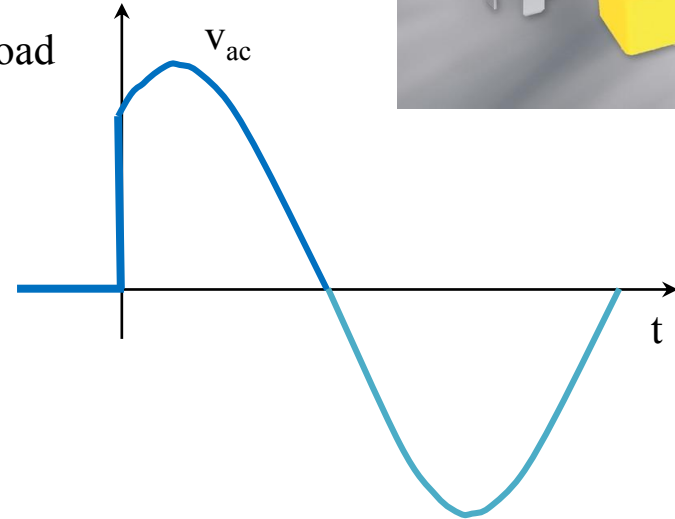
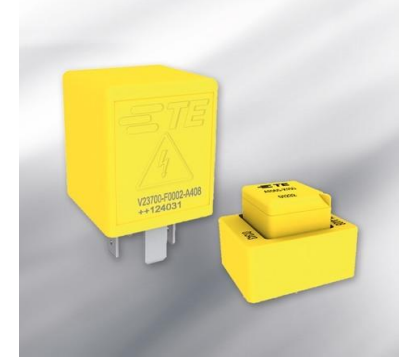
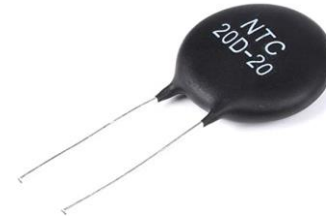
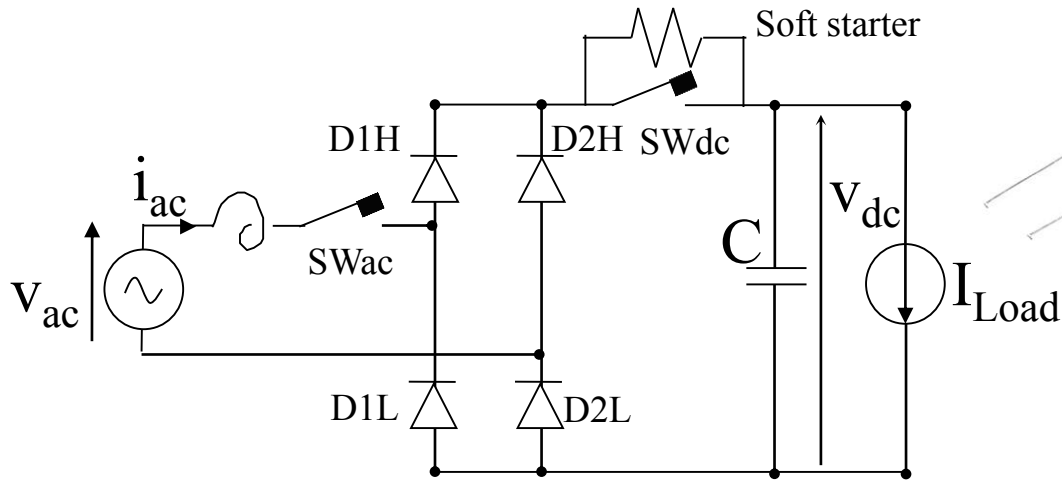
PFC (power factor correction) reduces the distortion (grid side), compensates the power factor and reduces the dc voltage ripple.

It needs more components.

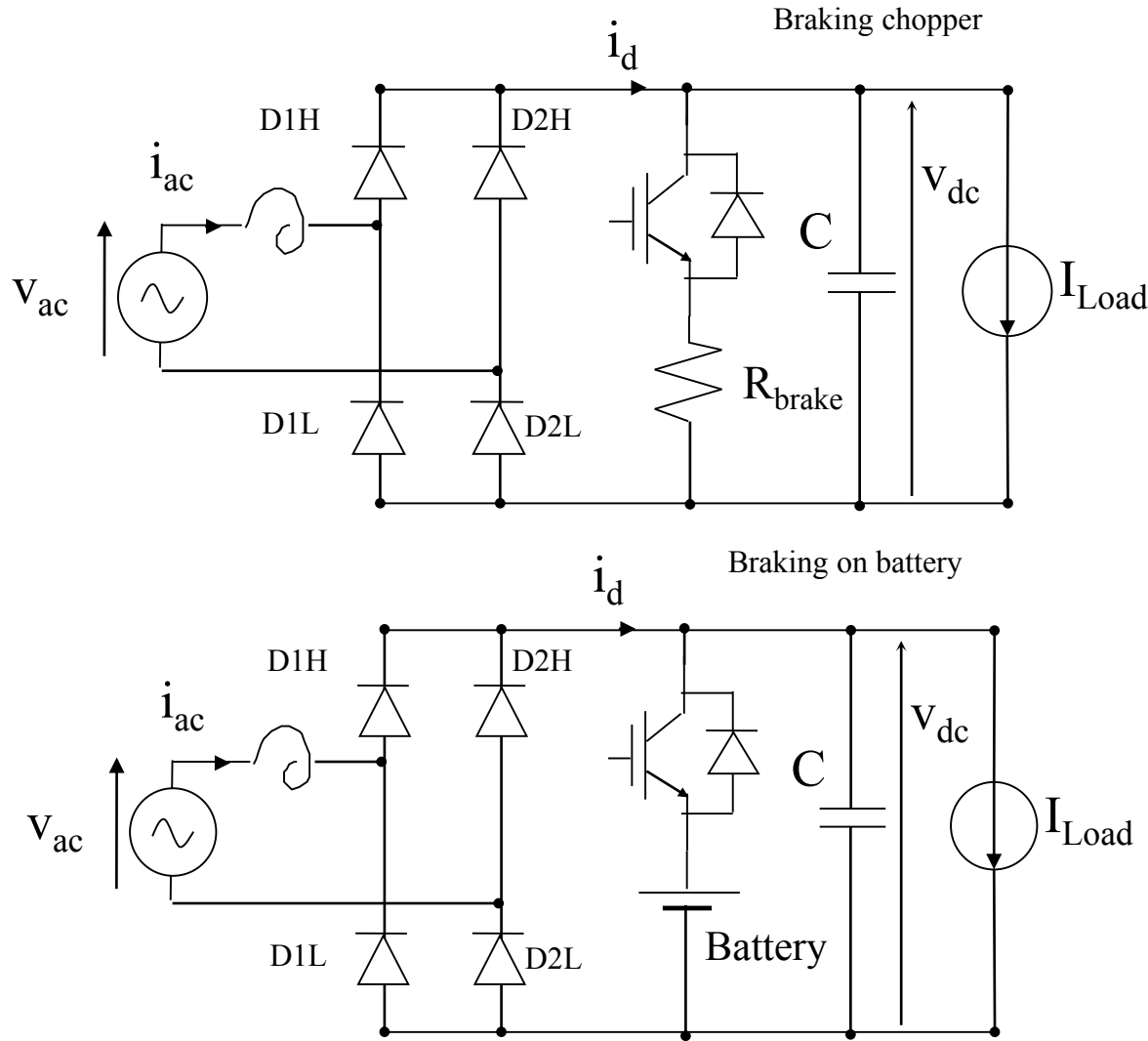


Diodes bridge and PFC require a soft-start device

Inrush current and Soft-start device



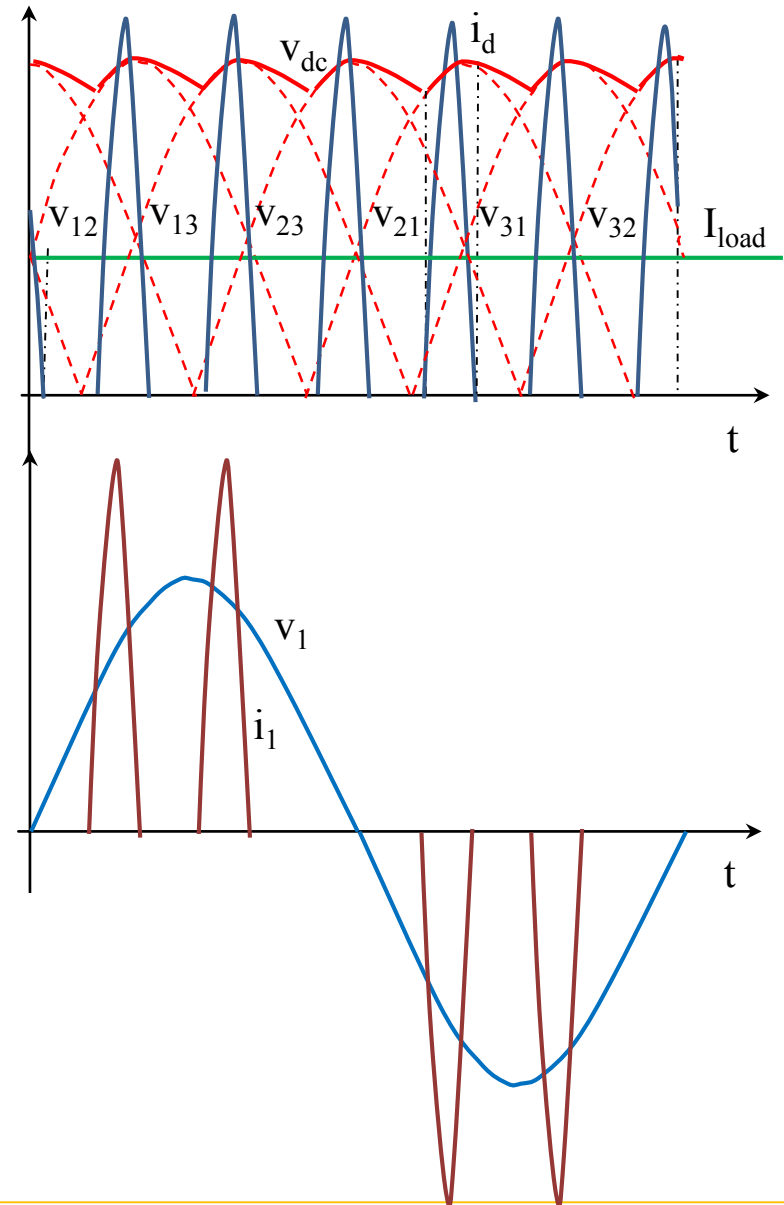
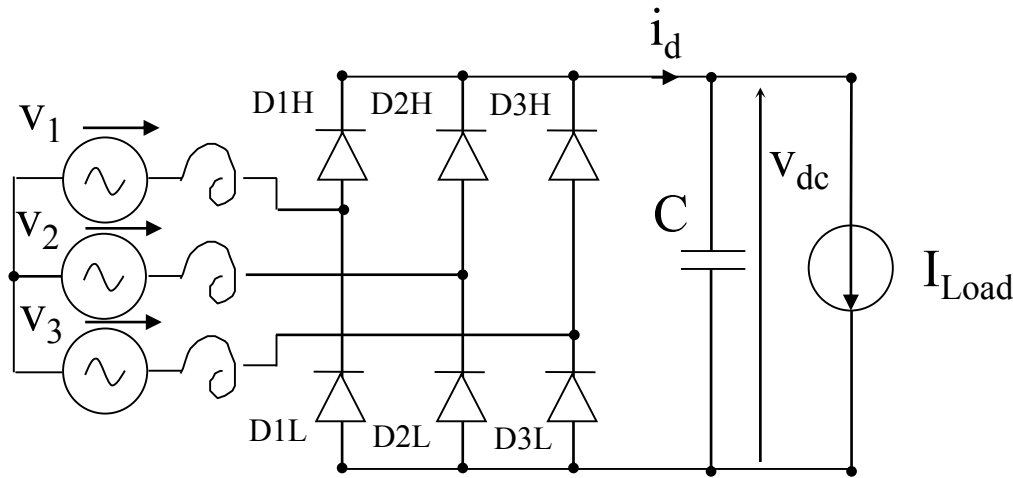
- Closing the switch on the ac side (SW_{ac}) without any pre-charging device, with the capacitor C discharged, the inrush current is very high, limited only by the impedance of the grid.
- In order to limit the inrush current, a soft-starter device has to be used: an NTC alone or a resistor with a contactor (SW_{dc}) as a bypass



During the braking phase the kinematic energy, stored in the mass/inertia of the mechanical load, are converted into electric energy. The instant power flow changes its sign, this means that I_{load} becomes negative.

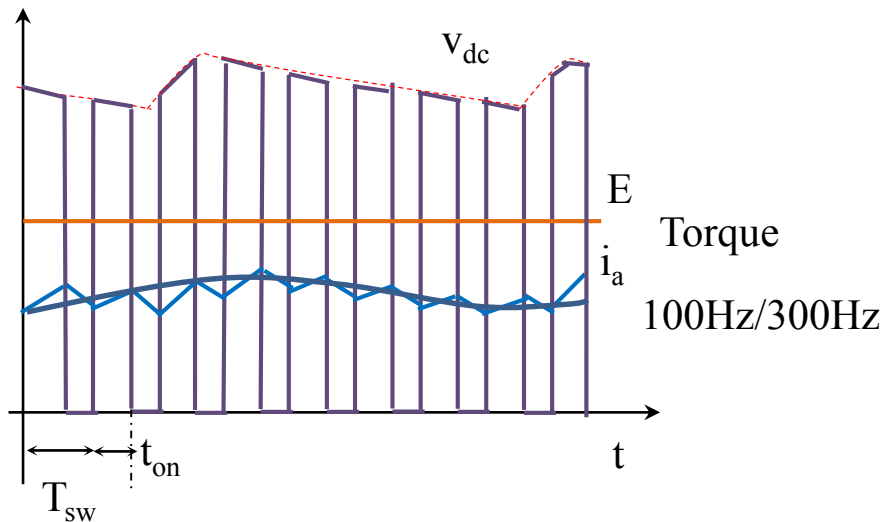
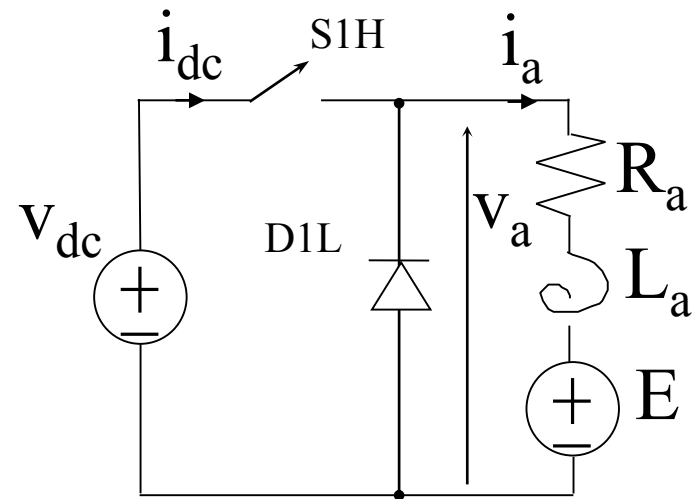
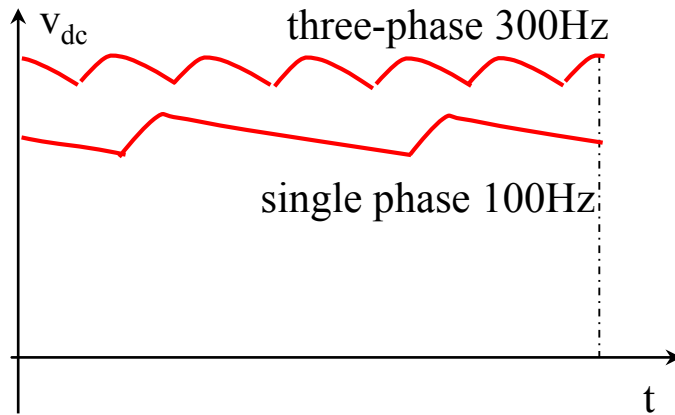
Diode bridge assures that $i_d \geq 0$, so the energy has to be dissipated or stored in a battery.

Three-phase diode bridge

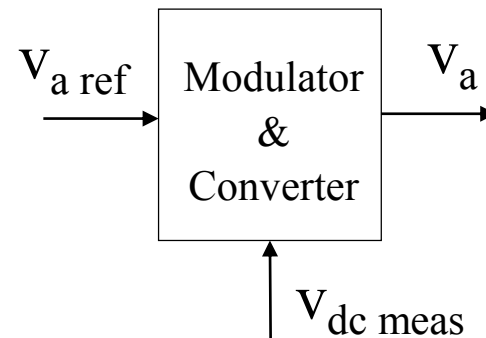


- Between N diodes with a common cathode, it conducts the one with the highest anode voltage
- Between N diodes with a common anode, it conducts the one with the lowest cathode voltage
- EMI filter

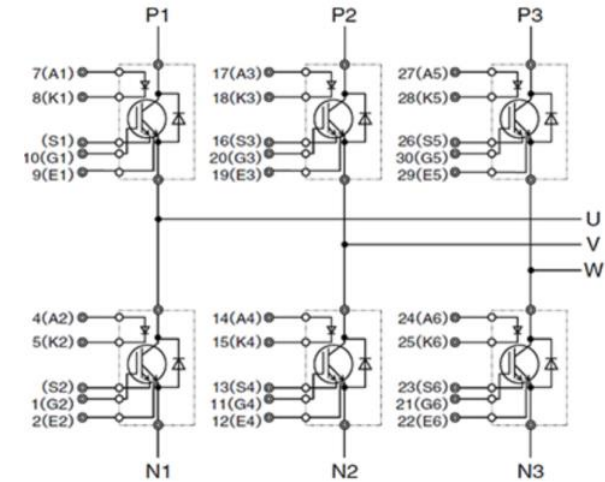
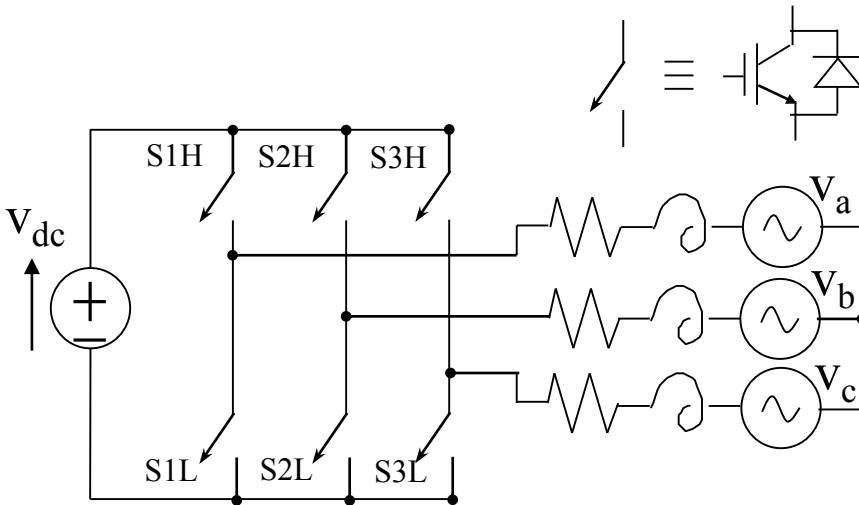
Compensation of the dc bus voltage oscillation



Power Supply



DC/AC: three-phase voltage source inverter (VSI)



Sinusoidal PWM with third harmonic injection (inscribed circle)

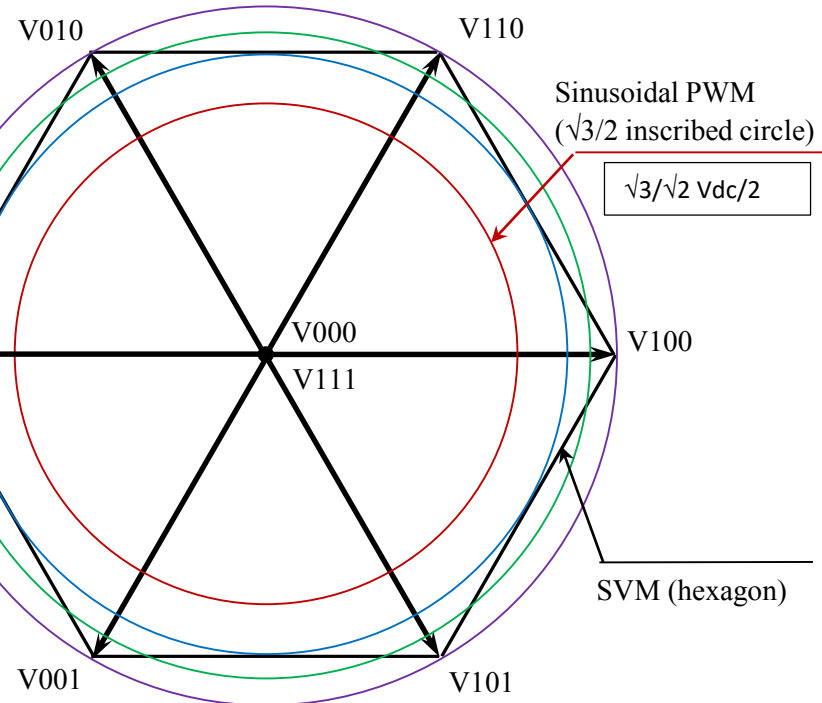
$$1/\sqrt{2} V_{dc}$$

Circumscribed circle

$$\sqrt{2}/\sqrt{3} V_{dc}$$

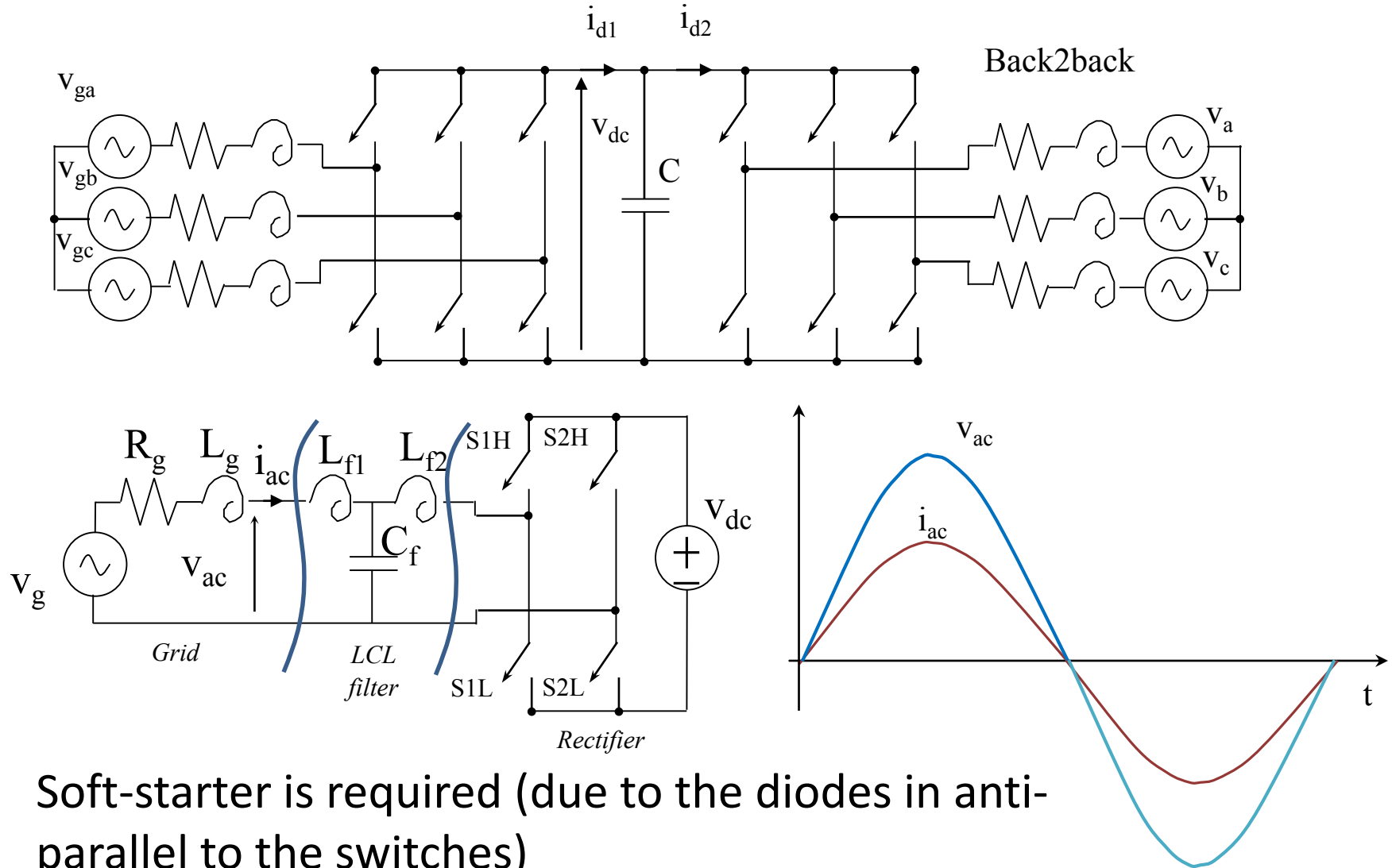
$$2/\pi \sqrt{3}/\sqrt{2} V_{dc}$$

Square wave first harmonic ($4/\pi \sqrt{3}/2$ inscribed circle)



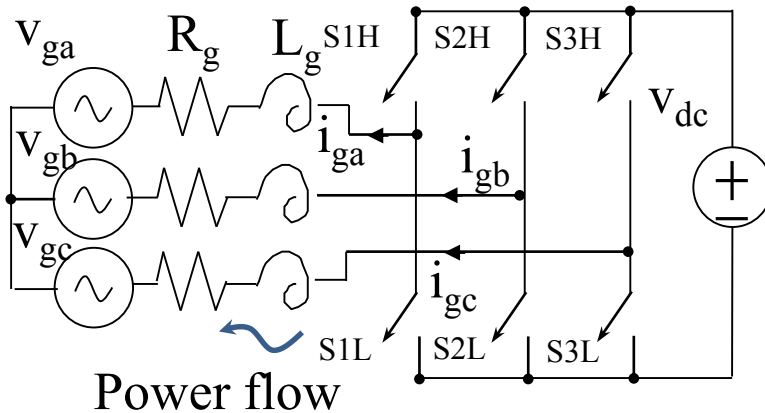
Back-to-back configuration

Called also “Regenerative rectifier” or “Active Front End” (AFE)



Soft-starter is required (due to the diodes in anti-parallel to the switches)

Regeneration

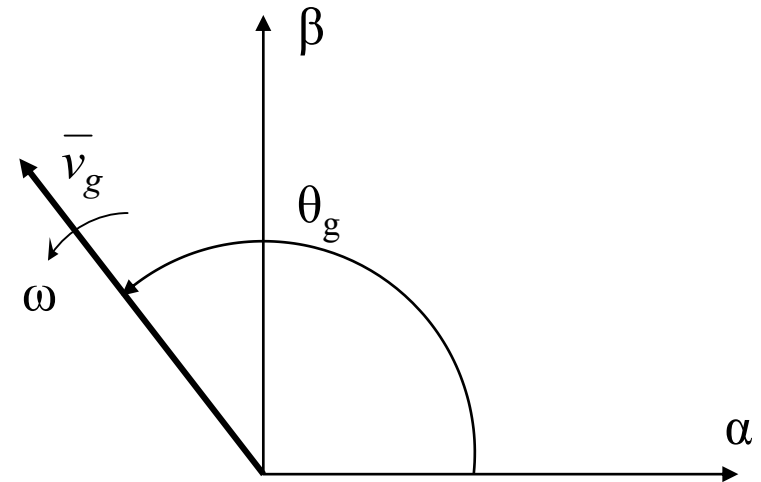


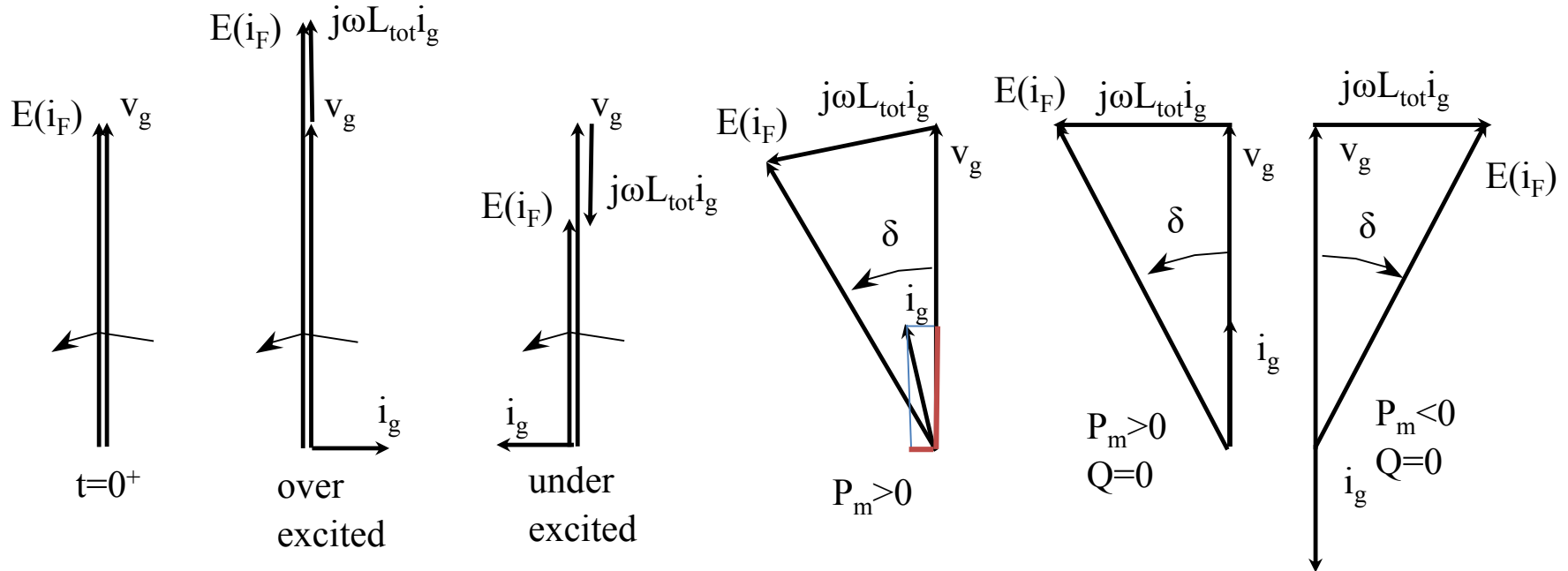
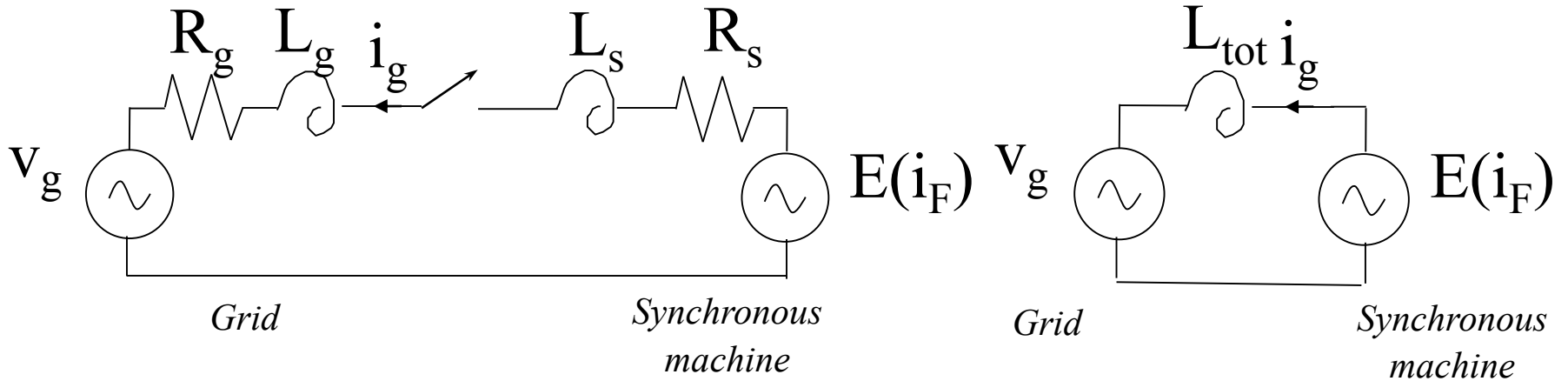
The space phasor formula may be applied to the grid voltage v_g . In a stationary frame ($\alpha\beta$), it results ($v_{ga} = V_{ph\ max} \cos(\theta_g)$):

$$\overline{v_g} = V_g e^{j\theta_g}$$

Where V_g is the rms value of the line-line voltage, while θ_g the angle between the space phasor v_g and the α -axis

The grid has an equivalent circuit similar to that of an ac machine. The main difference is that the voltage and the frequency are quite constant (in an ac machine the emf and the frequency are quite proportional to the mechanical speed)





Control of power flow into the grid

Call v_i the voltage produced by the VSI (limited to the hexagon operating area). Choose a reference frame fixed with the grid voltage, where the imaginary axis (q) is aligned with the space phasor v_g . It results:

$$\bar{v}_i = R_g \bar{i}_g + L_g p \bar{i}_g + j \dot{\theta} L_g \bar{i}_g + \bar{v}_g$$

The components along d and q axes are:

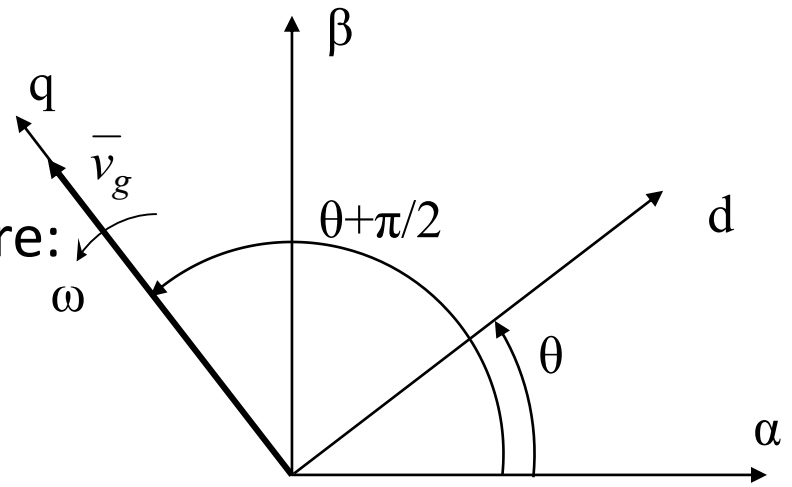
$$v_{id} = R_g i_{gd} + L_g p i_{gd} - \dot{\theta} L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + L_g p i_{gq} + \dot{\theta} L_g i_{gd} + v_g$$

The active and reactive power fed into the grid are:

$$P_g = \text{Re}(\bar{v}_g \underline{i}_g) = v_g i_{gq}$$

$$Q_g = \text{Im}(\bar{v}_g \underline{i}_g) = v_g i_{gd}$$



Control of active power fed into the grid

The derivative of the angle θ is the frequency ω .

Consider a steady state condition and a grid voltage constant in amplitude and frequency: $\overline{v_g} = V_g e^{j\omega t}$. Looking from the reference frame (dq), all the quantities are constant. Therefore:

$$\dot{\theta} = \omega$$

$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

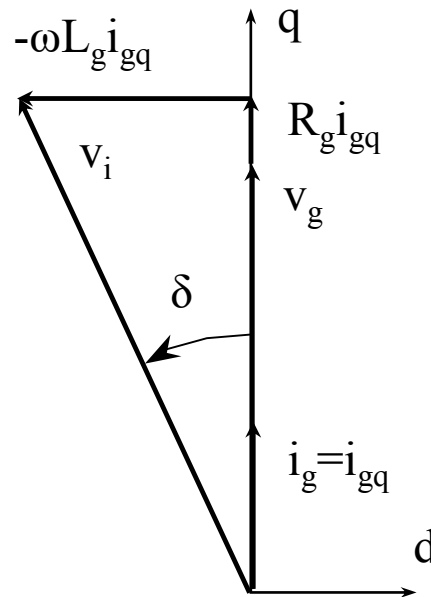
$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

If the current i_{gd} (component along the d axis) is fixed to zero, the phasor diagram is:

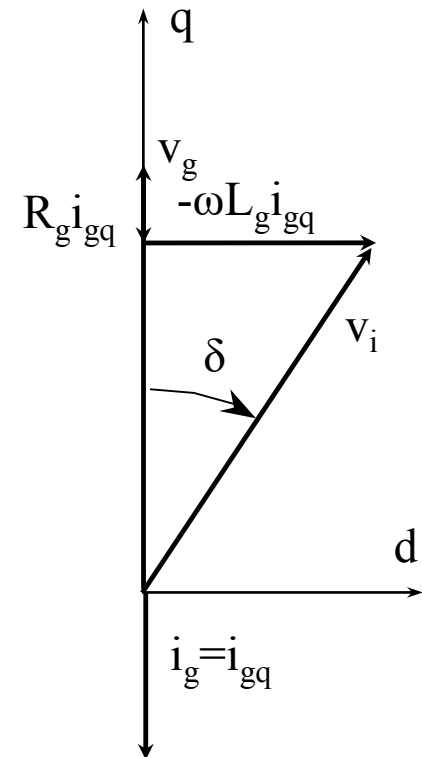
$$v_{id} = -\omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + V_g$$

Regenerative & $Q_g=0$



Rectifier & $Q_g=0$



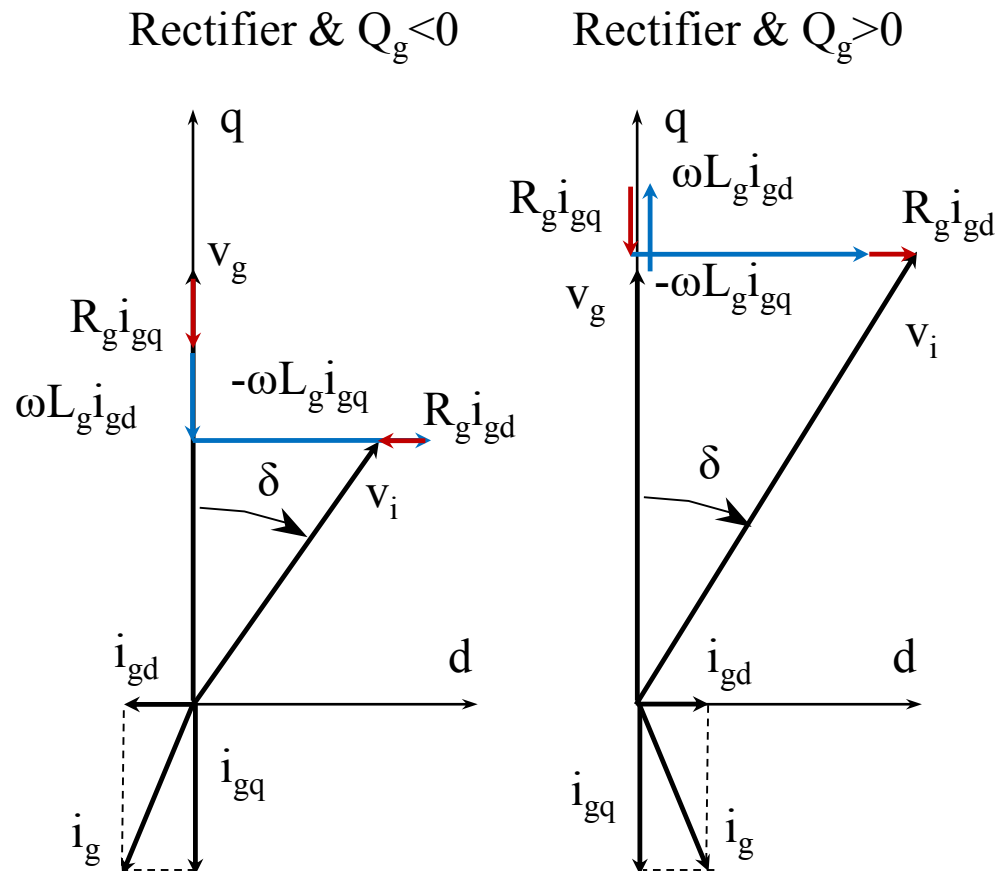
Control of active/reactive power fed into the grid

Controlling the current i_{gd} , it is possible to drive the reactive power: a positive value means that the inverter looks like a capacitor (and the grid like an inductor); vice versa for a negative value.

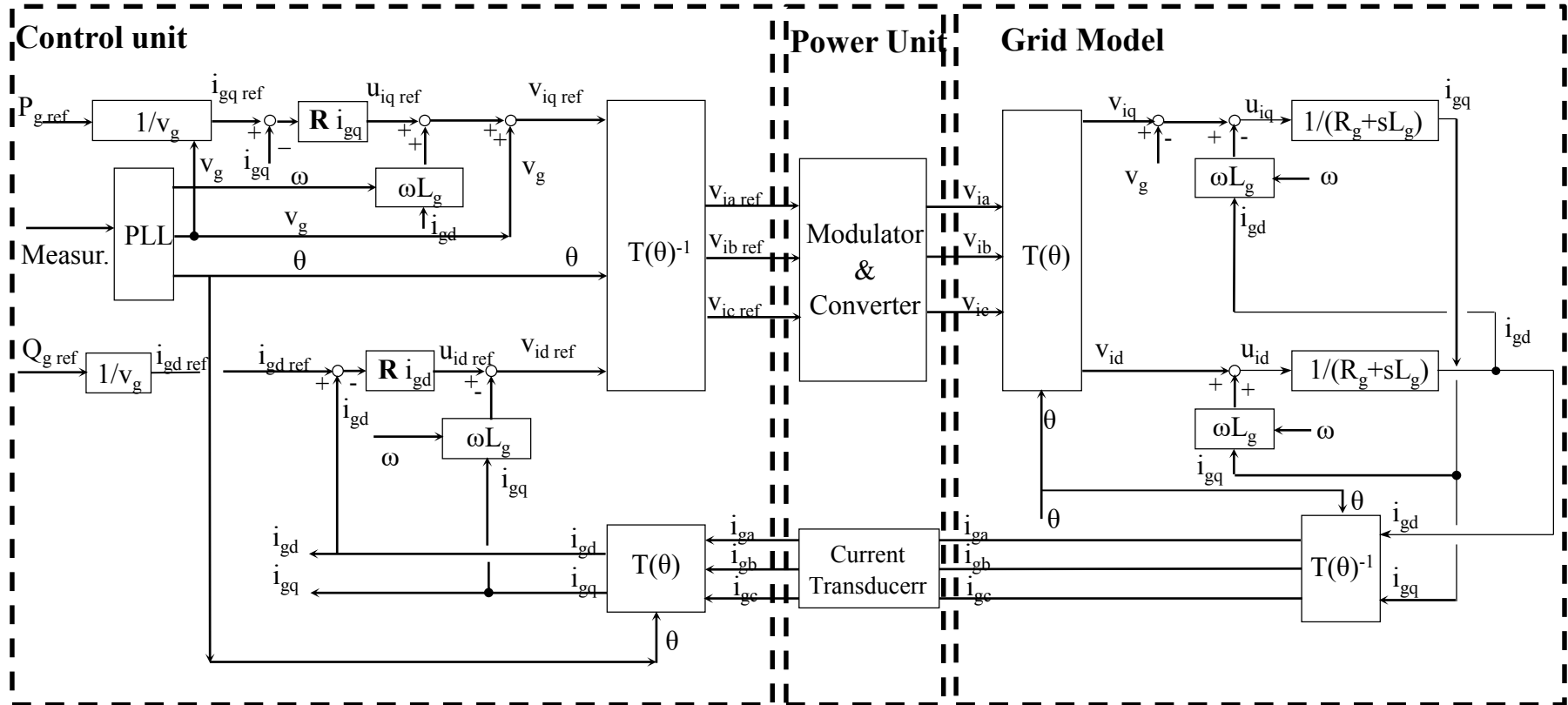
$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

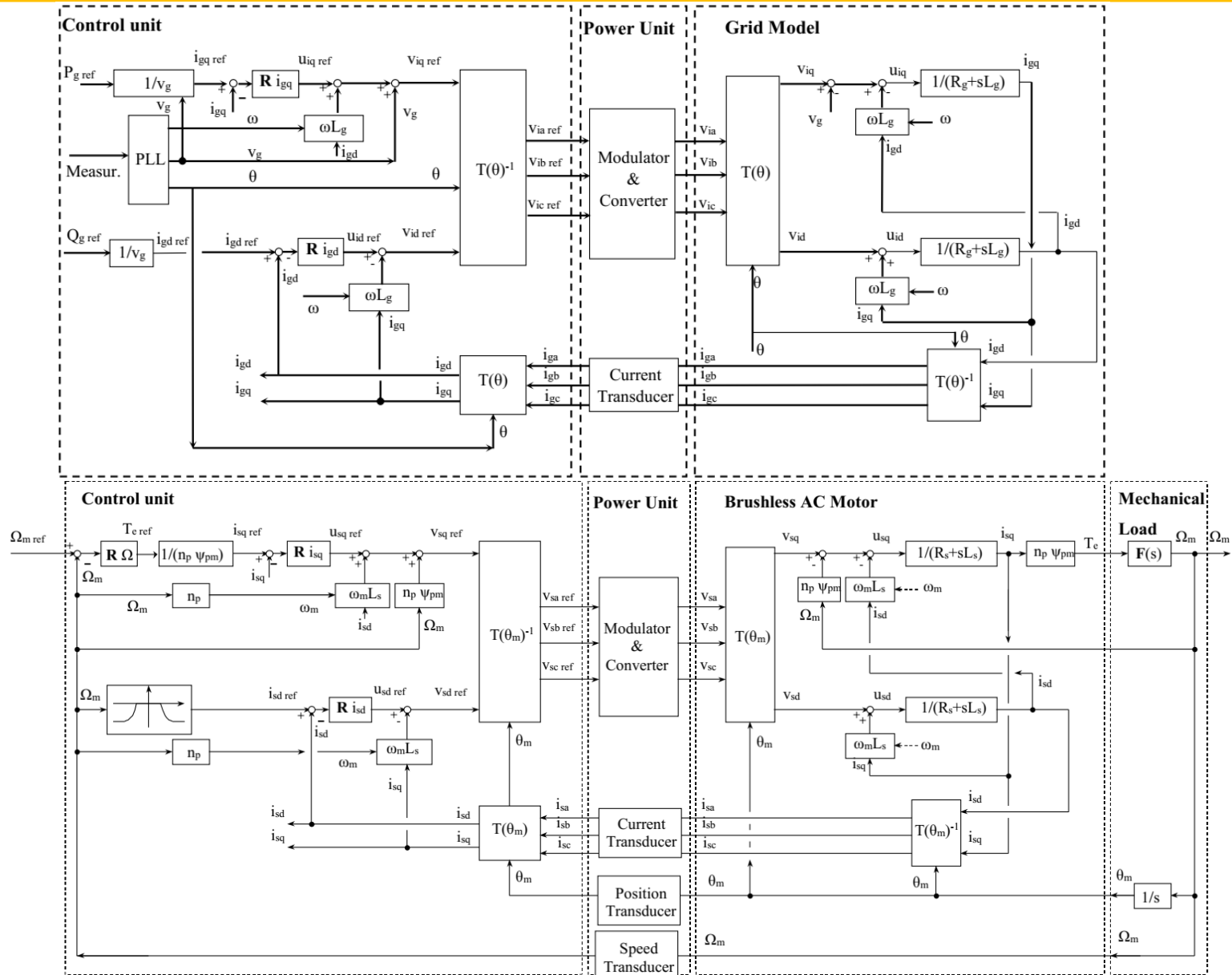
The figure represents the conditions when the active power is negative (operation as rectifier). Similar approach may be done for positive value of the active power (regeneration)



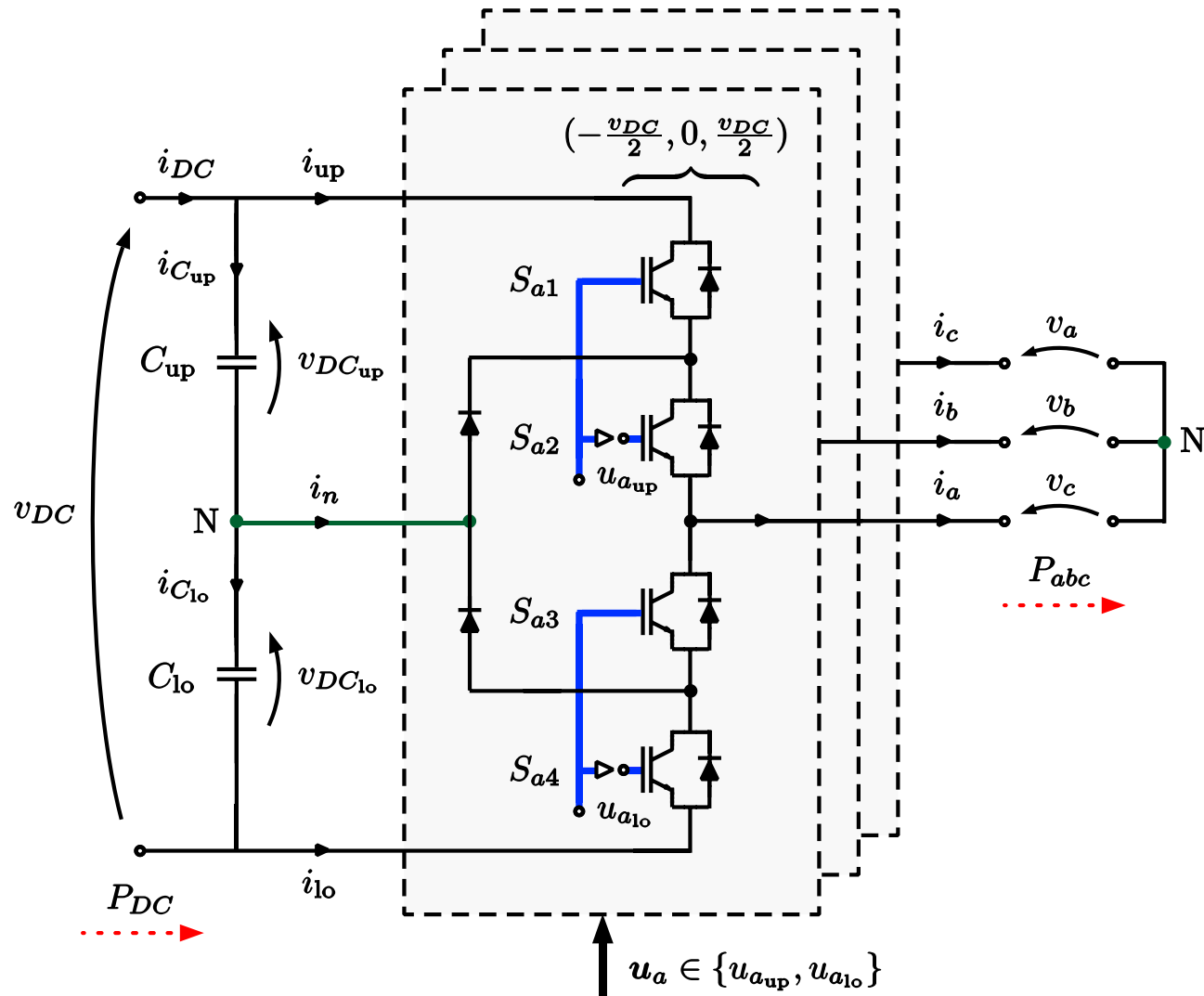
Control schemes



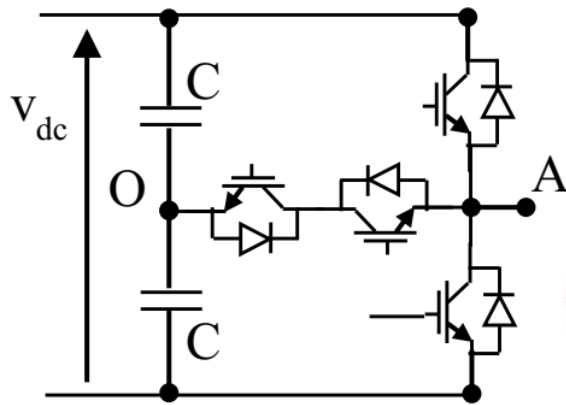
Control schemes: comparison with ac brushless



Multilevel converter: 3L-NPC

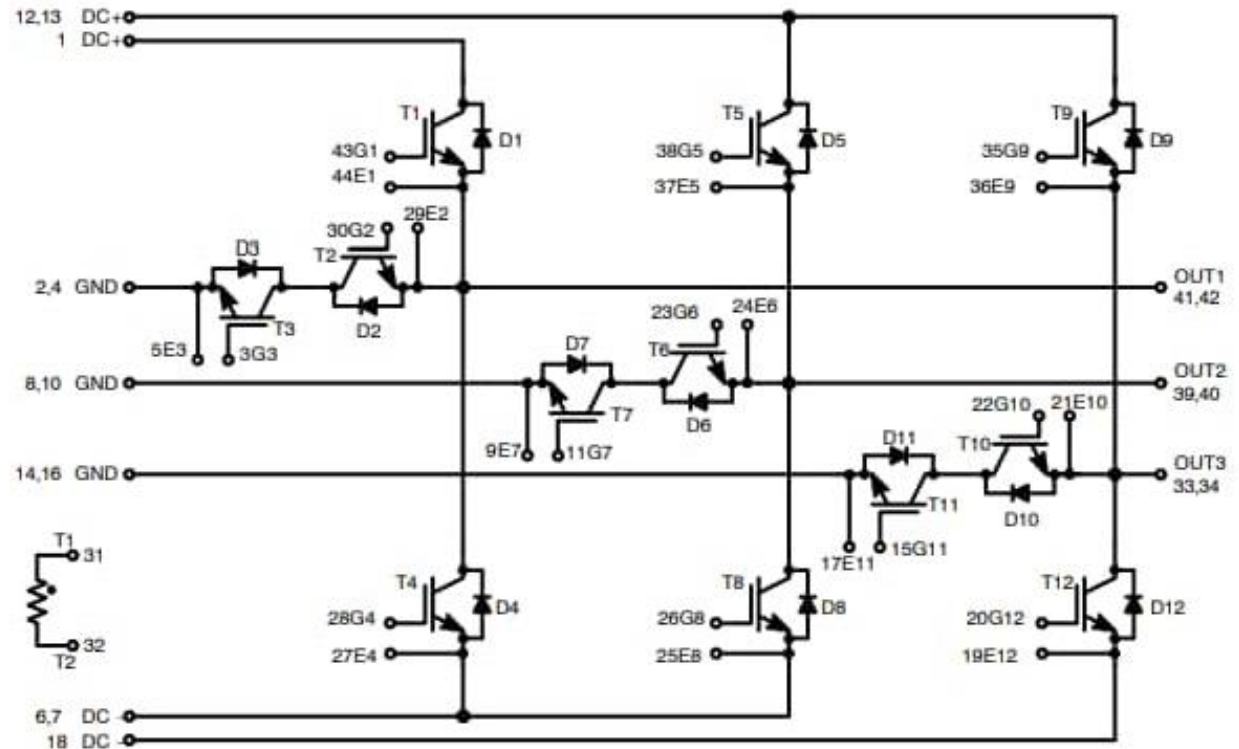


Multilevel converter: 3L-TNPCC



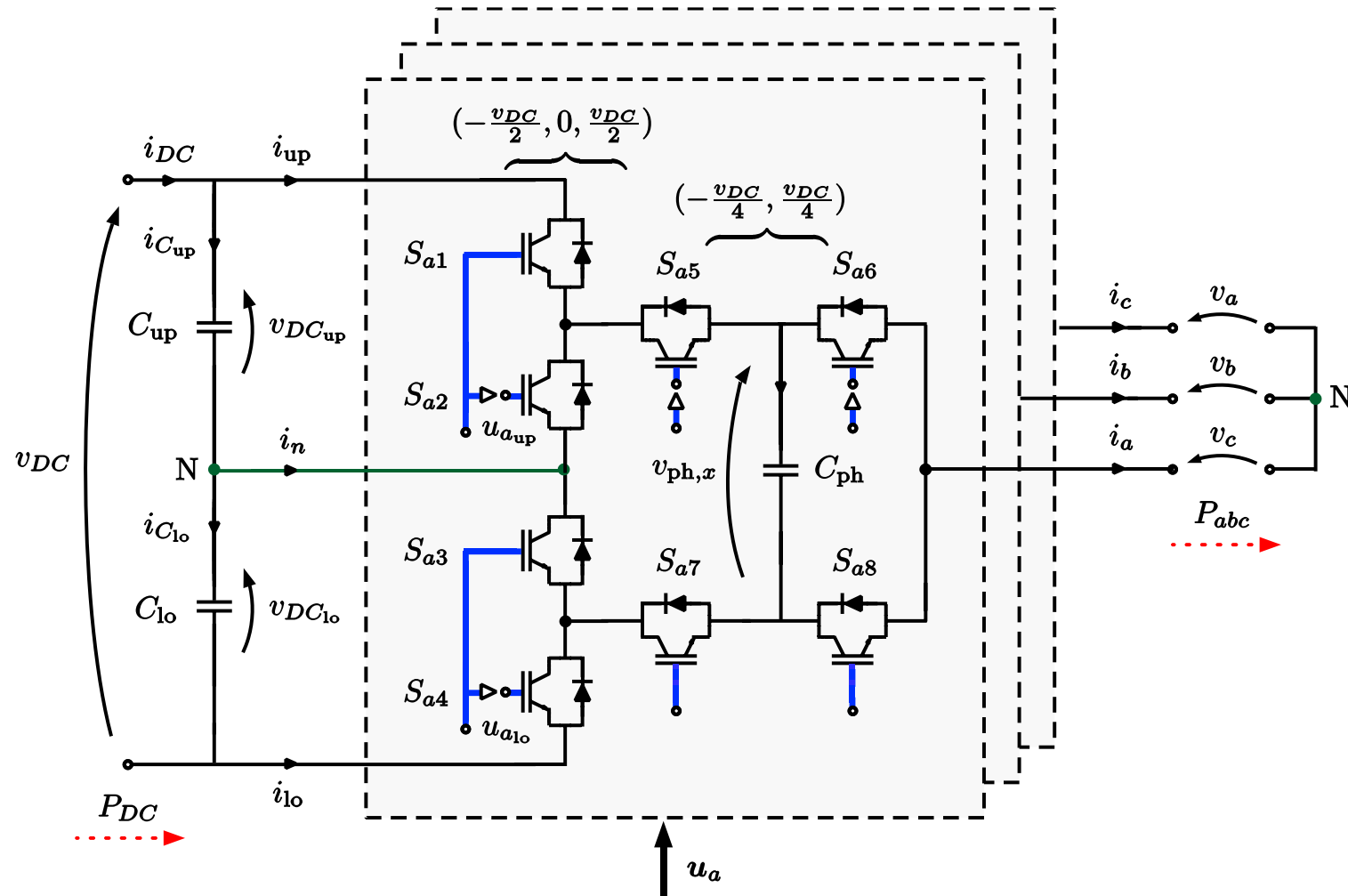
One leg

Three legs

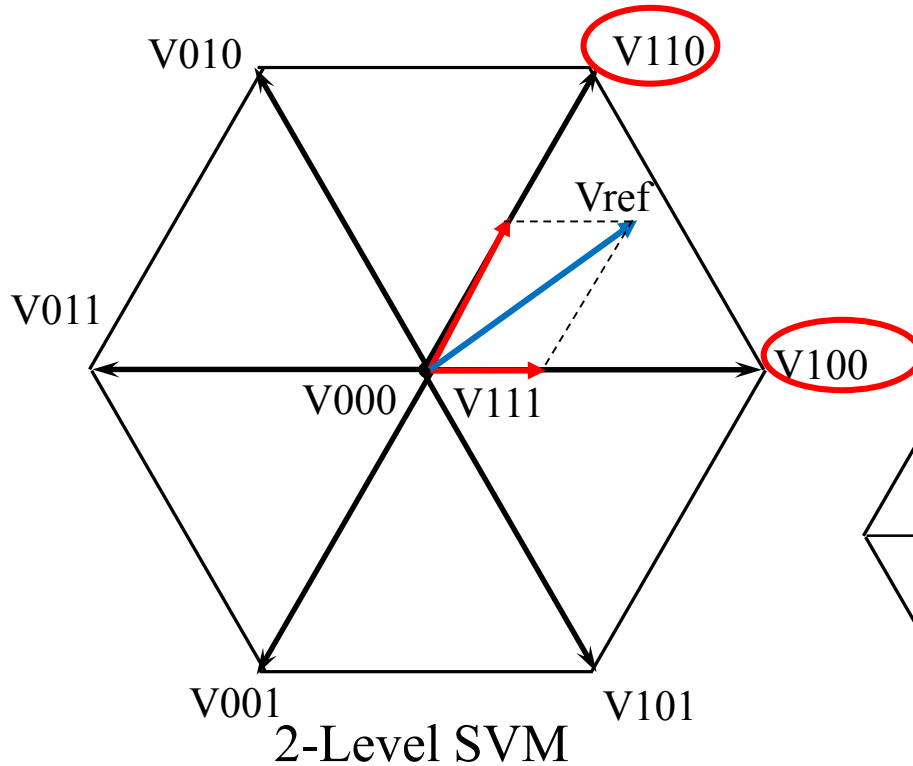


ON Semiconductor NXH40T120L3Q1 3-Phase TNPC Module

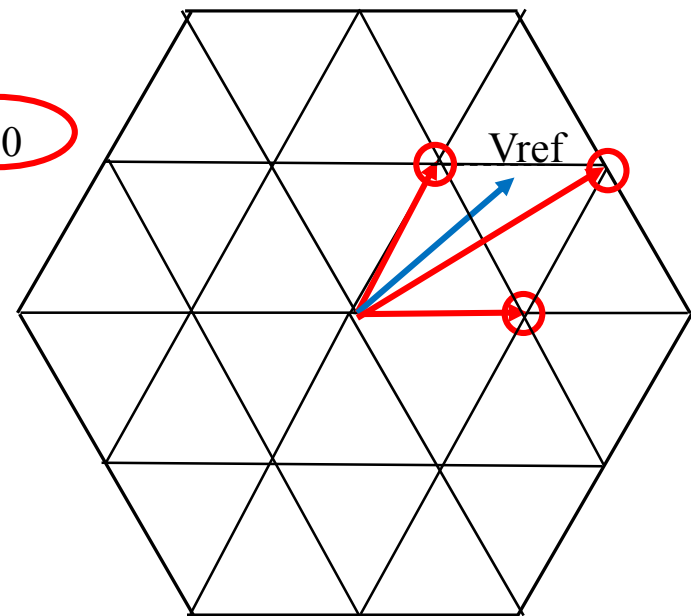
Multilevel converter: 5L-NPC



SVM for a 2L-VSC and 3L-NPC



3-Level SVM



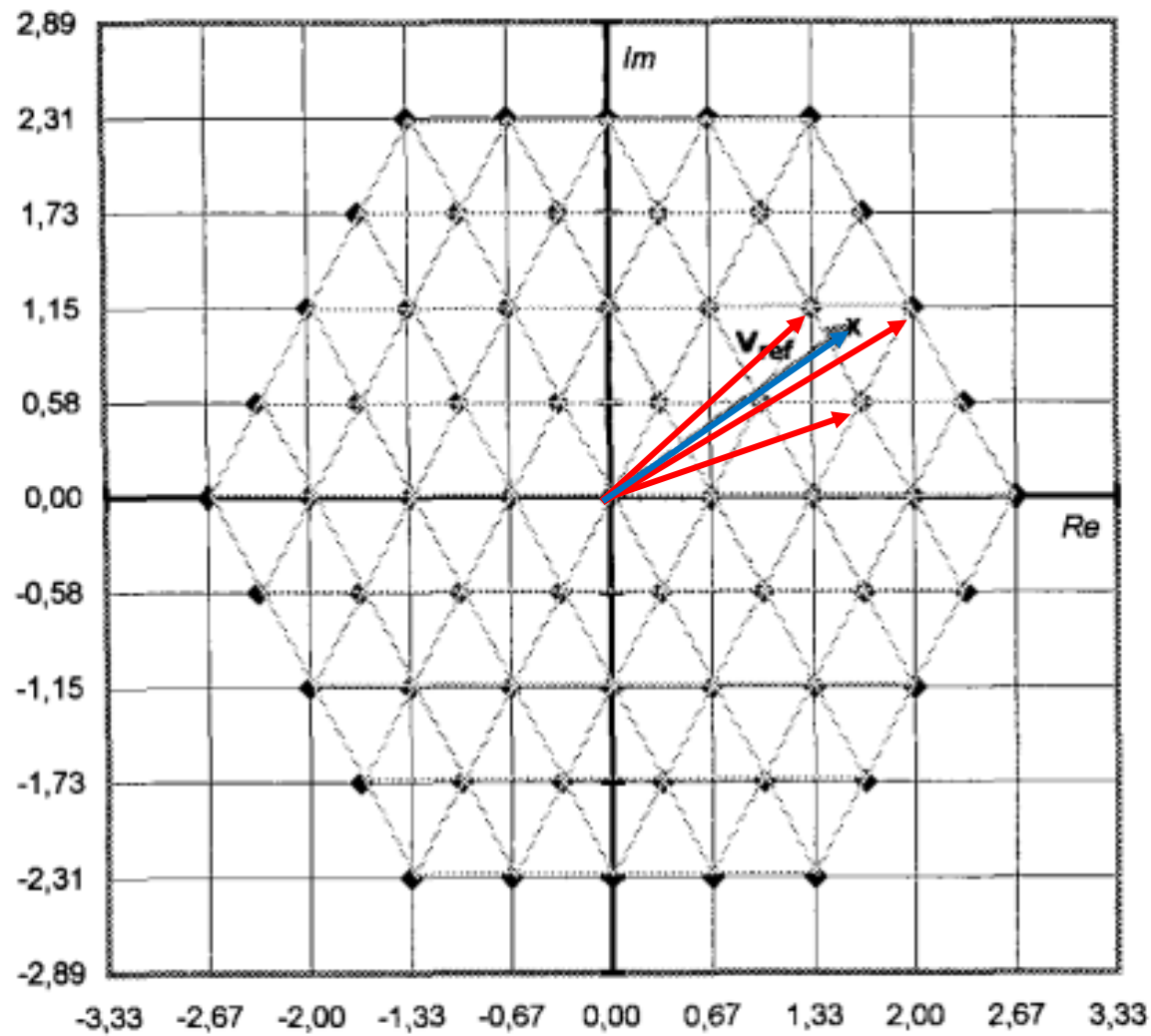
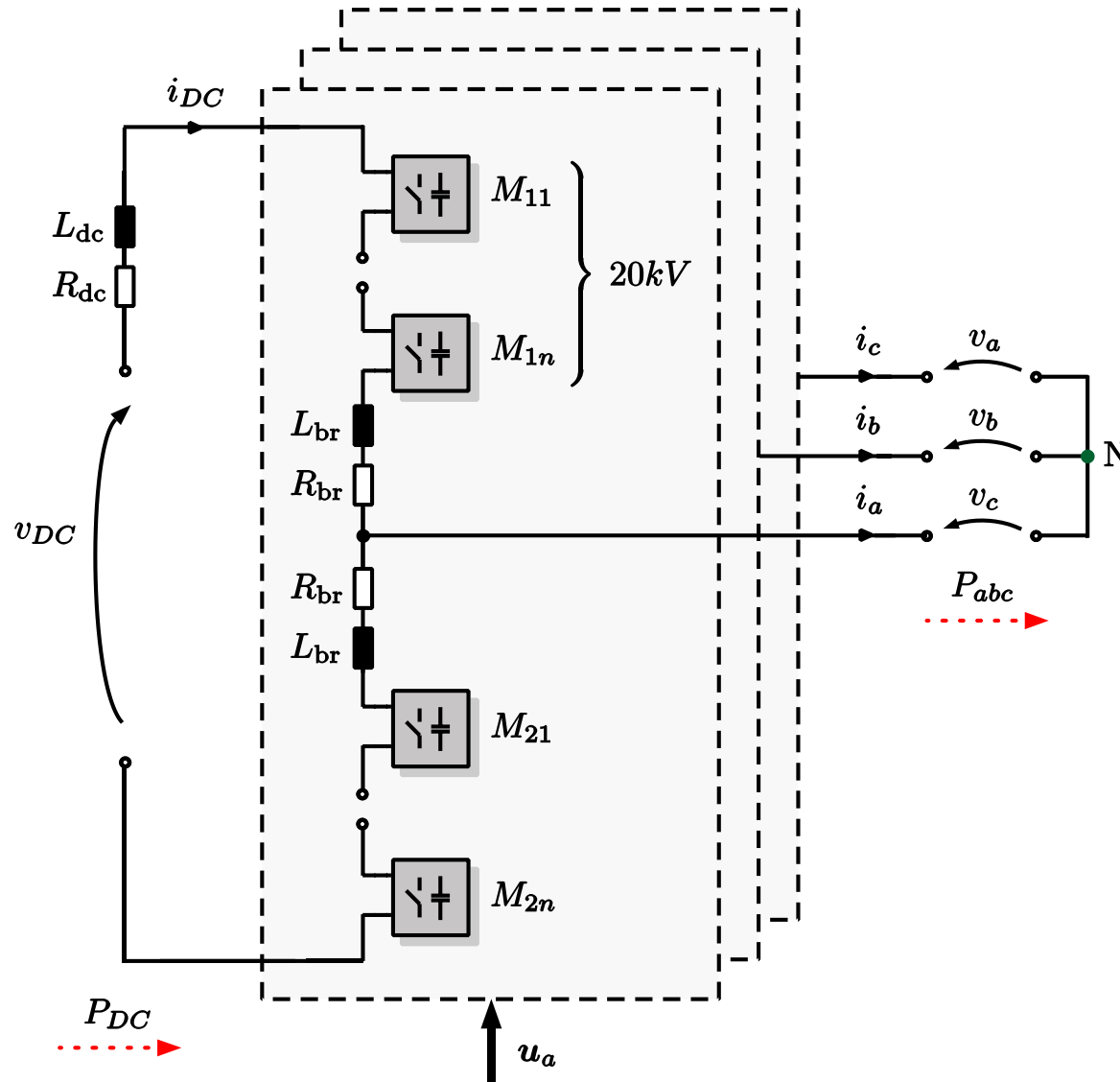


Fig. 3 Space-vector diagram of a 5-level converter

Multilevel converter: MMC



Control of active power fed into the grid

The derivative of the angle θ is the frequency ω .

Consider a steady state condition and a grid voltage constant in amplitude and frequency: $\overline{v_g} = V_g e^{j\omega t}$. Looking from the reference frame (dq), all the quantities are constant. Therefore:

$$\dot{\theta} = \omega$$

$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

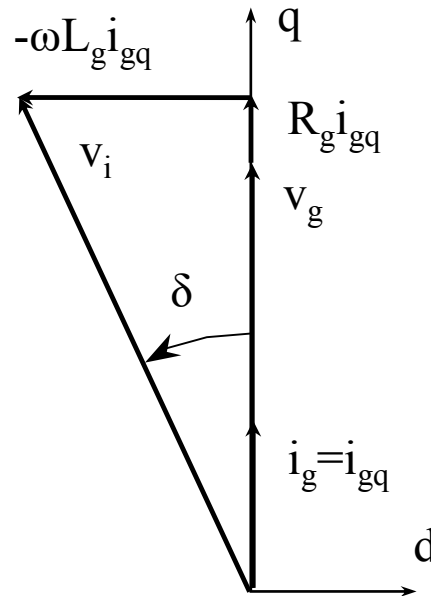
$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

If the current i_{gd} (component along the d axis) is fixed to zero, the phasor diagram is:

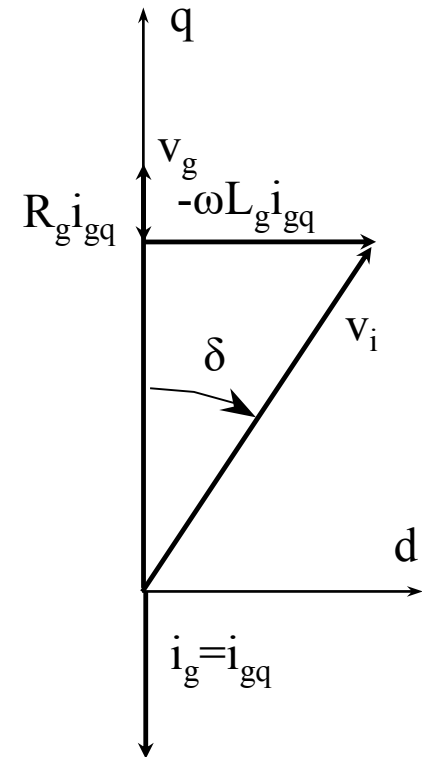
$$v_{id} = -\omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + V_g$$

Regenerative & $Q_g=0$



Rectifier & $Q_g=0$



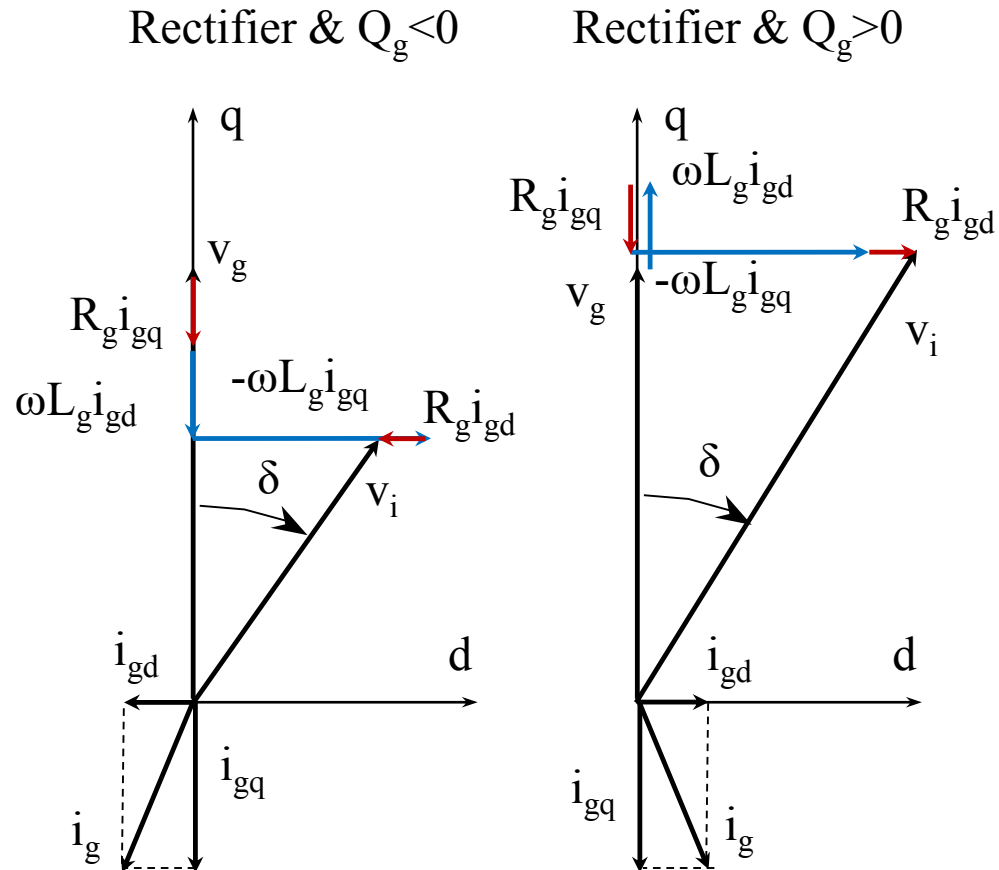
Control of active/reactive power fed into the grid

Controlling the current i_{gd} , it is possible to drive the reactive power: a positive value means that the inverter looks like a capacitor (and the grid like an inductor); vice versa for a negative value.

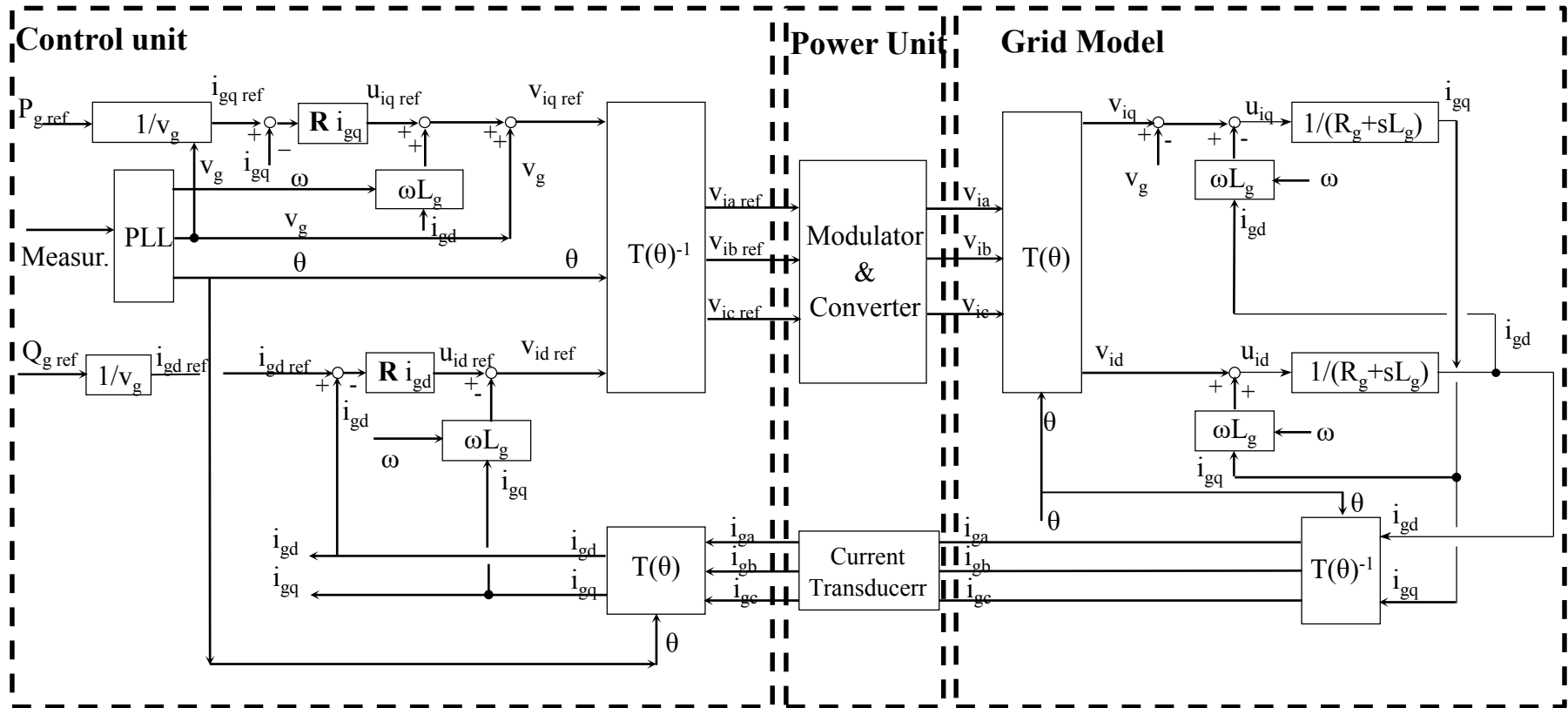
$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

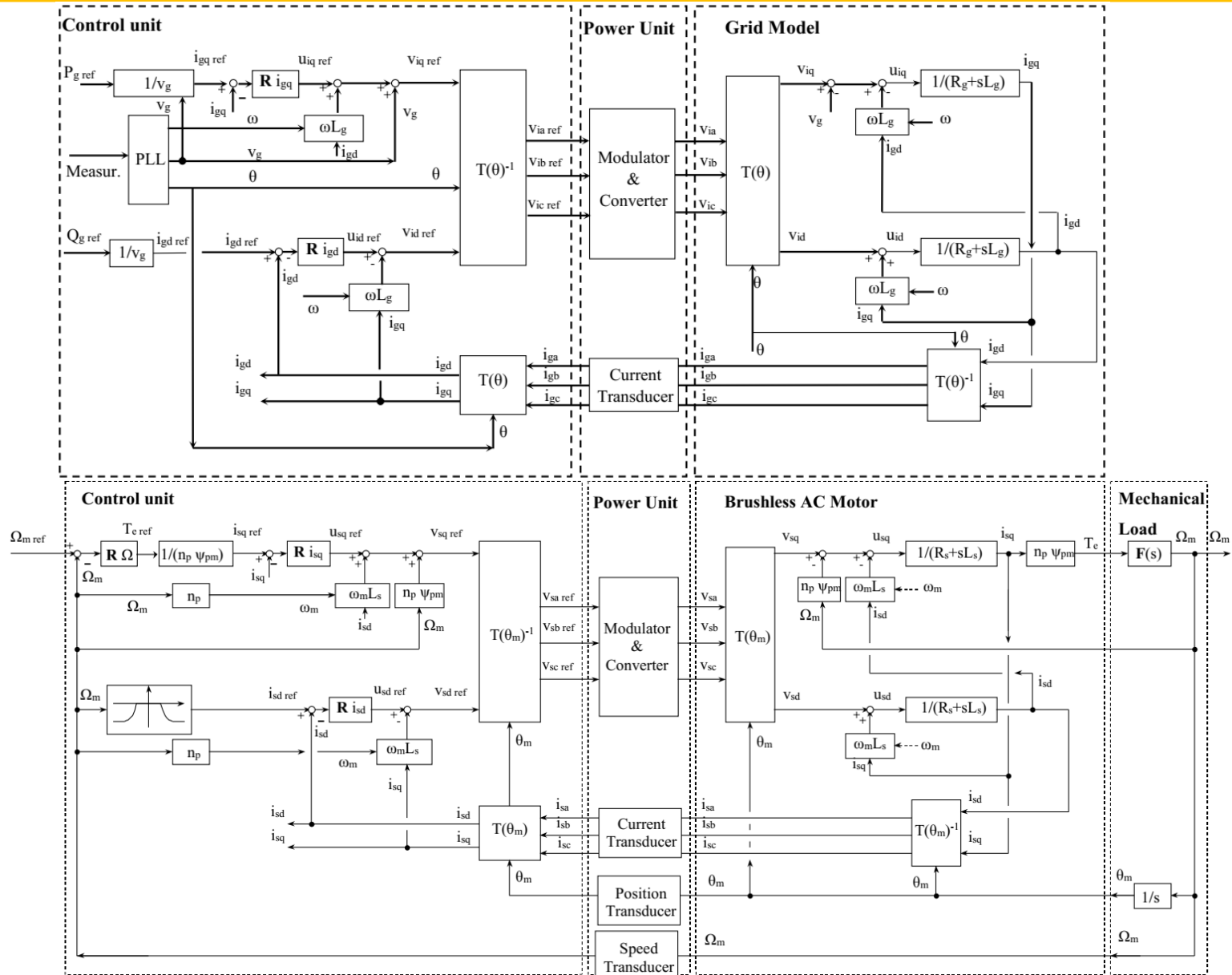
The figure represents the conditions when the active power is negative (operation as rectifier). Similar approach may be done for positive value of the active power (regeneration)



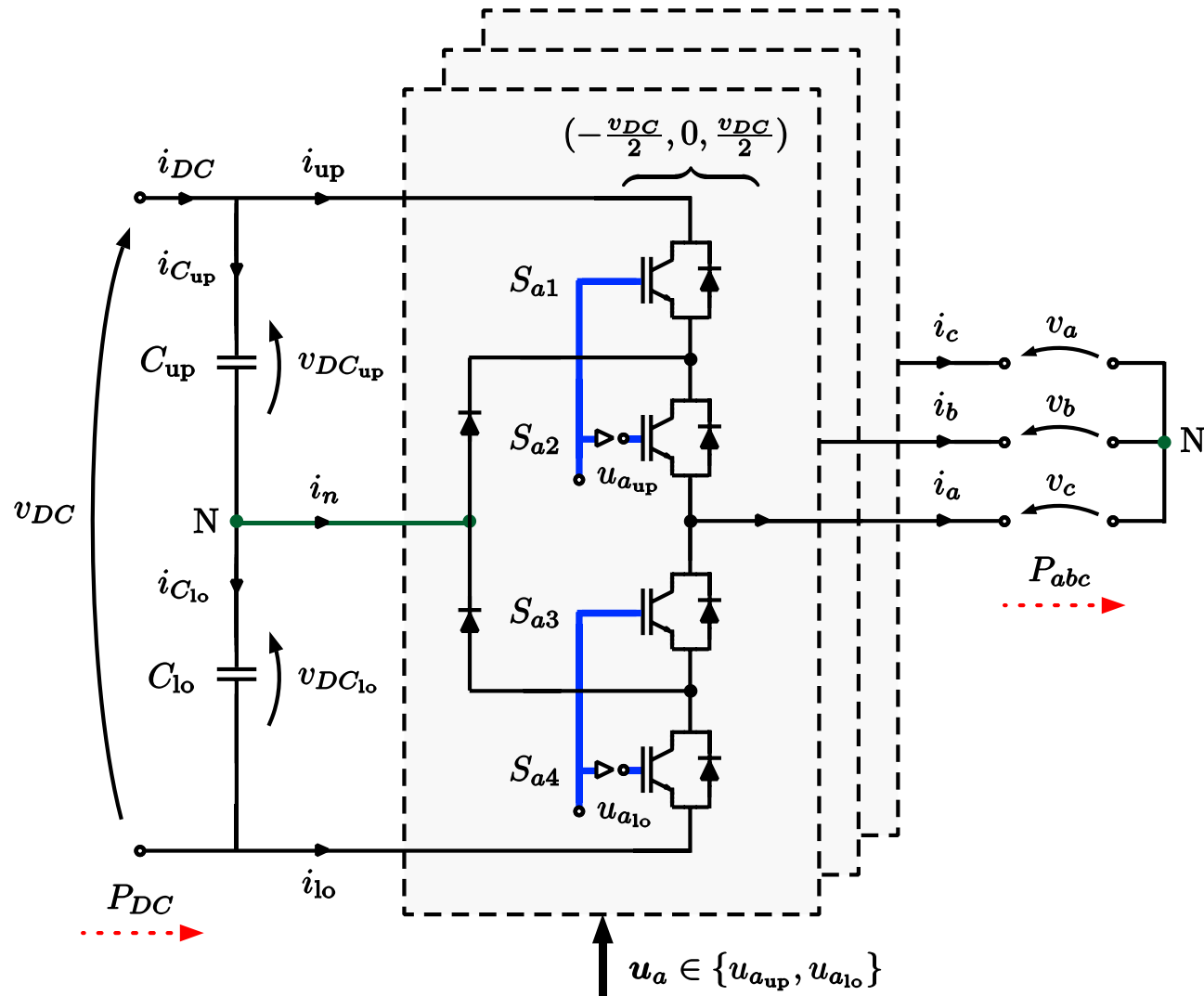
Control schemes



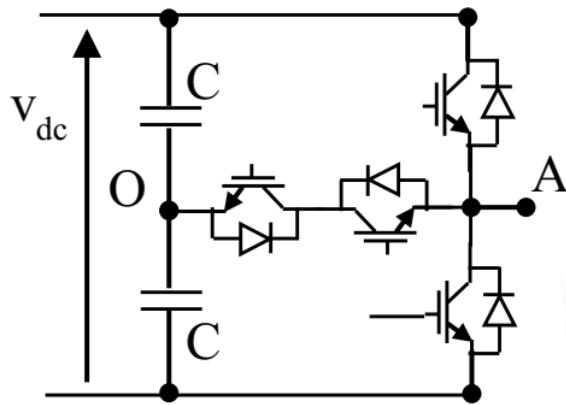
Control schemes: comparison with ac brushless



Multilevel converter: 3L-NPC

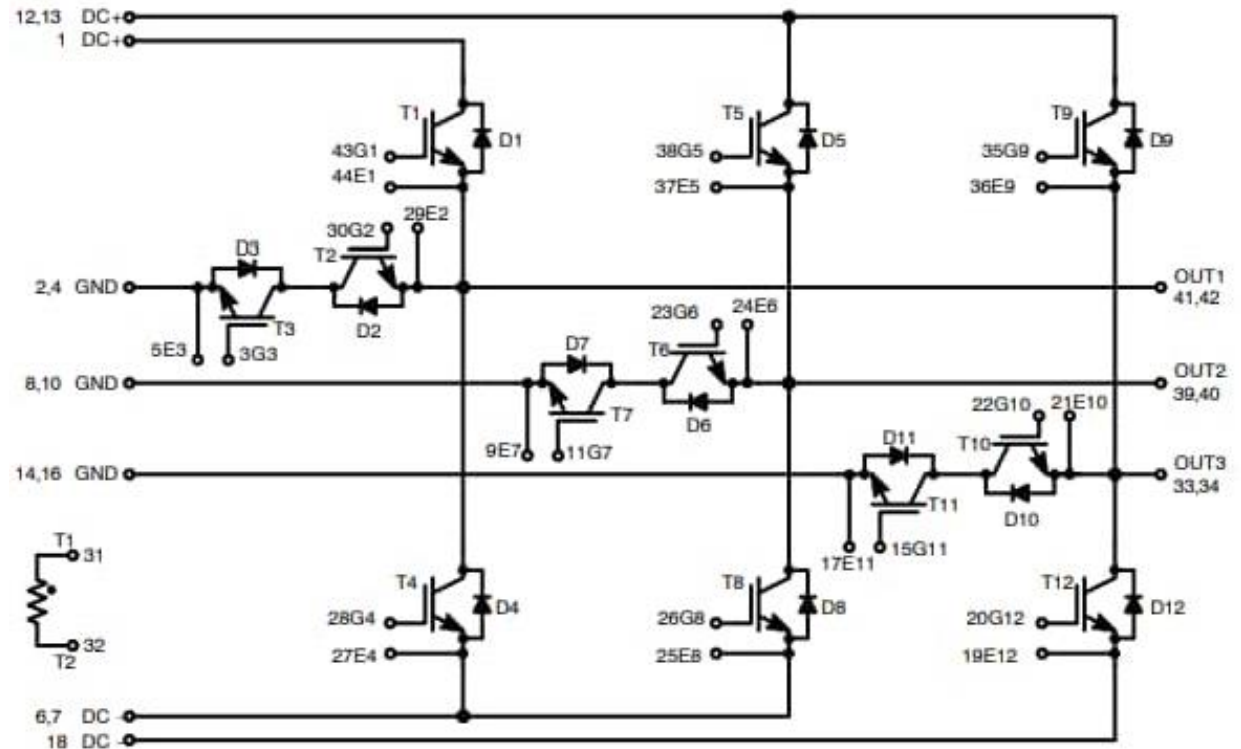


Multilevel converter: 3L-TNPCC



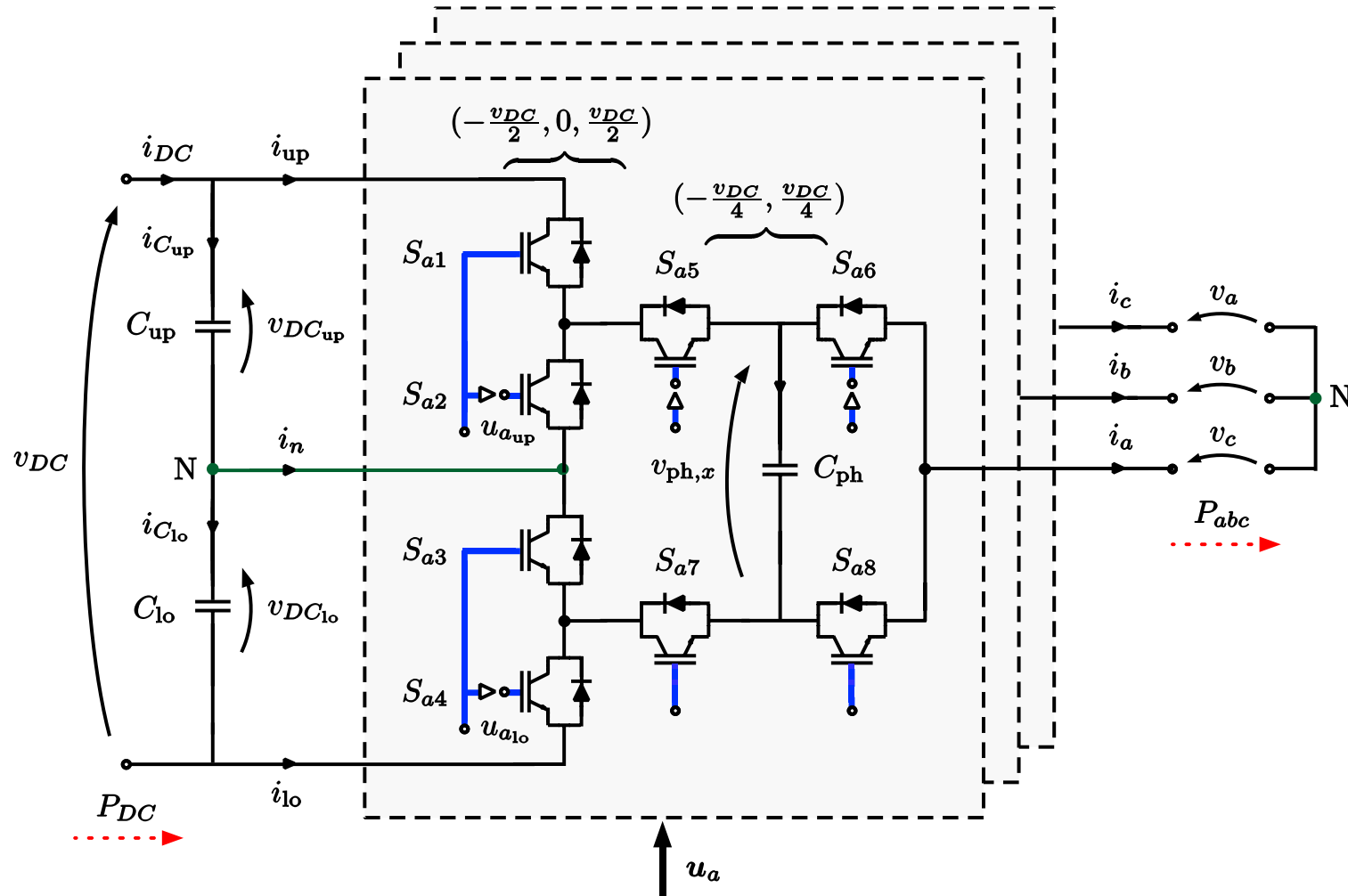
One leg

Three legs

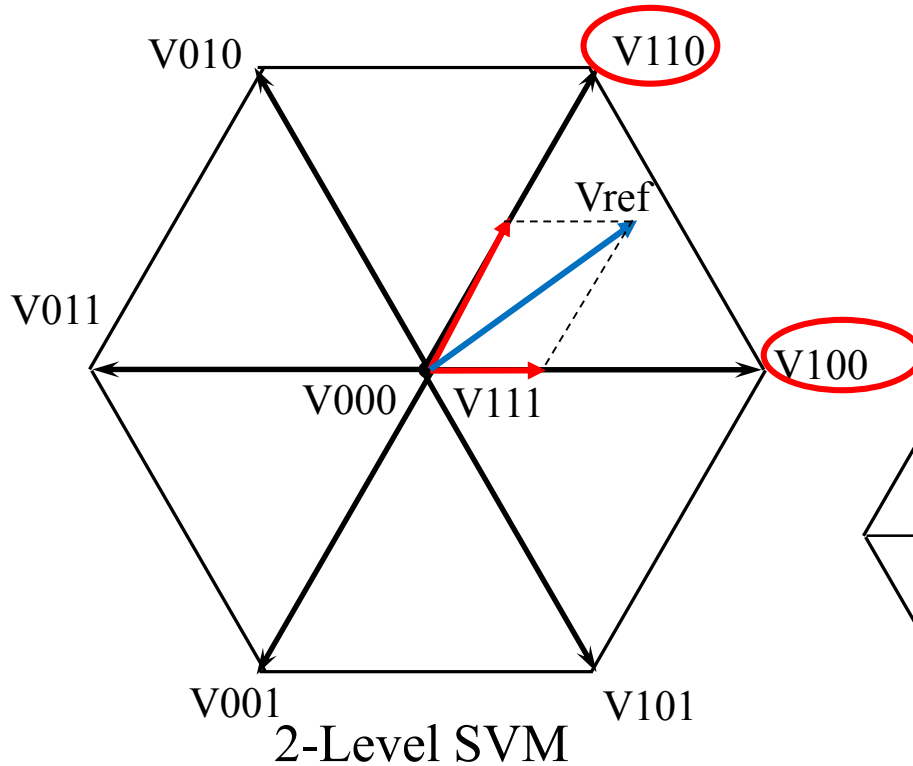


ON Semiconductor NXH40T120L3Q1 3-Phase TNPC Module

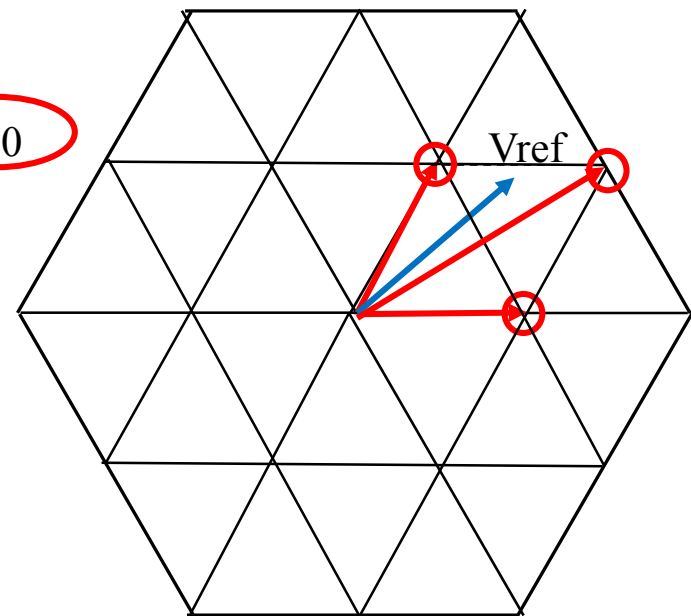
Multilevel converter: 5L-NPC



SVM for a 2L-VSC and 3L-NPC



3-Level SVM



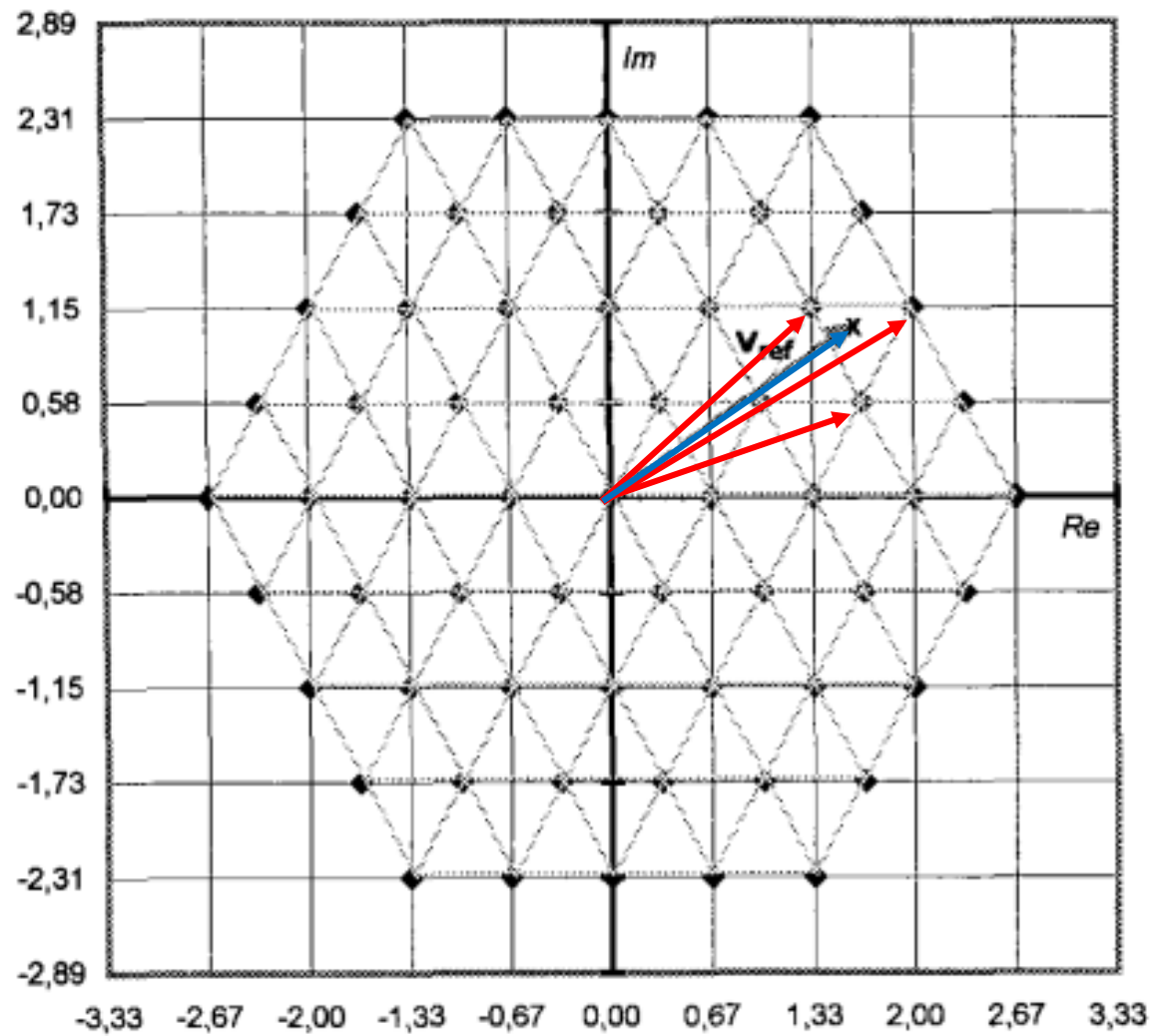
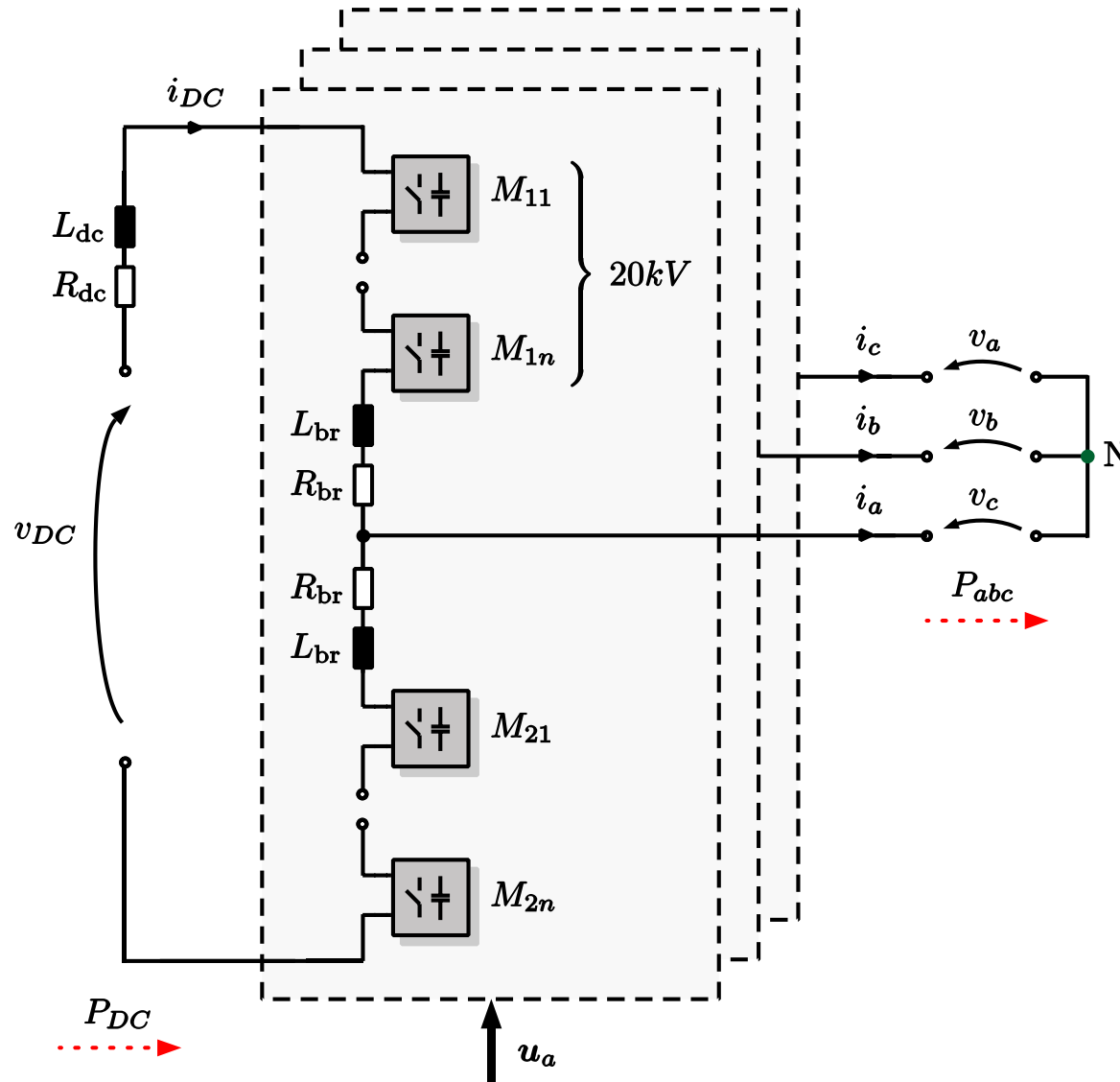
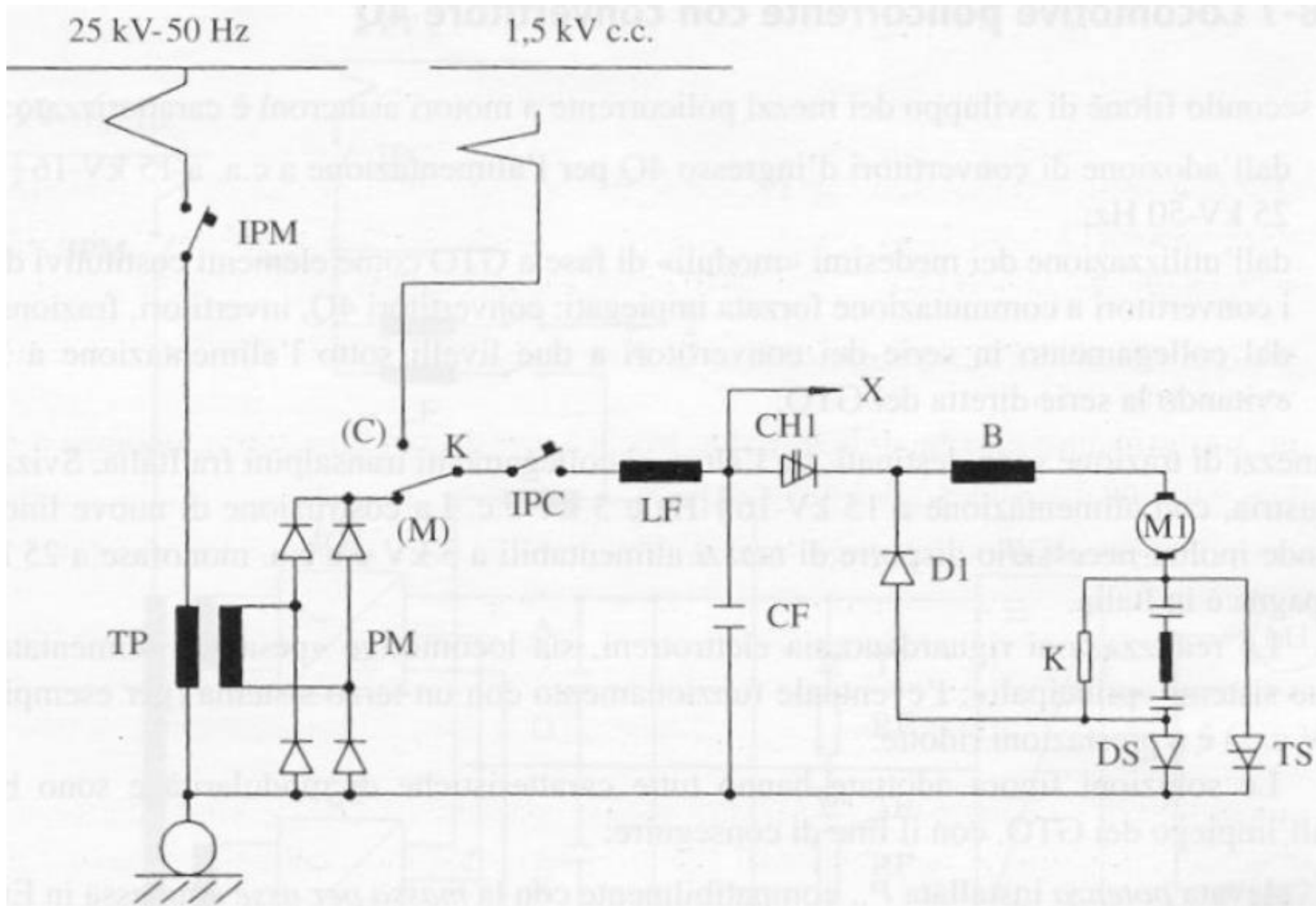


Fig. 3 Space-vector diagram of a 5-level converter

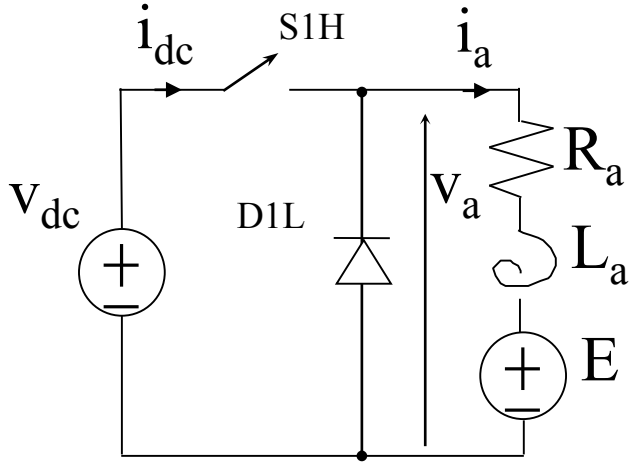
Multilevel converter: MMC



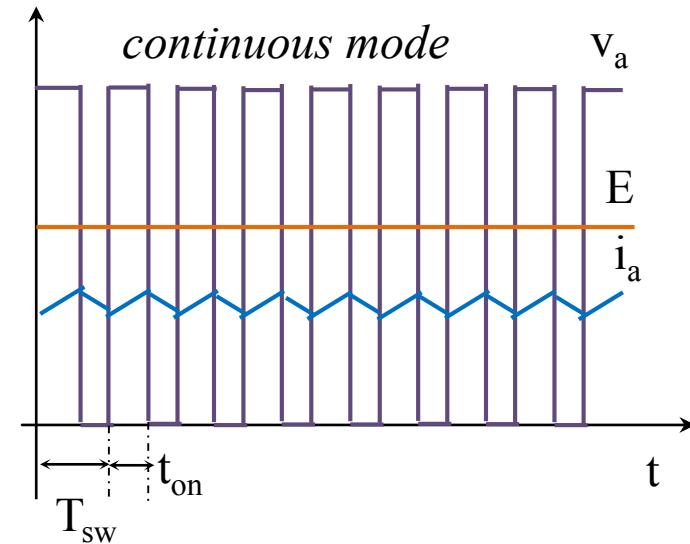
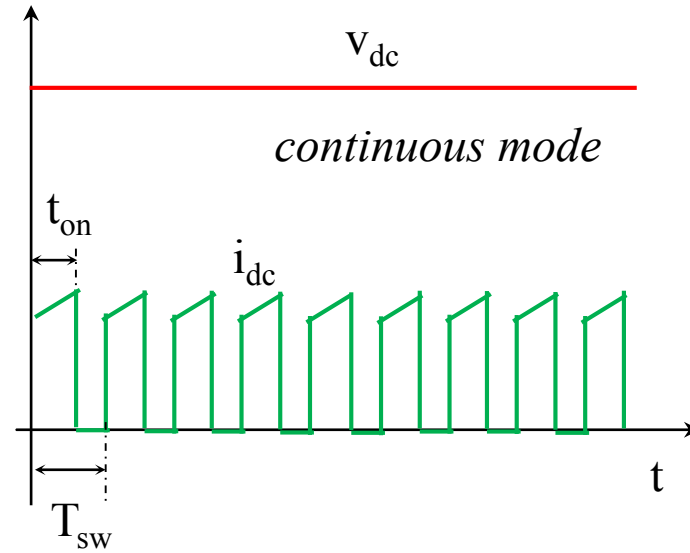
DC/DC: Chopper

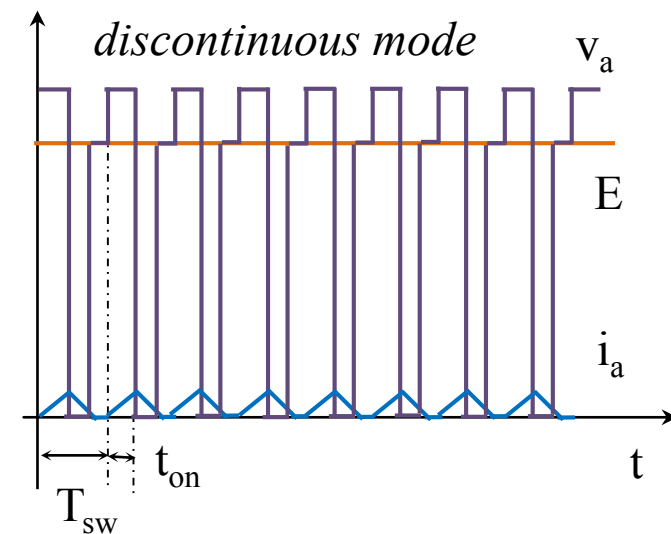
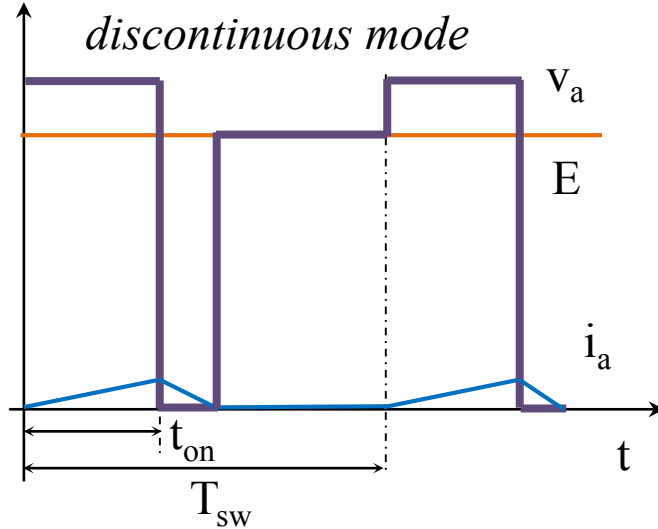
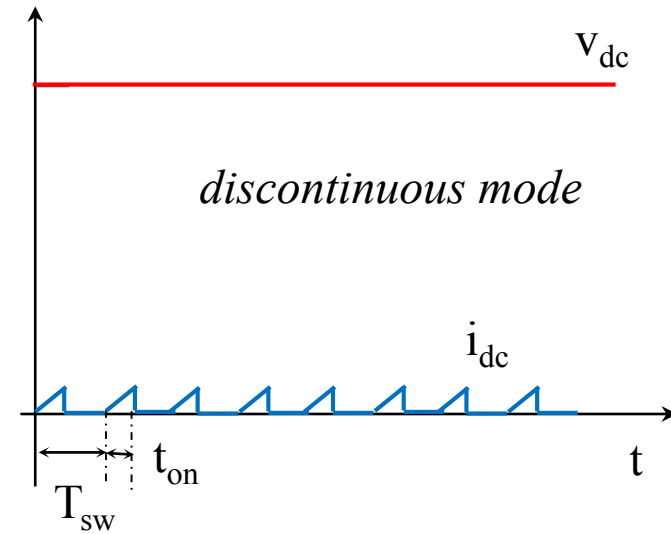
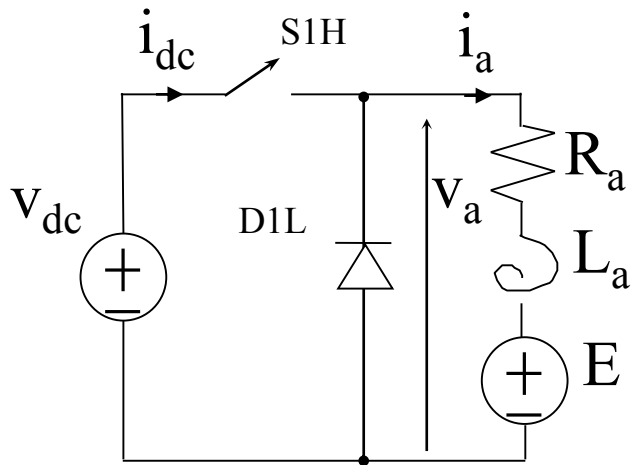


CH1: static switch, D1 free-wheeling diode



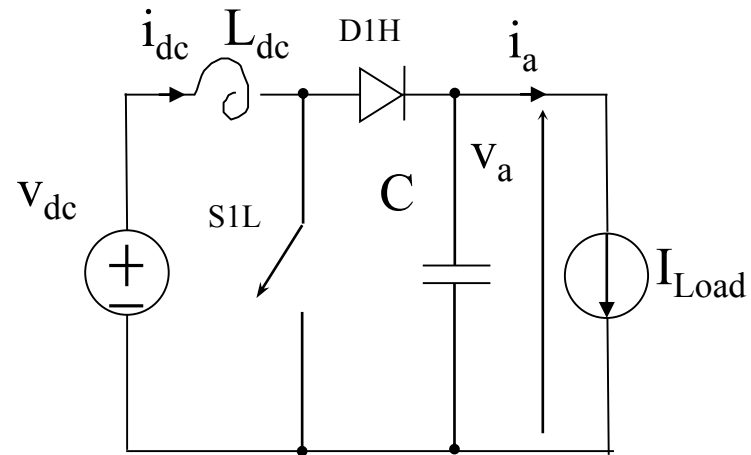
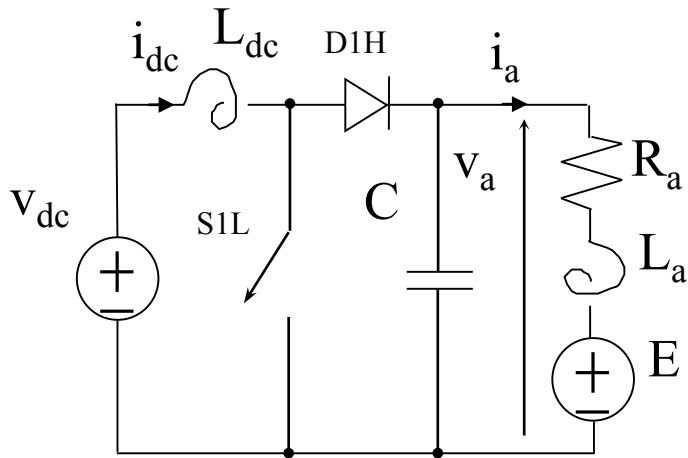
- Buck
- Step-down
- Chopper
- 1-quadrant



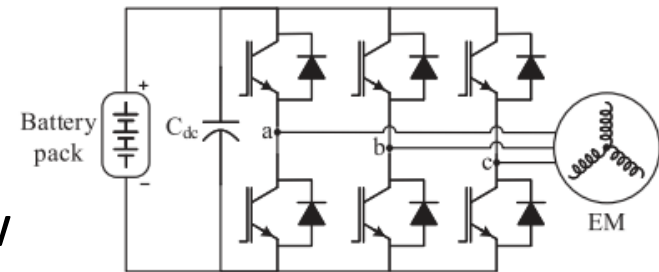


The output voltage is **not proportional** to the duty cycle

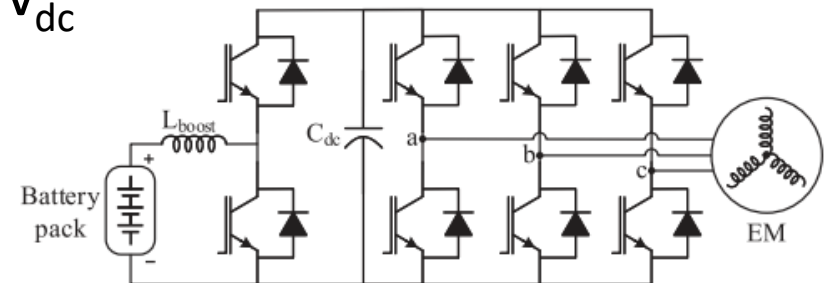
DC/DC: boost converter



- Also called step-up
- Used when the battery voltage is low
- The output voltage is higher than v_{dc}
- A soft-starter is needed
- PFC has similar characteristics
- Operating region

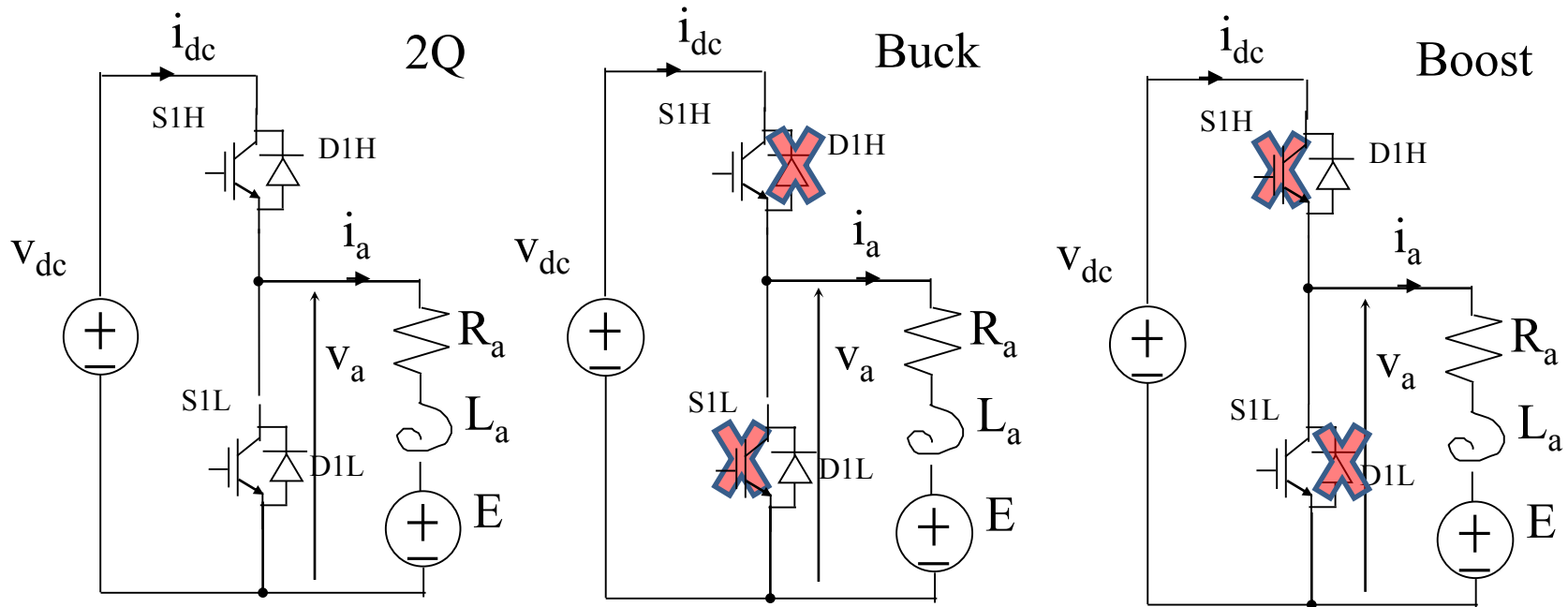


(a) VSI.



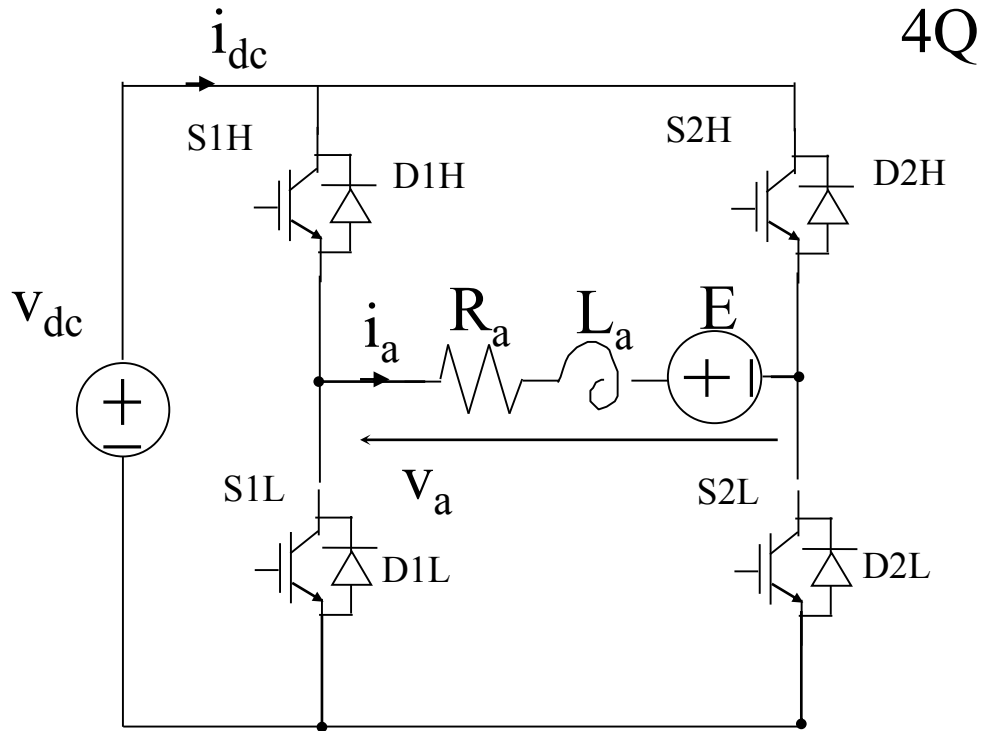
(b) VSI combined with a DC/DC boost converter.

DC/DC: half bridge



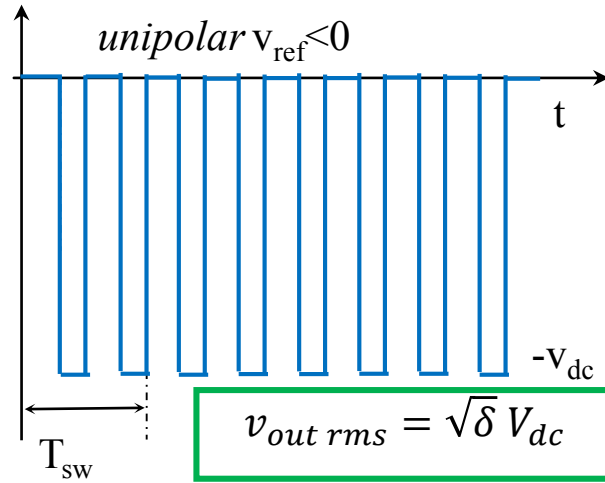
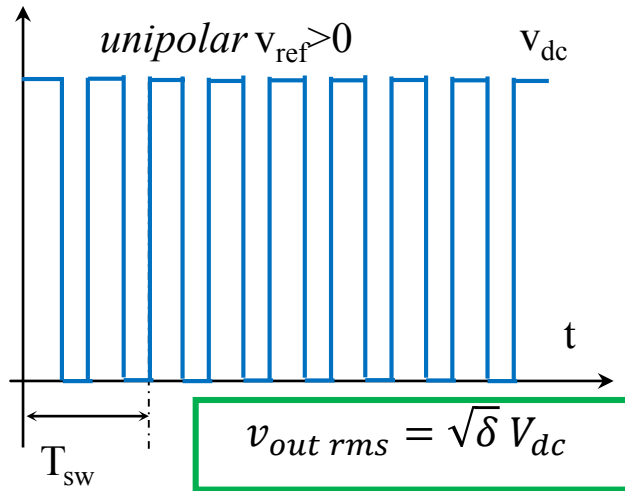
- Two quadrants converter: v_a always positive, i_a positive and negative
- From the mechanical point of view: speed always positive, torque positive (motor) or negative (brake)

DC/DC: Full bridge (H-bridge)

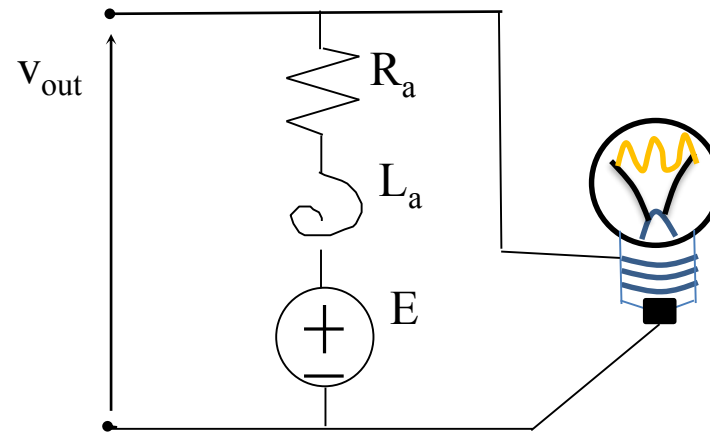
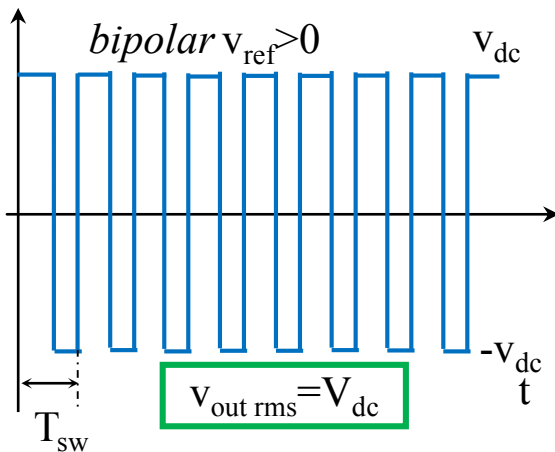


- Four quadrants converter (4Q): v_a always positive and negative, i_a positive and negative
- From the mechanical point of view: speed positive and negative, torque positive (motor) or negative (brake)

Modulation strategies: unipolar or bipolar



$$\Delta I_{uni} \cong \frac{1}{4} \Delta I_{bip}$$



Isolated DC/DC converters

- Recharging device (station or on-board system)
- Auxiliary supply (trains)

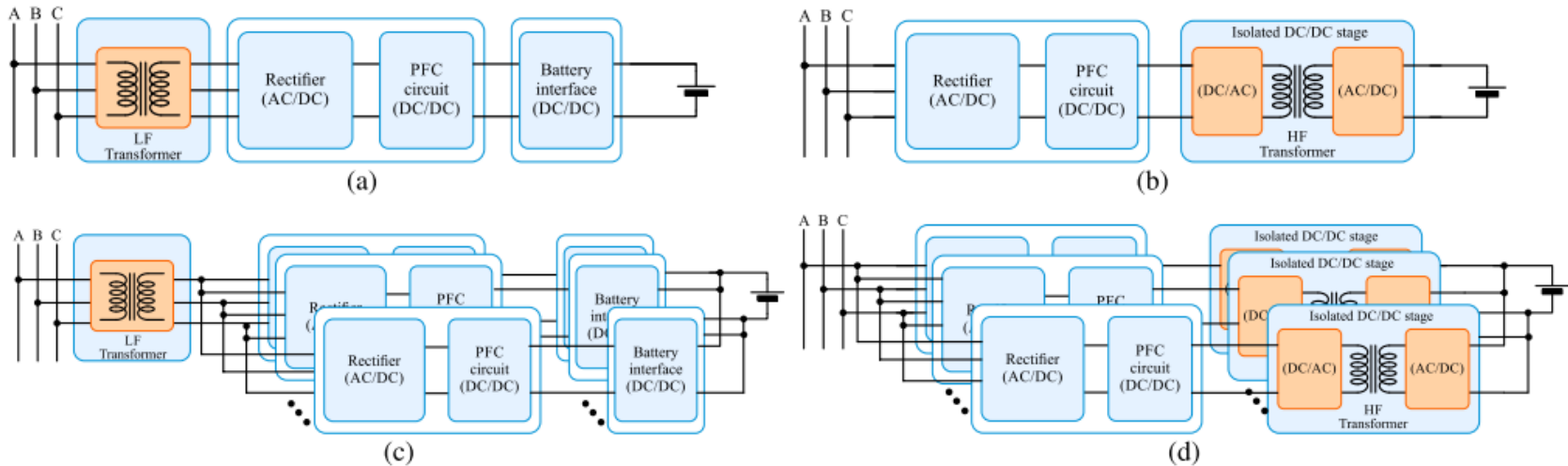
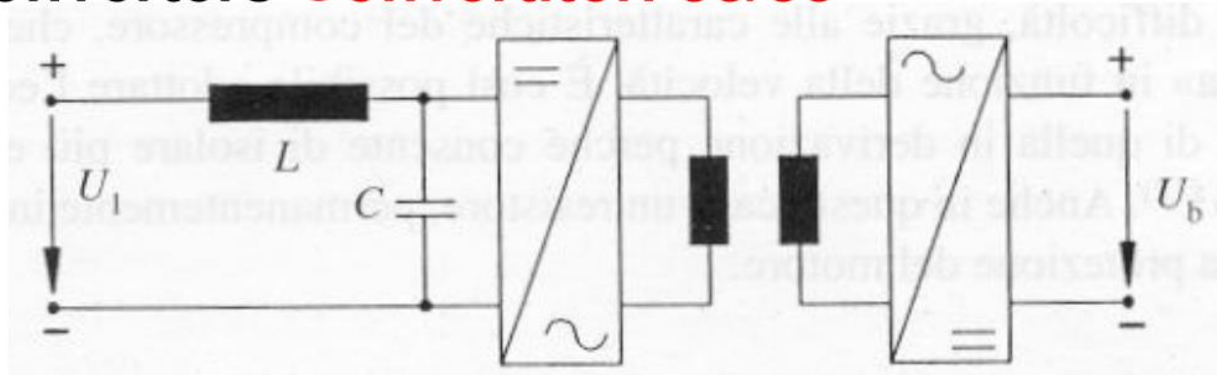
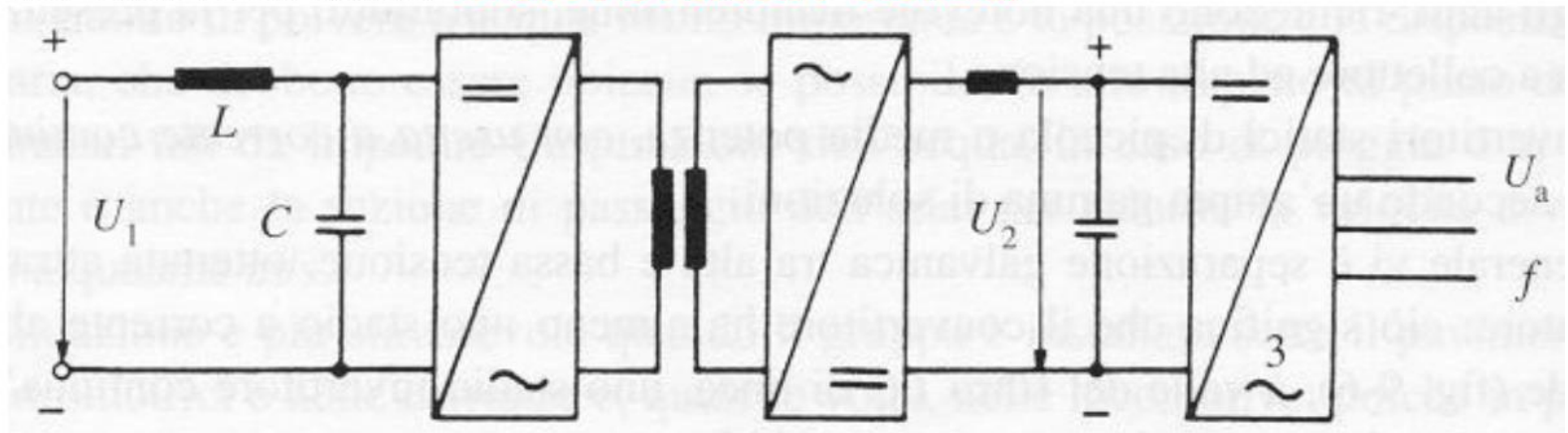


Fig. 1. Simplified block diagram of conventional dc fast charger power conversion systems. (a) Single-module charger with a nonisolated dc/dc converter. (b) Single-module charger with an isolated dc/dc converter. (c) Multiple paralleled modules shown in (a). (d) Multiple paralleled modules shown in (b).

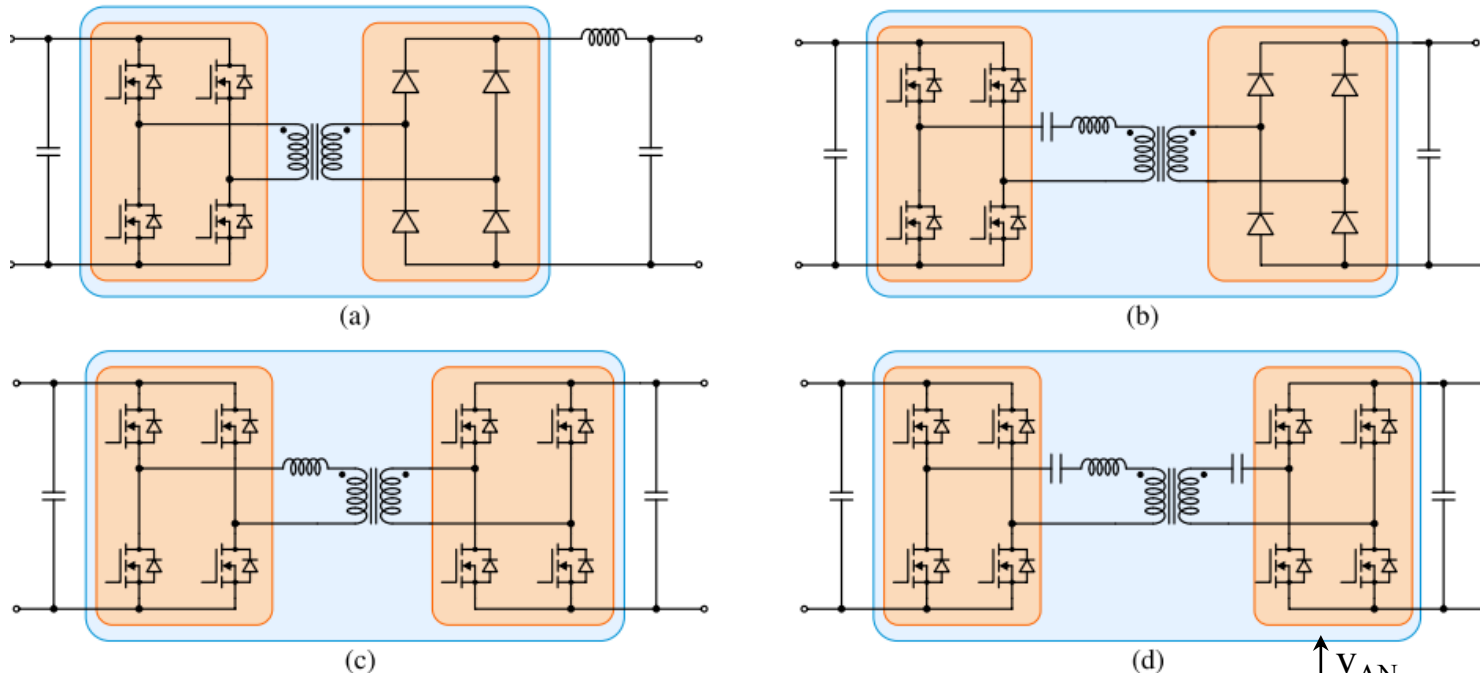
DC/DC converters **Convertitori cc/cc**



DC/AC converters **Convertitori cc/ca**

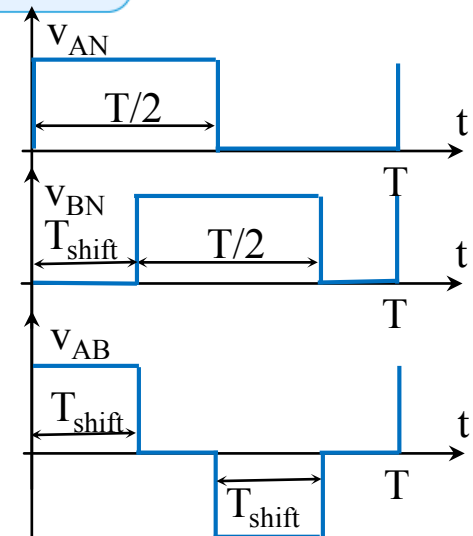


SST: solid-state transformer

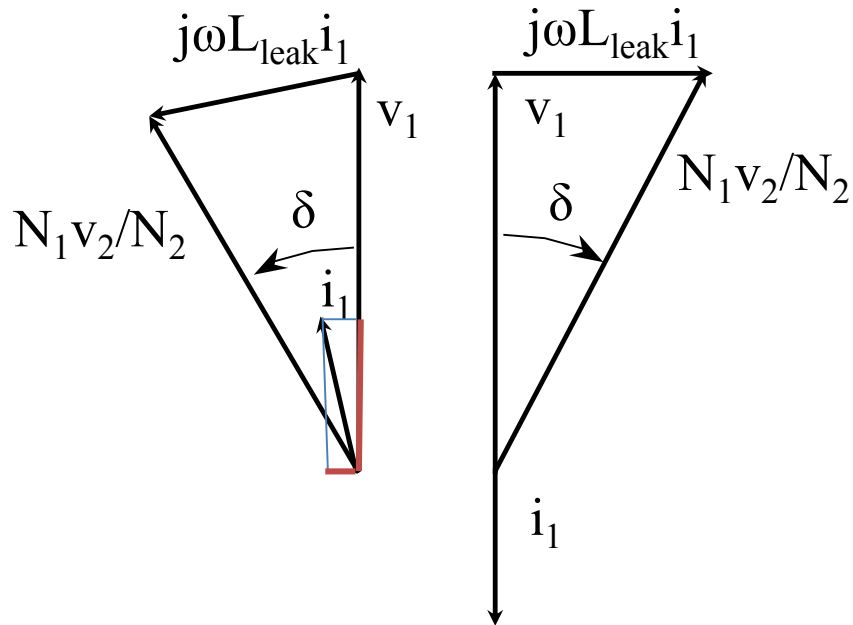
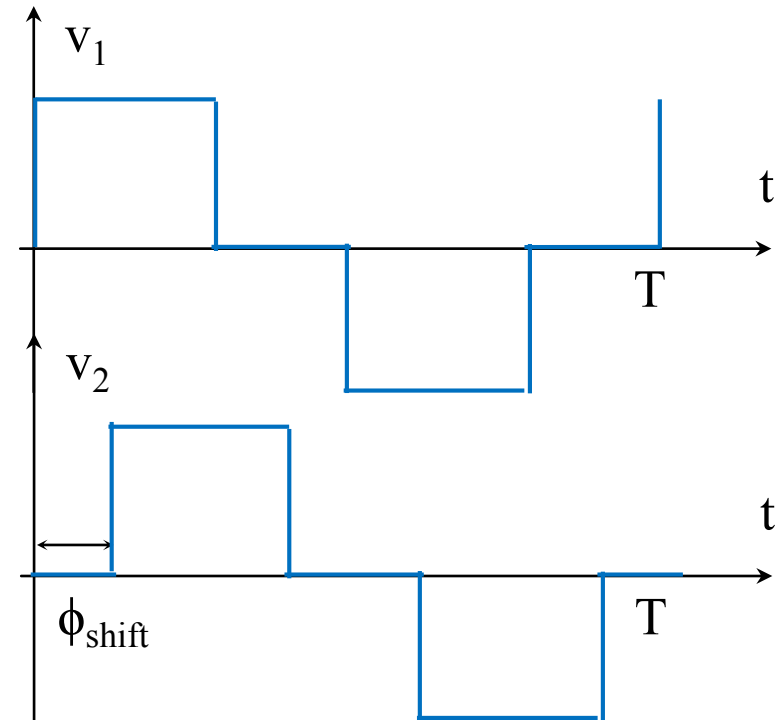
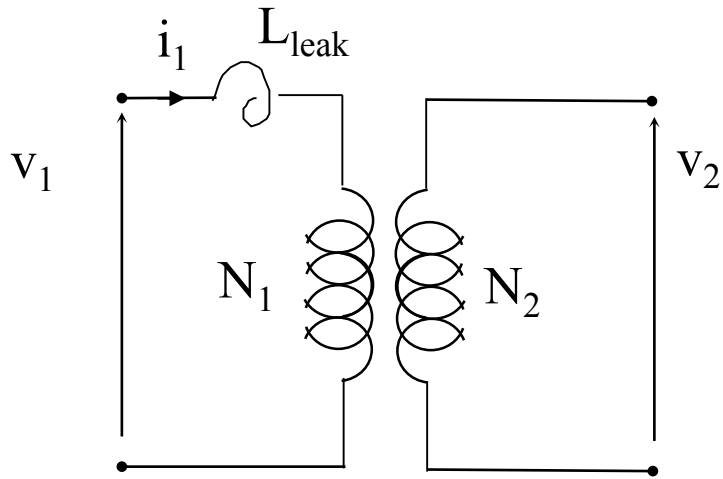


Isolated dc-dc converter topologies for dc fast chargers

- Phase-shift full-bridge (PSFB)
- LLC resonant converter
- Dual active bridge (DAB)
- CLLC resonant converter



Dual active bridge control technique



High power configuration

